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Stimulus-cleavable chemistry in the field of controlled drug delivery

Yufei Xue,^{†abc} Hua Bai,^{†a} Bo Peng,^{*ab} Bin Fang,^a Jonathan Baell,^{id b} Lin Li,^{*a}
Wei Huang^{*af} and Nicolas Hans Voelcker^{id *abcde}

Stimulus-cleavable nanoscale drug delivery systems are receiving significant attention owing to their capability of achieving exquisite control over drug release via the exposure to specific stimuli. Central to the construction of such systems is the integration of cleavable linkers showing susceptibility to one stimulus or several stimuli with drugs, prodrugs or fluorogenic probes on the one hand, and nanocarriers on the other hand. This review summarises recent advances in stimulus-cleavable linkers from various research areas and the corresponding mechanisms of linker cleavage and biological applications. The feasibility of extending their applications to the majority of nanoscale drug carriers including nanomaterials, polymers and antibodies are further highlighted and discussed. This review also provides general design guidelines to incorporate stimulus-cleavable linkers into nanocarrier-based drug delivery systems, which will hopefully spark new ideas and applications.

^a Frontiers Science Center for Flexible Electronics, Xi'an Institute of Flexible Electronics (IFE) and Xi'an Institute of Biomedical Materials & Engineering, Northwestern Polytechnical University, 127 West Youyi Road, Xi'an 710072, China. E-mail: iambpeng@nwpu.edu.cn, iamlli@nwpu.edu.cn, provost@nwpu.edu.cn

^b Commonwealth Scientific and Industrial Research Organisation (CSIRO), Clayton, Victoria 3168, Australia

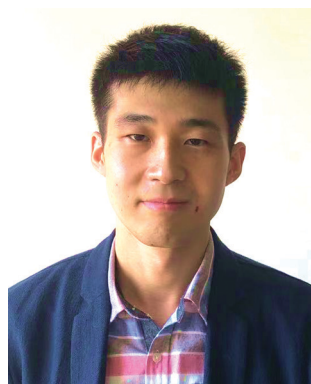
^c Drug Delivery, Disposition and Dynamics, Monash Institute of Pharmaceutical Sciences, Monash University, Parkville, Victoria 3052, Australia. E-mail: nicolas.voelcker@monash.edu

^d Melbourne Centre for Nanofabrication, Victorian Node of the Australian National Fabrication Facility, Clayton, Victoria 3168, Australia

^e Department of Materials Science & Engineering, Monash University, Clayton, Victoria 3168, Australia

^f Key Laboratory of Flexible Electronics (KLOFE) & Institute of Advanced Materials (IAM), Jiangsu National Synergetic Innovation Center for Advanced Materials (SICAM), Nanjing Tech University (Nanjing Tech), 30 South Puzhu Road, Nanjing, 211816, P. R. China

[†] Co-first author.



Yufei Xue

Yufei Xue received his BSc degree (2017) in Pharmaceutical Engineering from China Pharmaceutical University, Nanjing, and he is now a PhD student under the supervision of Professor Nicolas Voelcker. His current area of research includes, amongst others, design and synthesis of stimulus-cleavable linkers for controlled drug release from porous silicon nanoparticles. He is also interested in modifying the surface chemistry of porous silicon-based nanostructures for biosensing.



Hua Bai

Hua Bai received her PhD in Toxicology from the Fourth Military Medical University in 2011. In 2019, she obtained her current assistant professor position in Northwestern Polytechnical University (China). She has a long experience of studying redox biological signaling in human diseases, with a special focus on redox regulation in liver regeneration. Currently, her research interests involve mitochondrial dynamics and quality control in diseases associated with mitochondrial dysfunction.

1. Introduction

Over the past few decades, significant progress has been achieved in the field of drug discovery, affording new therapeutic candidates with high efficacy and safety for treatment of different diseases.^{1–6} However, traditional pharmaceuticals, especially small molecular drugs follow a non-specific distribution path within human body when administrated *via* traditional drug formulations, and therefore may induce unacceptable toxicity at healthy tissues, causing severe side effects and narrowing the therapeutic index.^{7,8} Moreover, several other drawbacks also prevent promising drug candidates from moving to the clinical applications, such as hydrophobicity,^{9–12} development of multidrug resistance (MDR)^{13–16} and low cell/tissue (barrier) permeability.^{17–20} The problems associated with traditional pharmaceuticals have given an impetus to the discovery of drug delivery systems (DDSs), a method or device designed to decrease exposure of healthy organs or cells to

the drugs^{21,22} and improve the pharmacokinetic (PK) parameters of drugs, such as bioavailability^{23–25} and plasma clearance.²⁶ For instance, anticancer drugs loaded nanoscale materials (e.g. engineered nanoparticles and polymers–drug conjugates (polymer–DCs)) showed promising accumulation at tumour sites by exploiting the enhanced permeability and retention (EPR) effect, leading to increased specific targeting ability and reduced cytotoxicity to normal cells.^{27,28}

Although substantial advances have been made in the development of DDSs, poorly controlled drug release behaviour from DDSs emerged as a main issue, which causes undesired premature drug release and exposure to healthy tissues, and thus impairs the therapeutic effect.²⁹ In order to achieve an ideal concentration of drugs at target sites, the concept of controlled drug release was incorporated into the DDSs.^{30,31} Early efforts have been devoted to exploring materials capable of releasing the



Bo Peng

Bo Peng received his BSc degree in Clinical Pharmacy (2011) from Huazhong University of Science and Technology, his MRes degree in Medicinal Chemistry at the University College London (2012) and his PhD degree in Chemical Biology (2017) at National University of Singapore under the supervision of Professor Shao Q. Yao. In 2017 Bo joined the research group of Professor Nicolas Voelcker as a postdoctoral research fellow at Commonwealth Scientific and

Industrial Research Organisation (CSIRO). Bo's research interest mainly focuses on the applications of chemistry and biology in lab-on-a-chip platforms and nanomedicine.



Lin Li

Lin Li received his BSc and PhD in Chemistry from Anhui University (China) in 2004 and 2009, respectively. He carried out postdoctoral studies with professor Shao Q. Yao in NUS (Singapore). In 2015, he obtained a professor position in Nanjing Tech University (China). In 2018, he obtained his current professor position in Northwestern Polytechnical University (China). His research group focuses on synthetic small-molecule indicators for monitoring biological/chemical species in physiological environments.



Wei Huang

Wei Huang received his PhD degree from Peking University (1992). He began his postdoctoral research in National University of Singapore (NUS, 1993); moved to Fudan University where he founded the Institute of Advanced Materials (IAM, 2002); was appointed as the Deputy President of Nanjing University of Posts and Telecommunications (NJUPT, 2006); was elected as Academician of the Chinese Academy of Sciences (CAS) in 2011. He assumed his duty as the

President of Nanjing Tech University (NanjingTech, 2012); and was appointed as Deputy President & Provost of Northwestern Polytechnical University (NPU, 2017). His research interests include organic/flexible electronics, nanomaterials, and nanotechnology.



Nicolas Hans Voelcker

Nicolas Voelcker received his BSc at the University of Saarland (1993), his MSc at the RWTH Aachen (1995) and his PhD degree (1999) at the DWI Leibniz Institute. He received postdoctoral fellowships at the Scripps Research Institute in La Jolla. In 2001, he became a Lecturer at Flinders University, an Associate Professor in 2006 and a full Professor in 2008. He was a Professor at the University of South Australia (2012–2017). Since 2017, he is the Scientific Director of the

Melbourne Centre for Nanofabrication, Professor at Monash University and Science Leader at the Commonwealth Scientific and Industrial Research Organisation (CSIRO).

drugs in a controlled manner, and numerous materials have attracted significant attention. In particular, nanomaterials, such as porous nanoparticles,^{32,33} liposomes^{34–36} and polymers,^{37–40} have been harnessed for their potential to achieve controlled drug release with promising results.^{41–43} However, the main problem suffered from these attempts is that the drug molecules are mainly loaded in the drug carriers *via* a non-covalent bond, which may cause drug leakage during the administration. Covalent bonding strategies were then found to provide a convenient means of achieving controlled drug release behaviour, either indirectly through building molecular gates^{44,45} or directly *via* tethering pharmaceuticals to the proper carriers,^{46,47} where the drug release is regulated by a stimulus-cleavable linker (SCL).

SCLs are defined as chemical structures with sensitivity to a single stimulus or multiple stimuli. Cleavage of the structure only occurs after the exposure to corresponding stimulus.^{48–50} This unique relationship between the linker and stimulus has attracted significant attention for realising controlled drug release in DDSs, where the SCLs bridge the drug molecules or capping agents to the drug carriers. The SCLs can remain stable in the circulation, while exposure of them to the corresponding stimulus can induce the cleavage to enable drug release. The stimuli to trigger the cleavage of the linkers can be categorised as exogenous stimuli and endogenous stimuli, both of which can be exploited for DDSs construction. The exogenous stimuli cover light,⁵¹ temperature,⁵² magnetic field,⁵³ high energy radiation⁵⁴ and ultrasound,⁵⁵ conveying tremendous advantages including ease of tuning and precision of control.⁵⁶ As a result of on-demand linker cleavage, DDSs featuring those linkers show a promising way to overcome inter-patient variability.⁵⁷ Endogenous stimulus-induced linker cleavage on the other hand exploits abnormal increases of enzymatic activity or abnormal micro-environments in cancers or other diseases,^{58,59} and DDSs based on endogenous stimulus-induced linker cleavage have excellent potential in clinical applications.^{60–62}

Compared with administration of free drugs, the combination of SCLs with nanoscale drug carriers offers a potential alternative to achieve highly controlled drug delivery. More importantly, linker chemistry affords simpler and more general mechanism of action in comparison with those DDSs that rely on a more complicated design.^{63–67} Of note, existing review articles introducing SCL systems, are mostly limited to the antibody–drug conjugates (ADCs) platform.^{68–70} There are two recent review papers summarised important features and mechanisms of stimulus-responsive nanoparticles, but more attention was paid to nanomaterials instead of SCLs.^{71,72} The review written by Yoon and Chen mainly highlighted the cancer-associated stimulus-responsive theranostic approaches.⁷³ A controlled release review paper on mesoporous silica nanoparticles (MSNs) predominantly focused on molecular machine mechanisms in terms of controlled release.⁷⁴ The only comprehensive review article featuring the application of stimulus-responsive linkers in DDSs⁷⁵ was published six years ago and a lot of advances have been realised since then which remain to be highlighted and compared.

The successful construction of DDSs with controlled drug release behaviour relies on three structural entities, including

drug carrier with excellent biocompatibility and biodegradability, an appropriate linker system and the active pharmaceutical. Herein, we systematically demonstrate the chemical structure and cleavage mechanism not only for DDSs,^{76–80} but also extend our scope to the field of chemical biology such as stimulus-cleavable fluorogenic probes^{81–83} and prodrugs^{84–86} that applied similar approaches in order to reduce background noise and off-target effects, respectively (Fig. 1). We further discuss the feasibility of extending the applications of SCLs to the majority of nanoscale drug carriers including nanomaterials, polymers, antibodies and so on, in the aspect of both chemistry and biology. Therefore, we hope this review will stimulate cross-collaborations between organic chemists, chemical biologist, nanomaterial scientists and biologist, and promote the development of higher drug payload, better controlled release and less off-target DDSs.

2. Mechanism of linker cleavage

2.1 Photochemistry

Photo-cleavable linkers provide an unique method for precise therapy, since photochemistry does not require any additional reagent, just light.⁸⁷ For this reason, photochemistry can be deployed in an on-demand fashion and a non-invasive manner.⁸⁸ Over the past few decades, a variety of photo-cleavable linkers have been developed for drug delivery purpose, where drug release is triggered by light with different wavelengths to enable photochemistry such as covalent bond cleavage, isomerisation and rearrangement.⁸⁹ The pioneering work of light-cleavable linkers mostly used short wavelength and high energy UV light ($\lambda = 190\text{--}400\text{ nm}$) to disrupt the chemical structure of the linkers and promote drug release.⁹⁰ Aromatic compounds like *o*-nitrobenzyl (ONB) groups and coumarin esters, found broad applications to directly regulate the drug release *via* UV irradiation. The review paper written by Klačín *et al.* has comprehensively summarised the chemical and biological applications of photo-cleavable linkers.⁹¹ Here we will briefly introduce and discuss the common photo-cleavable linkers in DDSs.

ONB groups have come in focus for drug delivery, since its aromatic ring allows versatile modifications for drug attachment and incorporation into DDSs.⁹² As summarised in Table 1, applications of ONB groups have been demonstrated with numerous drug carriers, including MSNs,^{93,94} organic nanoparticles (ONPs, *e.g.*, dendrimers,^{25,95} micelles^{20,96}) and gold nanoparticles (GNPs).⁹⁷ Drug and gating molecules, such as methotrexate (MTX, **1a**),⁹⁵ doxorubicin (DOX, **1b**),²⁵ camptothecin (CPT, **1c**),⁹⁶ 5-fluorouracil (5-FU, **1d**)⁹⁷ and β -cyclodextrin (β -CD, **1e**),⁹⁴ have been conjugated to ONB groups (Fig. 2A and Table 1).

As for the mechanism of ONB cleavage, upon UV irradiation (one-photon, $\lambda = 254\text{--}365\text{ nm}$), the nitro group on the aromatic ring undergoes an intramolecular photoreduction to *aci*-nitro tautomers, followed by the formation of benzoxazolidines and subsequent drug release from the benzylic position (Fig. 2A).⁹⁸ Further investigations found that two-photon ($\lambda = 710\text{ nm}$ ⁹⁹ and 750 nm ¹⁰⁰) excitation can also rupture the ONB groups, but

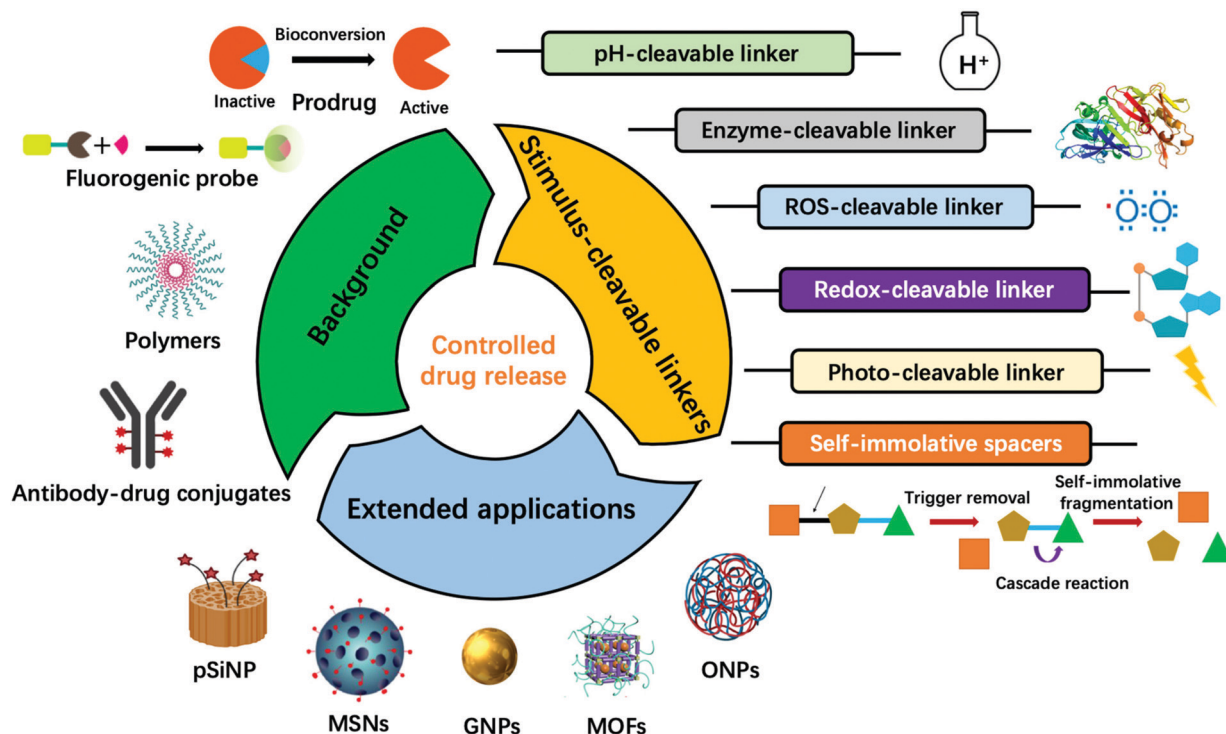


Fig. 1 Systematic illustration of various stimulus-cleavable linkers and their applications in potential nanocarriers for construction of DDSs. Abbreviations: pSiNPs, porous silicon nanoparticles; MSNs, mesoporous silica nanoparticles; GNPs, gold nanoparticles; MOFs, metal organic frameworks; ONPs, organic nanoparticles.

limited progress has been made in this regard because the relevant two-photon excitation mechanism is not completely elucidated yet.¹⁰¹ The benzylic site of ONB groups remains as the key structure for drug attachment. An array of molecules with different functional groups have been successfully articulated to ONB groups, comprising carboxylic acids,⁹¹ phosphoric acids,¹⁰² amines²⁰ and alcohols,¹⁰³ and further showed facile release upon UV irradiation (Fig. 2A).

The coumarin esters represent another attracting UV labile scaffold, which undergoes a photosolvolysis pathway to release the payload upon exposure to UV irradiation ($\lambda = 365 \text{ nm}$ ¹⁰⁴ and 475 nm ,¹⁰⁵ Fig. 2B). Noteworthy, the cleavage of coumarin esters requires assistance of nucleophiles, typically water.⁹¹ Coumarin esters have been widely used for prodrug formation, and successful examples include paclitaxel (PTX),¹⁰⁶ MTX,¹⁰⁷ chlorambucil (Cbl)¹⁰⁸ and tetracaine¹⁰⁹ (Table 1). Similar to ONB groups, the coumarin esters open the possibility of incorporating a wide range of drug molecules containing carboxylic acid (2a) and amine groups (2b and c) into the DDSs (Fig. 2B).¹¹⁰ Apart from the photosolvolysis mechanism, a light-triggered ($\lambda > 310 \text{ nm}$) [2+2] photodimerisation was found to occur between two neighbouring coumarin molecules, resulting in a cyclobutane dimer that can act as a molecular gate on MSNs.¹¹¹ Following the UV irradiation ($\lambda = 250 \text{ nm}$), the reversible photodimerisation leads to the gate opening to enable the release of biomolecules like cholesterol¹¹² from MSNs (Fig. 3). This approach has also been applied to realise light-controlled drug release behaviour on other drug carriers such as ONPs.¹¹³

Although UV-cleavable linkers had been widely used in various biological approaches, the low tissue penetration depth of UV light¹¹⁴ and potential toxic effects¹¹⁵ critically hindered their clinical applications. Therefore, the photo-cleavable linkers which can be cleaved under longer wavelength light source are urgently needed. Boron-dipyrromethenes (BODIPY) derivatives serve ideally to rapidly release the payload through one-photon excitation,¹¹⁶ and the central component in such derivatives is referred to π -extended systems that undergoes electronic transitions upon visible light irradiation.¹¹⁷ BODIPY-based cleavable linkers can be cleaved and subsequently release the drug molecules at different wavelengths (Fig. 4A). So far, drug molecules bearing carboxylic acid (4a)¹¹⁸ and amine groups (4b)¹¹⁹ have been successfully tethered to the *meso*-position of BODIPY linkers for prodrug formations (Table 1). Moreover, a recent study demonstrated that BODIPY oxime ester (shown in Fig. 4B) can also be cleaved upon blue-green light irradiation ($\lambda = 503 \pm 30 \text{ nm}$) to release carboxylic acid-containing drug molecules like valproic acid.¹²⁰

However, light with deeper tissue penetration will be more beneficial to *in vivo* studies. Another fluorophore, heptamethine cyanine (Cy), possesses the same π -extended systems as the *meso*-substituted BODIPY derivatives. The cyanine-based linkers can be cleaved upon red light irradiation and has shown potential to be utilised in prodrug formations.¹²¹ The photolysis of cyanines relies on a two-step mechanism: photooxidative cleavage (requires the access to singlet oxygen) and subsequent hydrolysis (Fig. 5A). Continuous efforts have been made in developing applications of Cy linker in DDSs, especially in ADCs,

Table 1 Summary of photo-cleavable linkers

Core structure	Cargo	Wavelength (nm)	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
ONB	Methotrexate	365	ONPs (dendrimers)	KB cells	—	95
	Doxorubicin	365	ONPs (dendrimers)	KB cells	—	25
	5-Fluorouracil	365	GNPs	MCF-7 cells	—	97
	Retinoic acid	365/750(TP)	Prodrugs	—	Zebrafish embryos	100
	cAMP	365	Prodrugs	Rat glioma C6 cells	—	129
	Camptothecin	365	ONPs (micelles)	MCF-7 cells	—	103
	Doxorubicin	365	MSNs	HeLa/MCF-7/A549/HepG2 cells	—	94
Coumarin	Glutamate	365/780(TP)	Prodrugs	—	Rat brain slices	104
	Chlorambucil	420, 800(TP)/≥365	MSNs/ONPs (liposomes/PNPs)	HeLa cells	—	108 and 130
	Methotrexate	405, 800(TP)/365	ONPs (PNPs)	A549/HeLa/L929/4T1 cells	—	107 and 131
	Paclitaxel	430	Prodrugs	—	—	106
	Tetracaine	400	Polymer-DCs	C2C12/PC12 cells	Rat	109
	Cholesterol	250	MSNs	—	—	112
BODIPY	Acetic acid	> 500	Prodrugs	Drosophila S2 cells	—	116
	Chlorambucil	530	Prodrugs	—	—	118
	Dopamine	545	Prodrugs	HEK 293T cells/rat hippocampal neurons	—	119
	Valproic acid	(473–533)	Prodrugs	HeLa cells	—	120
	Duocarmycin	690	ADCs	MDA-MB-468 (EGFR+) cells	MDA-MB-468-luc xenograft in nude mice	123
Cyanine	4-Hydroxycyclofen	690	Prodrugs	MCF-7/MEF cells	—	121
	Combretastatin A4	690	ADCs	MDA-MB-468/MCF-7 cells	A431 xenograft in nude mice	122
	Doxorubicin	650	MSNs	4T1/HeLa cells	4T1 xenograft in nude mice	124

Note: ONPs means organic nanoparticles; GNPs means gold nanoparticles; MSNs means mesoporous silica nanoparticles; PNPs means polymeric nanoparticles; polymer-DCs means polymer-drug conjugates; ADCs means antibody-drug conjugates; BODIPY means boron-dipyrromethenes; cAMP means cyclic adenosine monophosphate; ONB means *o*-nitrobenzyl; TP means two-photon excited; — means not mentioned

where combretastatin A4 (CA4) (**6a**)¹²² and duocarmycin (DUO) (**6b**)¹²³ were successfully installed on Cy linkers and efficient controlled drug release was further realised upon red light irradiation ($\lambda = 690$ nm, Table 1). Moreover, Cy linker also found applications in MSNs where gating agents like polyethylene (PEG) (**7**) were tethered to the Cy linker to regulate the release of drug (DOX, Fig. 5B).¹²⁴ These successful examples proved the concept of applying Cy linkers in DDSs, and showed advantages compared to UV-cleavable linker-based DDSs in *in vivo* applications where deeper tissue penetration of light and less light-induced damage are important.

Meanwhile, an interesting phenomenon known as photon up-conversion has attracted a wide range of researchers, where it effectively converts two or more low energy photons into one higher energy photon.¹²⁵ This provides a novel strategy to trigger the cleavage of UV-cleavable linker through the tissue-transparent window, which has been realised in up-conversion nanoparticles (UCNPs).¹²⁶ Most UCNPs contain rare-earth-based lanthanide- or actinide-doped transition metals to realise the ‘NIR to UV light’ conversion, and many UV-cleavable linkers like ONB linker¹²⁷ and coumarin ester¹²⁸ have been combined with UCNPs in the construction of DDSs. The utilisation of UCNPs in DDSs provides an alternative approach to realise controlled release at much longer wavelength of light with well-studied UV cleavable linkers.

In summary, light provides a powerful and remote stimulus to enable orthogonal drug release, which attracts attention from the therapeutic applications that require non-invasive activation of drug release. However, challenges still remain, as most reported photo-cleavable linkers are far from clinical applications. Therefore, the discovery of a photo-induced DDS with deeper tissue penetration, less energy requirement, decreased tissue damage and higher biostability should be the focus of future research. And if this problem is solved, we believe that photo-cleavable linkers hold great opportunities in DDS-based precision therapy.

2.2 Redox activation

2.2.1 Linker cleavage triggered by endogenous stimuli.

Reduction–oxidation (redox) reactions in living systems are critical for cell homeostasis, including energy metabolism, gene expression, cell growth and cell apoptosis.^{132–134} The redox status in cells is strongly associated with two redox couples, including NAD(P)H/NAD(P)⁺¹³⁵ and glutathione (GSH)/glutathione disulfide (GSSG).¹³⁶ As the most abundant redox couple in mammalian cells,¹³⁷ GSH/GSSG couple is mainly responsible for protecting the cellular components from reactive oxygen species (ROS), such as peroxides and free radicals.¹³⁸ Noteworthy, there is an obvious difference in GSH concentration between the extra-cellular and intracellular microenvironments.¹³⁹ Furthermore,

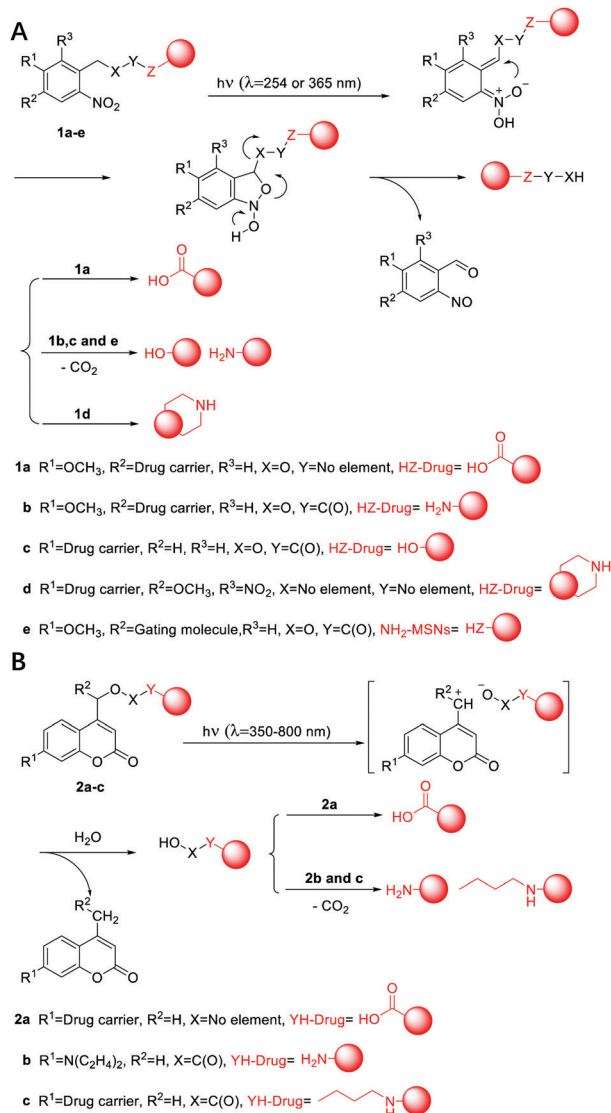


Fig. 2 ONB and coumarin based UV-cleavable linker cleavage. (A) Cleavage mechanism of ONB group upon UV irradiation; (B) mechanism of UV-induced coumarin-based linker cleavage.

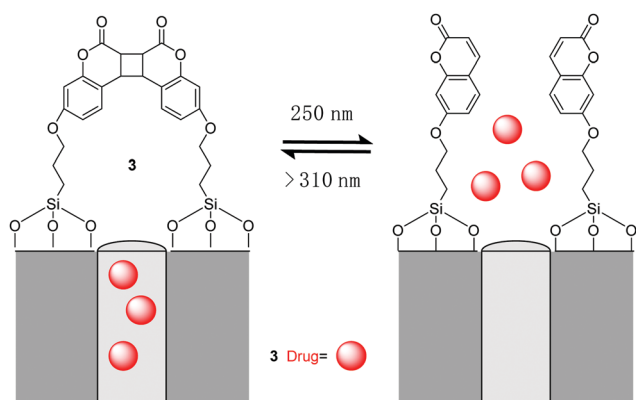


Fig. 3 MSNs capped with dimers of coumarin, with cargo released upon UV irradiation.

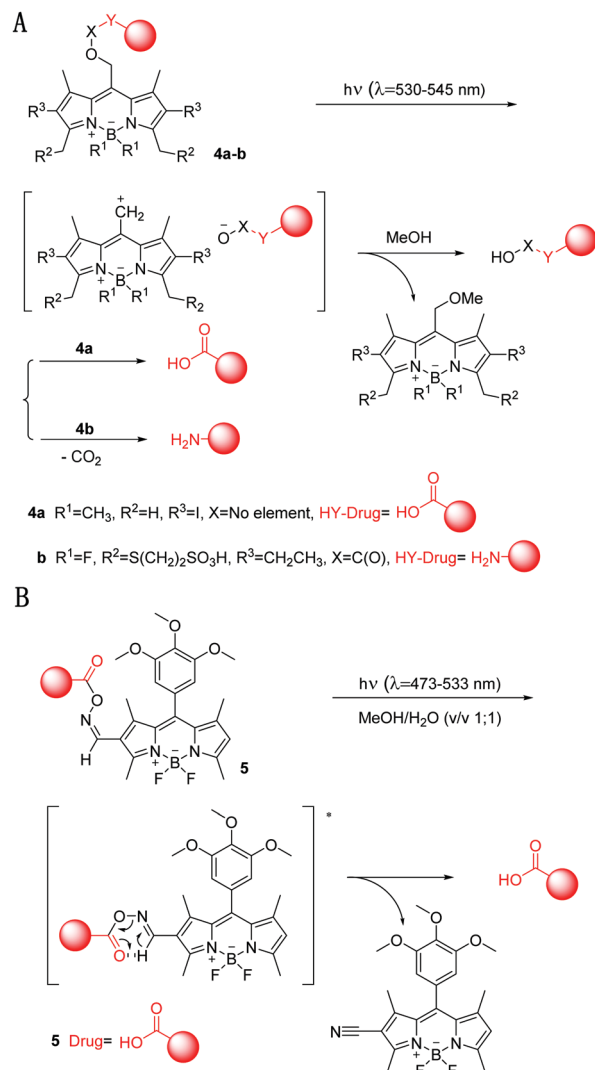


Fig. 4 (A) Chemical structure and mechanism of photolysis reaction of *meso*-substituted BODIPY group. (B) Proposed photo-uncaging mechanism of N–O bond cleavage of BODIPY oxime ester and the formation of 2-cyano-BODIPY as a byproduct.

elevated expression of GSH was observed in tumour tissues than that in healthy ones,¹⁴⁰ this discrepancy thus offering an incentive to develop GSH-cleavable linkers with the anticipation to realise tumour-specific drug release.

Disulfide linkers have been extensively explored for their high susceptibility to GSH,¹⁴¹ and are applicable to multiple drug carriers, such as peptides,¹⁴² antibodies,^{143–147} small molecules^{148–150} and ONPs (*e.g.*, dendrimers).¹⁵¹ Regarding the mechanism of disulfide bond cleavage, it is widely recognised that disulfide exchange reactions trigger the cleavage of disulfide linkers (Fig. 6A) to form the corresponding thiols.¹⁵² Drug release induced by disulfide linker cleavage has been achieved for a variety of drugs and dyes, including mitomycin C,¹⁴⁸ DOX,¹⁵³ PTX,¹⁵¹ gemcitabine (GEM),¹⁵⁰ maytansinoid,^{144,154} CPT,^{155,156} calicheamicin¹⁴⁵ and fluorescein dye¹⁵⁷ (Table 2). Of note, all the drug/gating molecules mentioned above lack the essential moiety (*i.e.* a free thiol group (**8a** and **8b**) to build

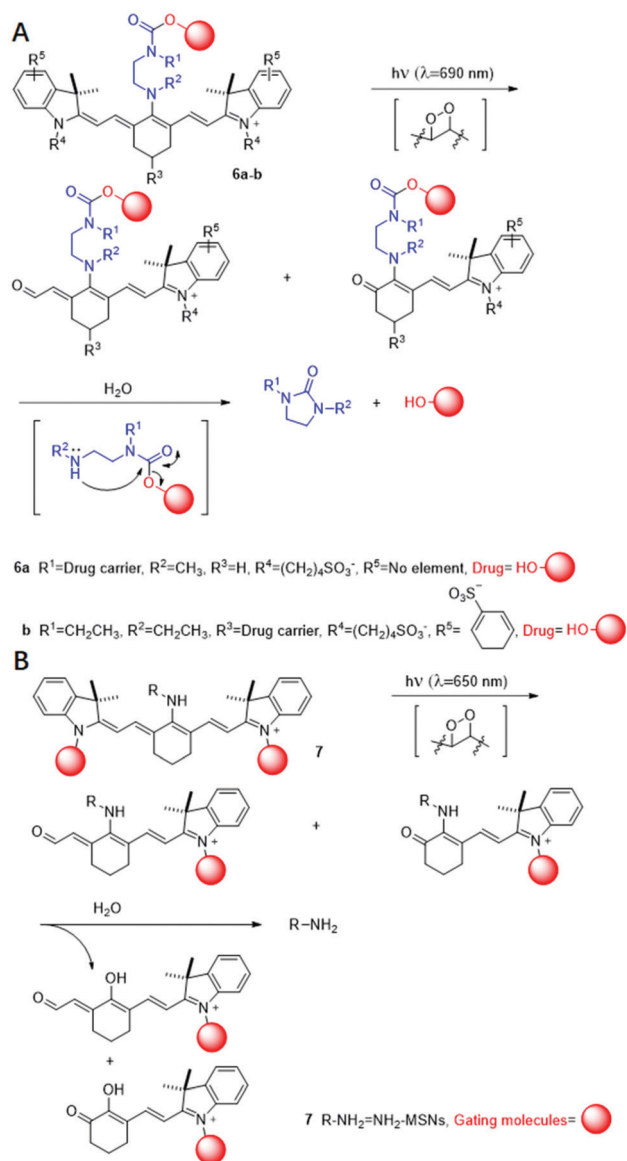


Fig. 5 Heptamethine cyanine (Cy) linker cleavage. (A) Cleavage mechanism of Cy upon red light irradiation for drug delivery; (B) removal of capping agents on MSNs in a red-light-dependent manner.

the linkage, therefore additional modification on them is required prior to linker construction. In order to construct the disulfide linker-drug conjugate, several strategies have been applied so far. For drug molecules bearing hydroxy (**9a**)¹⁵⁶ or amine groups (**9b**),¹⁵⁸ one can attach a disulfide-containing self-immolative spacer to form the respective ester, *i.e.* carbonate or carbamate esters (Fig. 6B). In the meanwhile, studies have also demonstrated the possibility of incorporating disulfide linkers into the backbone of ONPs to prepare GSH-degradable NPs for drug delivery purpose.¹⁵⁹

Apart from sulfur-based linkers, the diselenide versions also attracted attention for their high sensitivity to GSH. In contrast to disulfide linkers, diselenide ones exhibit lower redox potential, as the bond energy of the Se-Se bond (172 kJ mol⁻¹) and the C-Se bond (244 kJ mol⁻¹) are lower

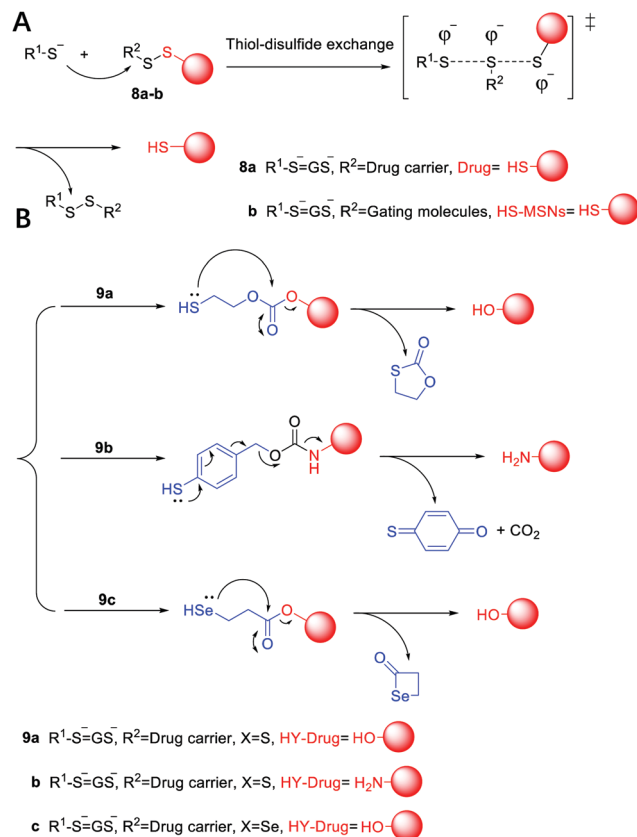


Fig. 6 Disulfide (diselenide) linker cleavage. (A) Cleavage mechanism of disulfide or diselenide linkage upon GSH exposure; (B) mechanism of spacer-assisted disulfide or diselenide linker cleavage.

than that of the S-S bond (268 kJ mol⁻¹).¹⁶⁰ Therefore, diselenide linkers show advantages in developing DDSs exploiting their higher GSH sensitivity compared with disulfide linkers.¹⁶¹ Recent studies revealed the possible application of diselenide linkers for DDS constructions for delivery of hydroxy group-containing drug molecules (**9c** in Fig. 6B) such as simvastatin¹⁶² and PTX.¹⁶³ Moreover, diselenide linkers can also be used as gating molecules to block the pores of MSNs, where GSH-triggered linker cleavage leads to cargo release.¹⁶⁴ However, as an emerging class of GSH-cleavable scaffold, diselenide linkers remain far less frequently investigated for drug delivery purpose compared to disulfide linkers. In addition, its dual sensitivity to both redox environments and reactive oxygen species (ROS) may cause drug release at unexpected sites.¹⁶⁵

Despite the fact that the high susceptibility of disulfide or diselenide linkers to GSH enables facile drug release, this feature is unfavourable in some clinical applications where sustained drug release over a few weeks is required.¹⁶⁶ Therefore, continuous efforts have been devoted to the development of GSH-cleavable adducts with expanded timescales of reduction, and thiol-maleimide adducts have emerged showing slower degradation rate in the presence of GSH (Fig. 7A).¹⁶⁷ Similar to disulfide linkages, thiol-maleimide adducts **10** can be reduced at a high concentration of GSH, based on the retro Michael-type

Table 2 Summary of reducible linkers

Core structure	Cargo	Stimulus	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
Disulfide linker	Mitomycin C	Bioconversion	SMDCs	—	M109/4T1 xenograft in nude mice	148
	Doxorubicin and paclitaxel	GSH/DTT	SMDCs	—	—	153
	Paclitaxel	Bioconversion/GSH, DTT	ADCs/ONPs (dendrimers)	A431 cells, L2987/PC-3 cells	A431 xenograft, L2987/PC-3 xenograft in nude mice	67 and 151
	Gemcitabine	GSH	SMDCs	KB/A549 cells	KB/A549 xenograft in nude mice	150
	Maytansinoid	Bioconversion/DTT	ADCs	A375/COLO 205 cells	COLO 205 xenograft in nude mice	144 and 154
	Calicheamicin	Bioconversion	ADCs	MX-1 cells	MX-1 xenograft in nude mice	145
	Camptothecin	GSH	ONPs (PNPs)	HepG2/L929/TCA8113/PC12 cells	HepG2 xenograft in nude mice	155 and 156
Diselenide linker	Fluorescein dye	GSH/DTT	MSNs	MCF-7 cells	—	157
	Camptothecin	GSH	ONPs (micelles)	B16F10/HeLa/4T1 cells	—	204
	Simvastatin	H ₂ O ₂	ONPs (PRINT nanoparticles)	Endogenous neural stem cells	APPswe/PS1dE9 double transgenic mouse	162
	Paclitaxel	H ₂ O ₂ /DTT	ONPs (PNPs)	KB/A549/4T1 cells	Rats and KB/4T1 xenograft in nude mice	163
Thiol-pyridazinedione	Ribonuclease A	H ₂ O ₂ /GSH	MSNs	HeLa/RAW264.7 cells	HeLa xenograft in nude mice	164
	Umbelliferone	GSH	SMDCs	—	—	79
DBS	SN-38	GSH	Prodrugs	B16F10/A2780/OCSC1-F2 cells	—	174
	Doxorubicin	GSH	Prodrugs/ONPs (micelles)	MCF7/V79/HEK293/MCF-7 ADR cells	MCF-7 ADR xenograft in nude mice	175 and 179
	Camptothecin	GSH	ONPs (liposomes)	HeLa cells	HeLa xenograft in nude mice	177
	Mechlorethamine	GSH	Prodrugs	A549/HeLa/HepG2/SMMC-7721/K562/HL60/786-O/293T	—	176
	1,8-Naphthalimide derivative	Cystine	Fluorescent probes	HeLa/MCF-7 cells	Mice tissue	205
<i>o</i> -(Azidomethyl) benzoyl group	7-Hydroxy-4-methyl-coumarin	H ₂ S	Fluorescent probes	HeLa cells	—	187
<i>p</i> -Azidobenzyl unit	2-(2'-Hydroxyphenyl)-benzothiazole	H ₂ S	Fluorescent probes	HeLa cells	—	188
	4-Amino- <i>N</i> -butyl-1,8-naphthalimide	H ₂ S	Fluorescent probes	MCF-7 cells	Mice	190
α,β -Unsaturated esters	2-(6-Methoxy-3-oxo-3 <i>H</i> -xanthen-9-yl)benzoic acid	H ₂ S	Fluorescent probes	COS7 cells	—	192
Azobenzene	Doxorubicin	Azoreductase	ONPs (dendrimers)	HepG2/Hep3B cells	Rat cardiomyocytes	196
	5-Aminosalicylic acid	Azoreductase	ONPs (dendrimers)	—	Homogenate of stomach or small intestine/Cecal segment of rats	197
	Rhodamine B	Nitrate reductase	MSNs	HeLa cells	—	198
	9-Aminocamptothecin	Azoreductase	Polymer-DCs	—	Homogenate of rat luminal and cecal contents and mucosa	206
	4'- <i>O</i> -Demethyl-4 β -(4''-aminoanilino)-4-desoxy-podophyllotoxin	Azoreductase	Prodrugs	HeLa/HepG2 cells	Mice	200
	<i>N,N</i> -Bis(2-chloroethyl)-1,4-benzenediamine	Hypoxia	Prodrugs	DU145/MDA-MB-231/A549/Huh7/normal human fibroblast cells	DU145 xenograft in nude mice	199

Table 2 (continued)

Core structure	Cargo	Stimulus	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
	<i>N,N</i> -Bis(2-chloroethyl)-1,4-benzenediamine and doxorubicin	Hypoxia	Prodrugs	4T1/B16 cells	4T1 xenograft in nude mice	201
	Paclitaxel	Hypoxia	ONPs (PNPs)	HeLa cells	HeLa xenograft in nude mice	202
<i>p</i> -Azidobenzyl unit	Doxorubicin	Triphenylphosphine-3,3',3''-trisulfonic acid trisodium salt	Prodrugs	A431 cells	—	203

Note: SMDCs means small molecule drug conjugates; ADCs means antibody–drug conjugates; ONPs means organic nanoparticles; MSNs means mesoporous silica nanoparticles; PNPs means polymeric nanoparticles; polymer–DCs means polymer–drug conjugates; GSH means glutathione; DTT means dithiothreitol; DBS means 2,4-dinitrobenzenesulfonyl; — means not mentioned.

addition followed by thiol exchange reaction.¹⁶⁸ Importantly, the cleavage rate of thiol–maleimide adducts is 10 to 100-fold lower than that of disulfide linkers, affording longer-term drug release in reducing environments.¹⁶⁹ Although the thiol–maleimide derivatives have long been studied for their GSH-induced decomposition,¹⁷⁰ limited progress has been made for drug delivery purpose. Novel thioethers, like thiol–pyridazinedione (PD), have been recently noticed by researchers for their responses to GSH.⁷⁹ The proposed mechanism of thiol–PD cleavage is shown in Fig. 7B, where GSH acts as a nucleophile to induce the addition–elimination reaction to liberate the drug with a sulfur terminal. Similar to the synthesis of disulfide linker–drug conjugate, self-immolative spacers with thiol terminal can be introduced to the thiol–PD framework (Fig. 7B) to promote the release of cargo such as umbelliferone (**11**).⁷⁹ Moreover, this pioneering study also highlighted the possibility

of modifications on the PD framework for further conjugation (e.g., click chemistry) to drug carriers.⁷⁹

In addition to GSH, other biothiols such as cysteine (Cys) and homocysteine (Hcy) are also of importance to maintain the redox balance in microenvironments.¹⁷¹ A deficiency of Cys can lead to various diseases, including liver damage, leukopenia and hematopoietic dysfunction.¹⁷² Existing GSH-cleavable linkers, thiol–PD, have also been studied for their sensitivity to Cys. Based on the same GSH-induced retro-Michael deconjugation mechanism, Cys can trigger the release of the payload from a thiol–PD motif.¹⁷³ Although the thiol–PD adduct exhibits time-dependent degradation that enables sustained drug release, its poor selectivity could cause undesired drug release in healthy tissues where Cys and GSH co-exist in the microenvironments. In this respect, the 2,4-dinitrobenzenesulfonyl (DBS) unit has recently emerged owing to its high specificity to Cys. DBS finds wide utilisations in prodrug formations, where hydroxy group-containing drug molecules like SN-38 (**12a**)¹⁷⁴ and DOX (**12b**)¹⁷⁵ can be tethered directly to DBS, and where nucleophilic aromatic substitution and removal of SO₂ enable drug release (Fig. 8A and Table 2). Furthermore, the introduction of self-immolative spacers (Fig. 8B) into DBS linkers improves their ability to bear a broader palette of drug candidates such as mechlorethamine (**13a**)¹⁷⁶ and CPT (**13b**)¹⁷⁷ (Table 2). However, GSH was more often reported to trigger the release of the drugs mentioned before, instead of Cys. Recently, a DBS linker has been used for constructing Cys or GSH-responsive DDSs (e.g., ONPs), where the cleavage of it enables the drug release directly¹⁷⁸ or indirectly.¹⁷⁹

As a gasotransmitter, endogenous hydrogen sulfide (H₂S) plays an important role in regulating various physiological processes, such as cell survival/death, cell differentiation, cell proliferation/hypertrophy and cellular metabolism.¹⁸⁰ H₂S protects neurons from oxidative stress by facilitating the production of GSH, and by scavenging ROS in the mitochondria.¹⁸¹ Abnormal level of H₂S is strongly associated with diabetes,¹⁸² inflammation¹⁸³ and neurodegenerative diseases.¹⁸⁴ These pathological effects of H₂S have encouraged the development of H₂S-responsive linkers for target drug delivery. Taking advantage of the reducibility and nucleophilicity of H₂S, a series of chemical moieties have been discovered for their response to H₂S.¹⁸⁵ As the smallest thiol, H₂S is capable of selectively reducing the azides into amines.¹⁸⁶ The amines produced by reduction of azides serve

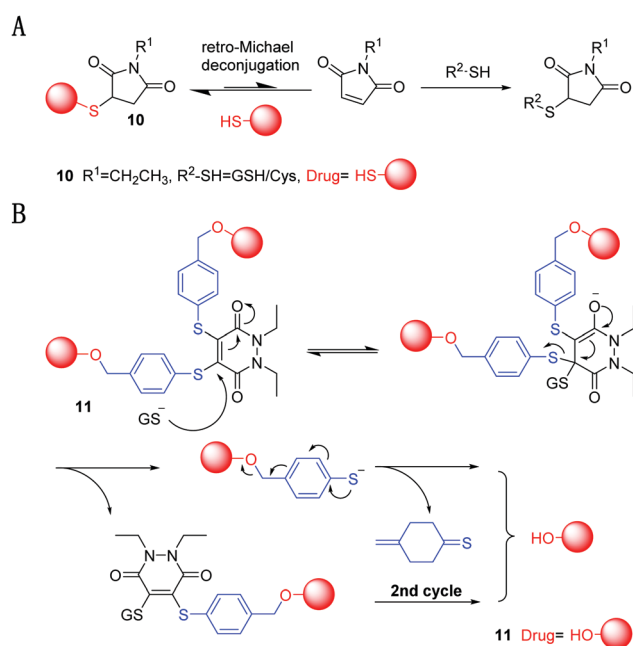


Fig. 7 GSH-triggered linker cleavage. (A) Degradation mechanism of thiol–maleimide adducts upon GSH exposure; (B) mechanism of cargo release from thiol–pyridazinedione (PD) derivatives in the presence of GSH.

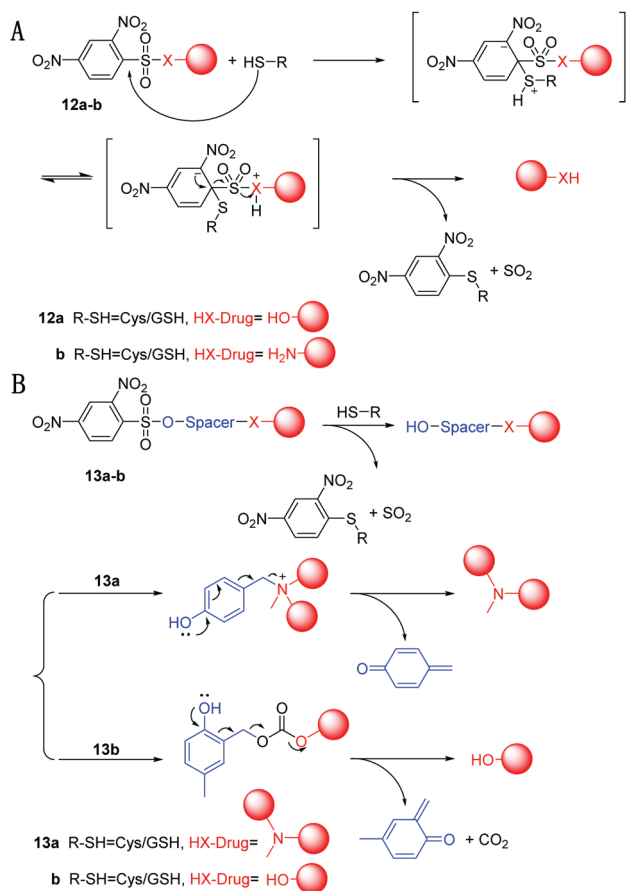


Fig. 8 Release of payload from novel GSH/Cys-cleavable linkers. (A) General mechanism of 2,4-dinitrobenzenesulfonyl (DBS) linker degradation triggered by GSH or Cys; (B) cargo release from spacer-assisted DBS linker cleavage.

as promising electron donors to induce a nucleophilic attack or rearrangement reaction to cause bond cleavage, which eventually results in drug release. Several self-immolative spacers bearing azide groups have been used to realise H₂S-controlled drug release, including the *o*-(azidomethyl) benzoyl group¹⁸⁷ and *p*-azidobenzyl unit.¹⁸⁸ The general mechanism of cargo release from *o*-(azidomethyl) benzoyl group refers to the intramolecular attack of free amine produced by azide reduction on the carbonyl ester,¹⁸⁷ while the release of payload from *p*-azidobenzyl unit relies on the 1,6-elimination reaction initiated by free amines (Fig. 9A and B).¹⁸⁹ However, the main application of both linkers mentioned above is in fluorescence-based H₂S sensing, where the linkers were modified with different fluorophores. For molecules bearing hydroxy groups, they can be tethered to the *o*-(azidomethyl) benzoyl group to form **14**,¹⁸⁷ whilst the possibility of linking hydroxy (**15a**)¹⁸⁸ or amine group (**15b**)¹⁹⁰-containing molecules to *p*-azidobenzyl unit was also demonstrated (Table 2).

On the other hand, H₂S exhibits unique double nucleophilic ability, and more H₂S-selective scaffolds have been discovered on the basis of this feature. To date, certain α,β -unsaturated esters (Fig. 9C) have attracted attention due to their high selective response to H₂S. Based on the tandem Michael addition reaction,¹⁹¹ where H₂S enables the initial nucleophilic attack on electrophilic centre of

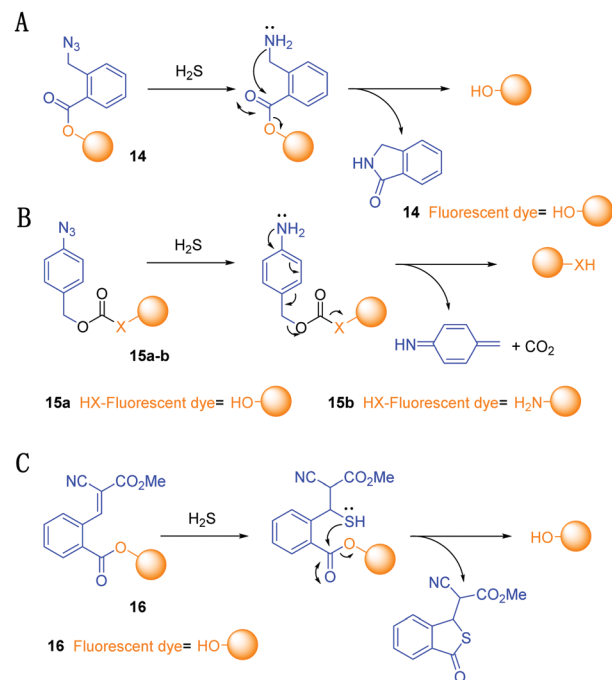


Fig. 9 H₂S-induced payload release. (A) General mechanism of cargo release from *o*-(azidomethyl) benzoyl group. (B) *p*-Azidobenzyl unit as a H₂S-responsive linker to trigger payload release. (C) α,β -Unsaturated ester-based linker cleavage.

a Michael acceptor to afford the corresponding thioacetal derivatives, this is followed by a intramolecular cyclisation reaction (second nucleophilic attack) that further trigger payload release (Fig. 9C). Unlike H₂S, other biothiols such as GSH and Cys, are not able to perform the second nucleophilic attack required for cyclisation, hence this Michael acceptor is very selective for H₂S over other biothiols. Similar to *o*-(azidomethyl) benzoyl group and *p*-azidobenzyl unit, α,β -unsaturated esters remain barely investigated for drug delivery purpose and the main application of them is in bioimaging field. As illustrated by Liu *et al.*, molecules featuring hydroxy groups can in principle be attached to the carboxylic acid site of Michael acceptor by forming the corresponding esters (**16**),¹⁹² but further studies are required to efficiently incorporate this linker into drug carriers.

2.2.2 Linker cleavage induced by exogenous stimuli. Despite the remarkable progress in the development of biothiol-cleavable linkers, the efficiencies of the linker cleavage depend on these endogenous stimuli, and their concentration differential in pathological tissue regions. However, unexpected drug release in normal tissue or microenvironment may hinder the therapeutic effect of DDSs, due to the existence of biothiols in such areas, although at a relatively low level. To circumvent this issue, the development of orthogonally cleavable linkers has moved into the spotlight. The approach using exogenous stimuli, which do not exist naturally in biological systems, opens up a possibility of realising on-demand drug release. Recently, a batch of reducible linkers that can be cleaved by exogenous stimuli have been extensively studied for drug delivery purpose, with azobenzene-based linkers being the most popular ones. The azobenzene

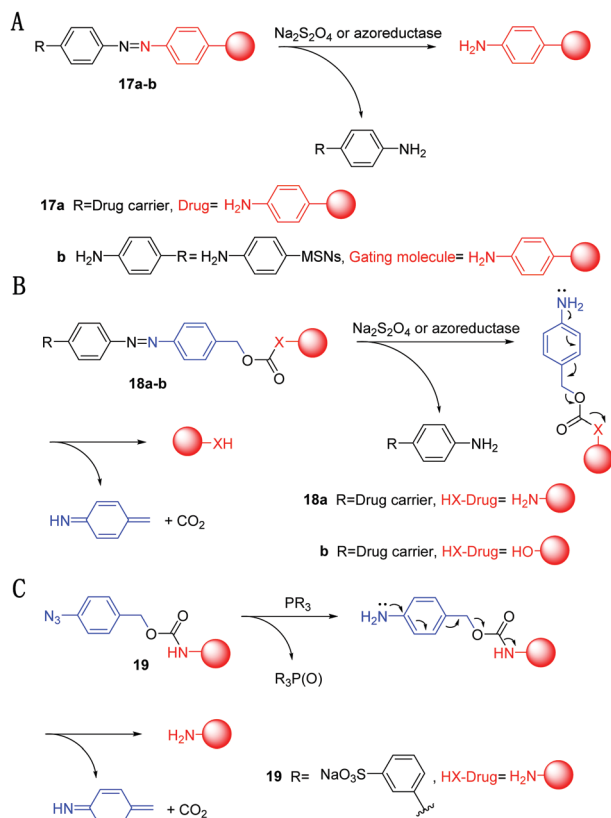


Fig. 10 Cargo release caused by azobenzene linker's reduction. (A) General mechanism of azobenzene reduction; (B) spacer-assisted cargo release from azobenzene linker; (C) *p*-azidobenzyl unit as a phosphine-responsive linker to trigger the payload release.

linker serves as a promising tool for modern proteomics, owing to its mild cleavage conditions.¹⁹³ Cleavage of an azobenzene scaffold was initially found to proceed in the presence of sodium dithionite,¹⁹⁴ which is a mild reducing reagent. Based on a two-step reduction mechanism, the dithionite ion provides four electrons to reduce aromatic azo linkers to afford the corresponding amines (Fig. 10A).⁷⁷ Moreover, with increased understanding to azobenzene scaffolds, their cleavage can also be engineered *via* a biological method, where it undergoes a bond scission reaction upon exposure to the NAD(P)H dehydrogenase [quinone] 1 (NQO1) and azoreductase under hypoxic conditions to liberate the free amines (Fig. 10A).¹⁹⁵ Representative examples of azobenzene scaffold application in the DDS field include tethering drug/gating molecules to drug carriers like ONPs (*e.g.*, dendrimers^{196,197}) and MSNs.¹⁹⁸ Aniline-containing drug/gating molecules (17a and b) can be directly attached to the azobenzene linker through an array of synthetic methods,^{199,200} whilst drug molecules with amine (18a)²⁰¹ and hydroxy (18b)²⁰² terminals can be indirectly tethered to the azobenzene scaffold *via* a self-immolative spacer (Fig. 10B), typically *p*-aminobenzyl alcohol (Table 2).

The *p*-azidobenzyl linker, the H_2S -responsiveness of which has been illustrated in the previous paragraphs, is also susceptible to exogenous stimuli like phosphine.²⁰³ The proposed mechanism of reduction is illustrated in Fig. 10C, referring to

Staudinger reaction where the phosphine reacts with azide functional group on *p*-azidobenzyl moiety (19) to form an azaylide intermediate, which subsequently hydrolyses to produce the corresponding phosphine oxide and primary amine. The resulting intermediate serves as a self-immolative spacer to rapidly release the drug attached to the *para*-position of aniline *via* a 1,6-elimination reaction. Although the aromatic ring in *p*-azidobenzyl linkers principally allows for versatile modifications to further install it onto the drug carriers, limited cases have been reported so far.

2.3 ROS activation

Reactive oxygen species (ROS) are referred to a type of single electron oxygen generated by the reduction of molecular oxygen (O_2), including hydrogen peroxide (H_2O_2), singlet oxygen ($^1\text{O}_2$), superoxide (O_2^-), hydroxyl radical ($\bullet\text{OH}$) and so on.²⁰⁷ Endogenous ROS are mainly produced by the mitochondrial metabolism and reactions catalysed by NADPH enzyme, while they can also be generated exogenously through the application of UV light or xenobiotic compounds.²⁰⁸ ROS play a vital role in maintaining homeostasis such as cell growth, migration and apoptosis,²⁰⁹ but elevated level of ROS will result in oxidative stress which can damage the nucleic acid, proteins, lipid and hydrocarbons.²¹⁰ Chronic oxidative stress has been considered as a pathophysiological indication of several types of diseases, including cancer,²¹¹ diabetes,²¹² cardiovascular diseases²¹³ and neurodegenerative diseases.²¹⁴

2.3.1 H_2O_2 -Cleavable linkers. As a key member of the ROS class, H_2O_2 is frequently used to assess the ROS-sensitive performance of the linkers, owing to its stability compared with all the other ROS.²¹⁵ Furthermore, higher concentration of H_2O_2 is present in inflamed²¹⁶ or cancerous cells²¹⁷ as a result of dismutation of O_2^- . The differential of H_2O_2 levels therefore distinguishes the pathological tissues from the healthy ones, which can be harnessed as a unique disease-related target for drug delivery. Accordingly, numerous types of chemical structures have been applied in developing the H_2O_2 -cleavable linkers, which typically include arylboronic esters,²¹⁸ thioketals,²¹⁹ aryl oxalate esters²²⁰ and proline oligomers.²²¹

Among the various types of linkers, the phenylboronic esters (PBE) can be specifically oxidised by H_2O_2 *via* the insertion of an oxygen atom into the C–B bond to produce the corresponding phenol and boronate ester.²²² PBE is commonly utilised for H_2O_2 sensing,^{223,224} prodrug formation^{225–228} and DDSs construction.^{229,230} PBE is usually formed by reacting the phenylboronic acid (PBA) with molecules containing diol group, for instance DOX (20a),²²⁹ lactobionic acid-functionalised chitosan (20b)²³¹ and 5'-deoxy-5-fluorouridine (5'-DFUR)²³² (Fig. 11A and Table 3). On the other hand, for drug/gating molecules without diol group, like 5-FU (21a)²²⁸ and MTX (21b),²³³ an alternative conjugation strategy was proposed by placing drug molecules at the *para*-position of boronic ester or boronic acid (Fig. 11B and Table 3). Upon exposure to H_2O_2 , the resulting phenols will undergo a 1,6-elimination reaction to recover the therapeutic effect of drug molecules. Moreover, PBE can also be utilised for preparation of H_2O_2 -responsive ONPs, where the cleavage of it in

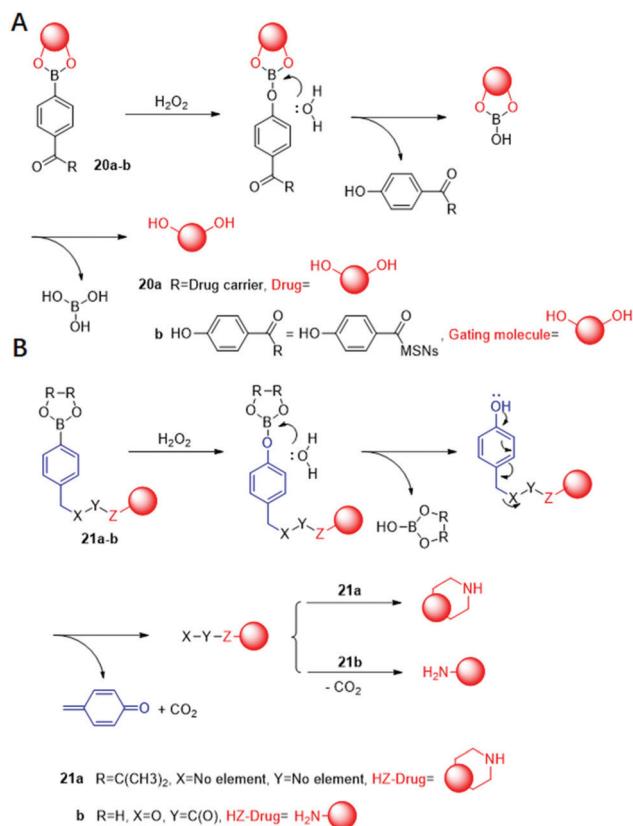


Fig. 11 Chemical structure and cleavage mechanism of phenylboronic ester (PBE)-based linker upon the treatment of H_2O_2 . (A) H_2O_2 -induced diol-containing cargo release from PBE linker. (B) Self-immolation-triggered drug release following the oxidation of PBE linker by H_2O_2 .

the presence of H_2O_2 can induce nanoparticle decomposition, resulting in payload release.²³⁰ Although the PBE moiety has been extensively exploited for drug delivery, problems still remain, particularly related to PBE instability under acidic environments,^{234,235} which may cause unexpected drug release at some organelles with acidic pH, like endosomes and lysosomes.²³⁶

To alleviate the stability issue of PBE, several efforts have been devoted to discovering and developing H_2O_2 -cleavable moieties with higher specificity. Among them, thioketals attracted attention because of their improved stability under both acidic and basic environments. Studies have shown that thioketals can be cleaved by oxidative means to form the corresponding ketones and thiols, and their incorporation into DDSs for H_2O_2 -induced drug release has been realised.^{237–240} Similar to disulfide bond exchange, the oxidative cleavage of the thioketal structure (22) produces derivatives with a sulfur terminal, which may impair the therapeutic efficacy of non-sulfur containing drug molecules (Fig. 12A). To address this problem, self-immolative spacers were introduced into the thioketal framework, enabling intramolecular nucleophilic attack of free thiol group generated through thioketal cleavage to trigger drug release (Fig. 12B and C).²⁴¹ Drug molecules such as DOX (23a),²³⁷ GEM (24)²⁴² and mitoxantrone (MTO)²³⁸ have

been successfully conjugated to thioketal linkers (Table 3), while gating molecules like methoxy poly(ethylene glycol) (mPEG) (23b)²⁴³ have been directly conjugated to thioketal to regulate drug release.

In addition to thiol-containing chemical structures, peroxalate esters (PO)²⁴⁴ also show high selectivity of H_2O_2 over other ROS. It has been widely recognised that the facile reaction between PO (25–26) and H_2O_2 forms 1,2-dioxetanedione, which subsequently degrades into carbon dioxide (Fig. 13A and Table 3). Given its high selectivity of H_2O_2 , PO has found extensive applications for H_2O_2 sensing^{245,246} and H_2O_2 -induced drug release from DDSs.²²⁰ For molecules possessing hydroxy groups, a coupling reaction between the hydroxy group and oxalyl chloride can be performed to form the core PO structure.^{247,248} On the other hand, modification with 2-(hydroxymethyl)-2-methylpropane-1,3-diol²⁴⁹ (Fig. 13B) has been demonstrated to provide the aldehyde-containing drug molecules (26) with a hydroxy terminal for construction of PO linker (Table 3). Following the oxidative degradation of PO linker, the acid-triggered hydrolysis of acetal-containing intermediate results in drug release. Well-defined DDSs engineered with PO-drug conjugates have enabled drug release from multiple drug carriers, especially ONPs (e.g., micelles²⁵⁰). Despite the fact that the PO group exhibits high selectivity for H_2O_2 , the strong reactivity of it with H_2O_2 usually results in burst drug release, which hampers applications in sustained drug release.²⁵¹

2.3.2 Other ROS-cleavable linkers. Apart from H_2O_2 , $^1\text{O}_2$ and $\cdot\text{OH}$ components of the ROS have also received increasingly interest for their high reactivity. The $\cdot\text{OH}$ exists as the three-electron reduction state of O_2 , which is mainly attributable to its high reduction potential.²⁵² As a very strong reductant, $\cdot\text{OH}$ can indiscriminately react with numerous biomolecules, including amino acids, lipids, sugars and nucleotides.²⁵³ Different from $\cdot\text{OH}$ as a radical, $^1\text{O}_2$ represents an excited state of O_2 where one of the unpaired electrons changes its direction of spin to generate two electrons spinning in opposite directions, thus offering high oxidative potential to $^1\text{O}_2$.²⁵⁴ Unlike H_2O_2 , $\cdot\text{OH}$ and $^1\text{O}_2$ have very short half life.²⁵⁵ $\cdot\text{OH}$ is usually generated through the *in vivo* decomposition of H_2O_2 catalysed by iron(II) or copper(I), also referring as the Fenton reaction;²⁵⁶ while the majority of $^1\text{O}_2$ for biological purpose is produced through a photosensitisation reaction²⁵³ through the application of photosensitisers.

Over the past decades, the strong reducibility of $\cdot\text{OH}$ has been studied for the design of cleavable linkers. Among these linkers, proline²⁵⁷ has recently come into the sight of researchers because of the relatively slower cleavage rate of proline oligomer compared with other ROS-sensitive groups (Fig. 14).²⁵⁸ With the anticipation that proline-based linkers can promote highly selective and time-prolonged drug release, this moiety has been extensively explored for the construction of $\cdot\text{OH}$ -responsive ONPs.^{221,257} The susceptibility of proline-based linkers (27) to $\cdot\text{OH}$ can be explained by the oxidation mechanism shown in Fig. 14, where the payload attached to the carboxylic acid site of proline structure is released.²⁵⁹ Given the carboxylic acid as the only site on proline for cargo attachment, molecules bearing free amines can be tethered to this site *via* amide coupling reactions.²⁶⁰

Table 3 Summary of ROS-cleavable linkers

Core structure	Cargo	Stimulus	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
PBE	Nitrogen mustard	H ₂ O ₂	Prodrugs	SR/NCI-H460/CAKI-1/SN12C cells; human primary lymphocytes	—	226
	5-Fluorouracil	H ₂ O ₂	Prodrugs	NCI-H522/MDA-MB-435/OVCAR-3/MCF-7/MDA-MB-468 cells	C57BL/6 mice	228
	Doxorubicin	H ₂ O ₂ /low pH	Organosilica nanoparticles/MSNs	HepG2/NIH 3T3 cells	—	229 and 231
	Naproxen	H ₂ O ₂	ONPs (PNPs)	—	—	230
	5'-Deoxy-5-fluorouridine	H ₂ O ₂	Prodrugs	A549/WI-38 cells	A549 xenograft; dissected organs and cryosections	232
PBA	Methotrexate and aminopterin	H ₂ O ₂	Prodrugs	Human/mouse liver microsomes; NCI-H460/MCF-7 cells	DBA/1J mice arthritis	233
	SN-38	H ₂ O ₂	Prodrugs	HCT-15/HT-29/MCF-7/MDA-MB-231/U87MG/U251 cells	U87MG xenograft in nude mice	275
Thioketal	Doxorubicin	¹ O ₂	ONPs (PNPs)	MDA-MB-231 cells	MDA-MB-231 xenograft in nude mice	237 and 239
	Mitoxantrone	KO ₂	ONPs (PNPs)	PC3, DU145, 22RV1 and LNCaP cells	LNCaP xenograft in nude mice	238
	Gemcitabine	¹ O ₂	Prodrugs	HeLa/H22 cells	H22 xenograft in nude mice	242
	Vancomycin	H ₂ O ₂	MSNs	MC3T3-E1 cells	Rat wound healing	243
PO	Camptothecin	H ₂ O ₂	ONPs (PNPs)	A549 cells	A549 xenograft in nude mice	220
	Tyrosol	H ₂ O ₂	ONPs (PNPs)	LPS-induced RAW 264.7 cells inflammation	—	247
	Vanillin	H ₂ O ₂	ONPs (PNPs)	LPS-induced RAW 264.7 cells inflammation	APAP-intoxicated mice	249
	Curcumin	H ₂ O ₂	ONPs (PNPs)	LPS-induced RAW 264.7 cells inflammation	APAP-intoxicated mice	248
	Camptothecin	H ₂ O ₂	ONPs (micelles)	4T1 cells	4T1 xenograft in nude mice	276
	Paclitaxel	H ₂ O ₂	ONPs (micelles)	HepG2 cells	Hepa 1–6 cells xenograft in nude mice	250
Vinyl ether	Ibuprofen	¹ O ₂	Prodrugs	—	—	267
Aminoacrylate	Estrone	¹ O ₂	Prodrugs	—	—	269
	SN-38	¹ O ₂	Prodrugs	MCF-7 cells	—	270
	Combretastatin A4	¹ O ₂	Prodrugs/SMDCs	Colon 26 cells	Colon 26 xenograft in nude mice	270 and 277
	Paclitaxel	¹ O ₂	Prodrugs/SMDCs	SKOV-3 cells	—	271 and 272
	Doxorubicin	¹ O ₂	MSNs	SCC-7 cells	SCC-7 xenograft in nude mice	273

Note: MSNs means mesoporous silica nanoparticles; ONPs means organic nanoparticles; PNPs means polymeric nanoparticles; SMDCs means small molecule drug conjugates; PBA means phenylboronic acid; PBE means phenylboronic ester; PO means peroxalate ester; H₂O₂ means hydrogen peroxide; ¹O₂ means singlet oxygen; LPS means lipopolysaccharide; APAP means acetaminophen; O₂^{•−} means superoxide; — means not mentioned.

An array of chemical structures have been explored for their sensitivity to ¹O₂, including vinyl ethers and aminoacrylates.²⁶¹ The early work of ¹O₂-cleavable linkers found that ¹O₂ could undergo a [2+2] cycloaddition reaction (Fig. 15A) with the alkene group of a vinyl ether, forming an active dioxetane derivative that spontaneously degrades into two carbonyl compounds.²⁶² Further studies revealed that diheteroatom-substituted alkene performs better than monoheteroatom-substituted alkene in response to ¹O₂, because the ¹O₂ may also undergo an Alder–Ene reaction with alkenes that can compete with [2+2] cycloaddition reaction.^{263,264} Vinyl ethers have also been incorporated into drug carriers such as ONPs (e.g., liposomes²⁶⁵ and micelles²⁶⁶), where the cleavage of vinyl ether leads to the decomposition of ONPs, which subsequently results in the drug release. The same chemistry has been applied in prodrug formations,²⁶⁷ which require suitable

drug molecules containing carbonyl group (28), thus narrowing the scope of suitable drug candidate (Table 3). However, the scarce synthetic methods²⁶⁸ for vinyl ethers also limit their wider application for controlled drug release.

Another class of novel ¹O₂-cleavable linkers, aminoacrylate, which is based on the β-enamino ester structure, showed higher reactivity than vinyl ethers in the [2+2] cycloaddition reaction with ¹O₂ (Fig. 15B),²⁶⁹ possibly due to the resonance between the lone pair of electrons from nitrogen with the C=C bond. Aminoacrylate has two advantages over vinyl ether, including less complicated synthesis and the compatibility to drug/gating molecules with hydroxy group(s). For drug/gating molecules bearing hydroxy groups, they can conjugate with the carboxylic acid site of aminoacrylate linker, while the amino site end can be exploited for installation of the drug-linker conjugate onto the photosensitisers (29a) or drug carriers (29b).

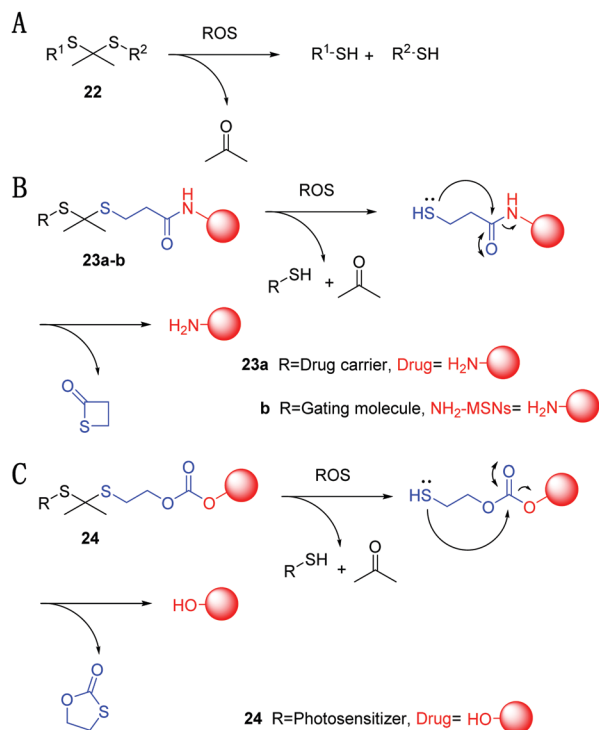


Fig. 12 Thioketal-based ROS-responsive linker. (A) General mechanism of thioketal degradation upon exposure to ROS; (B and C) spacer-assisted cargo release from thioketal.

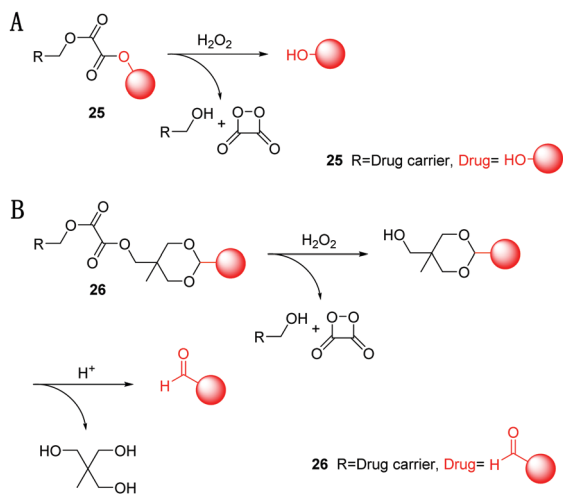


Fig. 13 Peroxalate ester (PO)-based ROS-responsive linker. (A) Hydroxy-group-bearing cargo release from PO linker upon ROS exposure; (B) carbonyl group-containing cargo release from PO linker.

And it has been widely applied for prodrug formations that covers a wide range of drug molecules, such as Estrone,²⁶⁹ SN-38,²⁷⁰ CA4²⁷⁰ and PTX,²⁷¹ the same motif was also used for the construction of fluorogenic probes²⁷⁰ (Table 3). Regarding the construction of 1O_2 -cleavable DDSs, aminoacrylate has been reported to be linked to folic acid²⁷² to construct small-molecule drug conjugate (SMDC), while gating molecules such as tetraxetan (DOTA, 29c)²⁷³

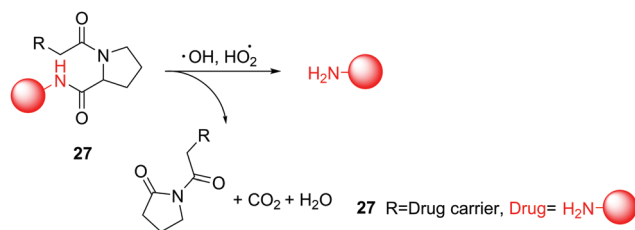


Fig. 14 Free radical-induced cleavage of proline-based linker, where the oxidation of proline enables the drug release from the C terminus of proline.

were tethered *via* aminoacrylate to MSNs to regulate the drug release. Of note, the 1O_2 required for cleavage of both vinyl ether and aminoacrylate linkers is usually generated by a photosensitizer (PS) that is used for photodynamic therapy (PDT) and PS can either be chemically tethered to the linker or be separated from the linker but localised to the same cells.²⁷⁴

2.4 pH-induced cleavage

pH is recognised as a physiological parameter in living organisms.²⁷⁸ As the smallest but extremely reactive ions, hydrogen ion (H^+) is produced through the balance between protonation and deprotonation of water, weak acids and weak bases. The regulation of pH is affected in particular diseased sites, such as cancerous²⁷⁹ and inflammatory²⁸⁰ tissues, due to their intensive respiratory CO_2 and lactic acid production.²⁸¹ The resulting excessively higher acidity of these pathological regions compared to blood and healthy tissues provides an appropriate endogenous stimulus for pH-cleavable drug delivery. In addition, the organelles that have relatively low pH such as endosomes and lysosomes are broadly explored by several DDSs for controlled drug release. Therefore, numerous efforts have been

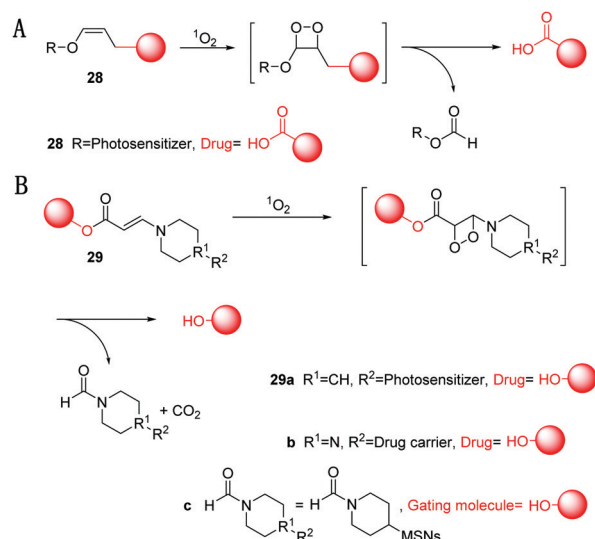


Fig. 15 Structures and cleavage mechanism of singlet oxygen (1O_2)-cleavable linkers. (A) [2+2] oxidation of vinyl ether-based linker by 1O_2 , and the subsequent cleavage of dioxetane. (B) Mechanism of drug or gating molecule release from aminoacrylate linker.

devoted to the design and development of chemical structures with high sensitivity to acidic microenvironments.

2.4.1 Schiff bases. Hydrazone linkers, a series of linkers involving acyl hydrazine,²⁸² alkoxy carbonyl hydrazone²⁸³ or benzenesulfonyl hydrazone²⁸⁴ as cores, find wide applications for drug delivery purpose as pH-cleavable linkers, which is attributable to their robust hydrolysis rate under acidic conditions and stability at neutral pH.²⁸⁵ Therefore, the incorporation of hydrazone linkers into the DDSs promises a stable DDS system in the circulation and rapidly drug release at acidic pH. Hydrazone linkers have been applied in a wide range of drug carriers, including ONPs (*e.g.*, dendrimers,^{286,287} micelles,^{288,289} liposomes^{290,291}), GNPs,²⁹² MSNs²⁹³ and antibodies^{294,295} (Table 4). The proposed mechanism of hydrolysis of hydrazone linkers is shown in Fig. 16A, where N¹ in the C¹=N¹-C structure is first protonated at acidic pH, followed by nucleophilic attack by water molecules. The resulting intermediate undergoes a facile internal H⁺ transfer to complete the release of drugs with carbonyl moiety while producing the residue terminated with amine group.²⁹⁶ Limitations still remain, the main of which is the limited flexibility in terms of linker conjugation reactions. As determined by the hydrolysis mechanism, the hydrazone linker is only applicable for drug/gating molecules containing carbonyl group (ketone, aldehyde), like DOX (30a),²⁹² auristatin E (30a)²⁹⁷ and hydrazide-modified sodium alginate (30b)²⁹⁸ (Table 4). Although hydroxy group-containing drug molecules like PTX (30c)²⁹⁹ can be modified with carbonyl group for construction of hydrazone linkage, further hydrolysis of the carbonyl group by esterase is required to generate the original structure of the drug (Fig. 16A). Notably, the hydrolysis product of hydrazone, also known as the hydrazide residue, showed a certain degree of cytotoxicity in several cases,³⁰⁰ which should be taken into consideration when designing a DDS with this cleavable linker type.

Another amine-containing chemical structure, the imine motif,³⁰¹ provides wider applicability for drug candidates (Fig. 16B) compared to hydrazines.³⁰² Drug/gating molecules which contain either a primary amine group (*e.g.* DOX (31a),³⁰³ MTX³⁰⁴ and poly(poly(ethylene glycol)methylether methacrylate-*co-p*-(2-methacryloxyethoxy)benzaldehyde) [P(PEGMA-*co*-MAEBA)] copolymers (31b)³⁰⁵ or a carbonyl group (*e.g.* gossypol (31c)³⁰⁶ can be conjugated to form imine bonds (Table 3). The formation of imine bonds occurs under mild reaction conditions, thus opening applicability to numerous drug carriers.³⁰⁷ Unfortunately, imines exhibit less hydrolytic stability than hydrazone in aqueous solution,³⁰⁸ since the increased negative-charge density on the C1 position of hydrazone reduces its electrophilicity.²⁹⁶ Therefore, premature drug release caused by the hydrolysis of imine bond in systemic circulation remains a challenge for applications in DDSs.

To address the instability problem of imine bond and realise time-prolonged drug release, considerable efforts have been devoted to oxime linkages owing to their greater stability than both imines and hydrazones (Fig. 16C).³⁰⁹ Because the high electronegativity of oxygen atom imparts high stability to the sp² nitrogen, aliphatic oxime exhibited 100 to 1000-fold higher resistance to hydrolysis than hydrazone.²⁹⁶ Inspired by their improved hydrolytic stability, oxime bond has been extensively

investigated for sustained drug release, including in prodrug formation^{310–312} and DDSs construction.^{313,314} However, similar to hydrazone, the application of oxime bond for drug conjugation has been limited to carbonyl group-containing drug molecules. For drug molecules bearing carbonyl group (32a), including DOX³¹⁵ and daunorubicin (DAU)³¹³ (Table 4), they can be directly reacting with alkoxyamine linker to afford the oxime bond, while the conjugation of hydroxy or amine group-containing drugs (*e.g.*, GEM (32b)³¹⁶ and AGD-0182 (32c)³¹⁷) requires additional modification to obtain an essential moiety to build the oxime linkage. Similarly, additional hydrolysis by esterases or proteases will further generate the drug molecules with their original structures (Table 4).

2.4.2 pH-cleavable esters. Apart from Schiff bases, carboxylic esters also received interest in the area of drug delivery for their general susceptibility to hydrolysis at acidic pH (Fig. 17A).³¹⁸ The carboxylic ester used for drug/gating molecule conjugation is usually formed through an esterification between the carboxylic acid and alcohol, and the latter exists widely in drug/gating molecules.³¹⁹ Despite the broad choice of drug molecules, this type of chemical structure tends to show relatively high stability in the weak acidic environment where the pH value ranges from 5 to 6, leading to slow hydrolysis rate constant.^{320,321} Therefore, carboxylic esters remain rarely explored as acid-cleavable linkers, but under basic environments they exhibit favourable hydrolysis rate,³²² which promotes growing interest to apply carboxylic ester in controlled drug release in basic organelles like the mitochondrial matrix (pH = 7.9–8.0)³²³ and peroxisomes (pH = 8.2).³²⁴ In addition to their susceptibility to pH, carboxylic esters can also be hydrolysed by esterases, such as aryl sulfatase, cholesteryl ester acid hydrolases, cathepsin and so on.³²⁵ The general responses of carboxylic esters to not only the pH value but also enzymes result in physiological instability and reduced specificity (see more detailed discussion of enzyme-induced cleavage in Section 2.5), which further causes premature drug release at undesired sites. To date, the carboxylic esters have been explored for the constructions of DDSs based on drug carriers like MSNs,³²⁶ antibodies^{67,86} and ONPs (*e.g.*, dendrimers³²⁷). Various types of gating and drug molecules have been incorporated, such as MTX (33a),³²⁷ tetra(ethylene glycol) (33b)³²⁶ and PTX (33c).⁶⁷ In the meanwhile, hydroxy group-containing drug molecules like cyclosporine A³²⁸ and erythromycin³²⁹ can also be modified with carboxylic acid end, in order to form the carboxylic ester with drug carriers bearing hydroxy group (Table 4).

Compared with carboxylic esters, orthoesters can be readily hydrolysed in mild acidic solution to form the corresponding esters and alcohols as shown in Fig. 17B.³³⁰ More importantly, further studies have revealed the strong relationship between the hydrolysis rate of orthoesters and their substituents,³³¹ indicating that the sensitivity of orthoesters to pH can be tuned by introducing different structural variations. Among these chemical modifications, six or five-membered ring substitution facilitate the hydrolysis rate of orthoesters, thus allowing for faster payload release.³³² The resulting cyclic orthoesters find

Table 4 Summary of pH-cleavable linkers

Core structure	Cargo	pH value	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
Hydrazone	Doxorubicin	5–6	ONPs (dendrimers)/GNPs/MSNs/ADCs	MDA-MB-231/MCF-7/4T1/luc/HepG2/HeLa/rabbit red blood cells	MDA-MB-231 xenograft in mice	292–294, 298 and 564
	Calicheamicin	5–6	ADCs	Carcinoma/HL-60/Raji (CCL86)/HEL92.1.7 cells	Colony formation; N87/LOVO/LNCaP/HL-60 xenograft in nude mice	565
	Paclitaxel	5–6	Polymer-DCs/GNPs	MACL MCF-7/MEXF 514/RXF 944/HepG2 cells	HepG2 xenograft in nude mice	299 and 566
	Auristatin E	5–6	ADCs/Prodrugs	H3396/Karpas 299/LXFL 1674 L/sw-620/CAL-27/RKO/MDA-MB-468/SK-OV-3/RCA colorectal carcinoma cells	A375/A2780/LXFA737/LXFE937 xenograft in mice	297 and 567
Imine	Doxorubicin	5–6	ONPs (PNPs/micelles)	B16F10/MC3T3-E1/HeLa/4T1 cells	B16F10/4T1 xenograft in nude mice	303 and 307
	Gossypol	5–6	Peptide-DCs	MCF-7/HeLa/WI-38 cells	—	306
	Methotrexate	5–6	ONPs (PNPs)	MCF-7/HeLa cells	HeLa xenograft in nude mice	304
	6-hydroxy-chromone-3-carbaldehyde	5–6	Fe ₃ O ₄ NPs	HeLa cells	—	568
Oxime	Doxorubicin	5–6	MSNs	HepG2 cells	—	305
	Betaxolol	Bioconversion	Prodrugs	—	Rabbits	310
	Propranolol	Bioconversion	Prodrugs	—	Guinea pig atria and trachea	311
	Timolol	Bioconversion	Prodrugs	—	Rabbits	312
	Daunorubicin	Bioconversion	Peptide-DCs	MCF-7/HT-29/LNCaP/A549/HCC1954/DU145/PC3 cells	Colon-26 xenograft in nude mice	313 and 314
	Doxorubicin	Bioconversion	SMDCs	MIA PaCa-2/MCF-7/Capan-2/A549/AtT-20 cells	—	315
	Alprenolol	Bioconversion	Prodrugs	—	Rabbits	569
Carboxylic ester	Gemcitabine	5–6	Peptide-DCs	A549/HCC1954/DU145/PC3 cells	Mice	316
	AGD-0182-30	5–6	ADCs	human hepatocytes/Caco-2/MDCKII-MDR1/HEK293/Karpas299/Molm-13 cells	MOLM-13 xenograft in nude mice	317
	Paclitaxel	Bioconversion	ADCs/ONPs (dendrimers)	A431/HBL-100/SK-OV-3/HCT-116/NIH 3T3/KB cells	A431 xenograft in nude mice	67 and 570
	SN-38	Bioconversion	ADCs/ONPs (dendrimers)	hMN-14/LoVo cells/HCT-116	Lung metastatic model of human colon Carcinoma in nude mice	86 and 571
Orthoester	Methotrexate	Bioconversion	ONPs (dendrimers)	KB cells	—	327
	Erythromycin	Bioconversion	ONPs (dendrimers)	RAW 264.7 cells; S. aureus strain	—	329
	Cyclosporine A	Bioconversion	ONPs (PNPs)	Splenic dendritic cells from naive BALB/c mice	BALB/c mice footpads dendritic cells injection	328
	[Ru(bipy) ₃]Cl ₂	Bioconversion	MSNs	HeLa/MCF-7 cells	—	326
	Cisplatin	5–6	ONPs (micelles)	HepG2/H22 cells	H22 xenograft in nude mice	335
β-Thiopropionate	Paclitaxel	5–6	ONPs (PNPs)	RAW 264.7/OVCAR-3 cells	—	340
	Camptothecin	5–6	ONPs (micelles)/silver nanoparticles	HeLa/HepG2 cells	—	341 and 572
	Doxorubicin	5–6	Fe ₃ O ₄ @mSiO ₂ nanocomposite	HeLa cells	—	342
Acetals or ketals	Podophyllotoxin and 5-fluoro-2'-deoxyuridine	5–6	Polymer-DCs	—	—	343
	Paclitaxel	5–6	ONPs (micelles)	MDA-MB-231/HeLa/A549/KB/MCF-7 cells	—	348, 573 and 574
	Curcumin	5–6	ONPs (micelles)	HepG2 cells	—	350
	Resveratrol	5–6	Prodrugs	—	Rat	344
	Raltegravir	5–6	Prodrugs	Primary rat/dog/human hepatocyte cells	Rat/dog	575
	Gemcitabine	5–6	ONPs (PNPs)	A2780 cells	A2780 xenograft in nude mice	349

Table 4 (continued)

Core structure	Cargo	pH value	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
	Doxorubicin	5–6	MSNs	HNE-1 cells	HNE-1 xenograft in nude mice	347
Silyl ether	Docetaxel	5–6	ONPs	A549 cells	A549 xenograft in nude mice	352
	Camptothecin	5–6	MSNs/ONPs (PNPs)	HeLa cells	HeLa xenograft in nude mice	353 and 576
	Gemcitabine	5–6	ONPs	LnCAP cells	—	355
	Monomethyl auristatin E	5–6	ADCs	NCI-N87/BT-474/MDA-MB-453/MCF-7/MDA-MB-231 cells	NCI-N87 xenografts in nude mice	356
Phosphoramidate linker	Levodopa	5–6	Prodrugs	—	—	359
	Monomethyl auristatin E	5–6	Prodrugs	—	—	360
<i>cis</i> -Aconityl moiety	Daunorubicin	5–6	Polymer-DCs	WEHI-5 cells	—	366
	Doxorubicin	5–6	ONPs (dendrimers)/GNPs/ADCs/MSNs	KB/U87MG/HTB119/A549/HeLa cells	—	50, 361, 363 and 364
	Camptothecin	5–6	ONPs (micelles)	MCF-7 cells	—	362

Note: ONPs means organic nanoparticles; GNPs means gold nanoparticles; MSNs means mesoporous silica nanoparticles; PNPs means polymeric nanoparticles; ADCs means antibody–drug conjugates; SMDCs means small molecule drug conjugates; polymer-DCs means polymer–drug conjugates; peptide-DCs means peptide–drug conjugates; AGD-0182-30 means microtubule disrupting agent and a stable noncleavable linker; — means not mentioned.

broad applications in the preparation of acid-degradable ONPs³³³ and prodrugs,³³⁴ where the hydroxy group-bearing compound is released as a result of linker cleavage. Although the facile hydrolysis of orthoester under mild acidic conditions can enable fast drug release to reach the therapeutic effective concentrations, studies on using orthoester-based linkers for the construction of DDSs are still in their infancy and limited success has been achieved in this field (34).³³⁵

In contrast to the high sensitivity of orthoesters to low pH, β -thiopropionates, which refer to carboxylic esters with a sulfide substitution at the β position (Fig. 17C), show relatively slow hydrolysis rate.³³⁶ It is widely acknowledged that the hydrolysis of β -thiopropionates starts with the generation of a partial positive charge on the carbonyl carbon atom owing to the inductive effect of the sulfur atom (Fig. 17C), and the neighbouring group participation by the sulfur atom leads to a relatively slow β -thiopropionate hydrolysis rate.³³⁷ Moreover, another advantage of this linker is that it can be prepared through a facile thiol-Michael addition reaction^{338,339} between thiol-containing compounds and α,β -unsaturated carbonyl compounds, in which cases the thiol-containing compounds can be used for the installation of linkers onto the drug carriers (35a), including ONPs³⁴⁰ and silver NPs.³⁴¹ Likewise, β -thiopropionates can be also used to tether gating molecules like PEG (35b)³⁴² to the surface of NPs to regulate the drug release.

2.4.3 pH-cleavable ethers. Acetals or ketals are commonly used in *in vivo* applications due to their high susceptibility to acidic microenvironment and stability under reductive and oxidative stresses.³⁴³ These properties therefore promote applications of acetals and ketals in a wide range of areas including the prodrug formation,³⁴⁴ degradable polymers construction³⁴⁵ and controlled release.³³³ The hydrolysis of acetals or ketals

induced by acidic conditions yields the corresponding drugs and carbonyl-containing products (Fig. 18A).³⁴⁶ Moreover, the hydrolysis rate of acetals or ketals can be tuned by changing the functional groups adjacent to the acetal or ketal carbon, where electron donating substituents play the predominant role in stabilising the cationic intermediates to facilitate hydrolysis.³⁴⁶ Taking advantages of the non-toxic hydrolysis side products and tunable hydrolysis rate, acetals or ketals have been widely accepted for drug delivery purpose. Although acetal or ketal-based linkers can deliver both carbonyl and hydroxy-containing drug molecules, conjugation with hydroxy-containing drug/gating molecules like poly(acrylic acid) homopolymer (PAA) (36a),³⁴⁷ PTX (36b)³⁴⁸ and GEM (36c)³⁴⁹ were more often reported (Table 4). While for molecules bearing phenolic hydroxy group, such as curcumin,³⁵⁰ they can be modified with an alcohol terminal prior to the reaction with vinyl ether to form the corresponding acetals or ketals.

As a novel pH-cleavable ether, silyl ethers are commonly used as protecting groups for alcohols and phenols in organic synthesis due to their tunable deprotection rate by altering the substituents on the silicon atom.³⁵¹ Most importantly, silyl ethers provide the advantage of maintaining stability upon exposure to variety of enzymes compared to ester-based linkers, thus ensuring more selective drug release.³⁵² Hydrolysis of a silyl ether generates the corresponding drug and a nontoxic silanol byproduct (Fig. 18B). Moreover, silyl ethers are prepared using a one-pot synthetic method.³⁵³ As a result, silyl ethers, typically trimethyl silyl ethers (TMS), triethyl silyl ethers (TES) and triisopropyl silyl ethers (TIPS), have been pursued in many areas including prodrug formation,³⁵² acid-labile biomaterials preparation³⁵⁴ and DDSs construction.³⁵⁵ The silyl ether is capable of delivering hydroxy-containing drug molecules such

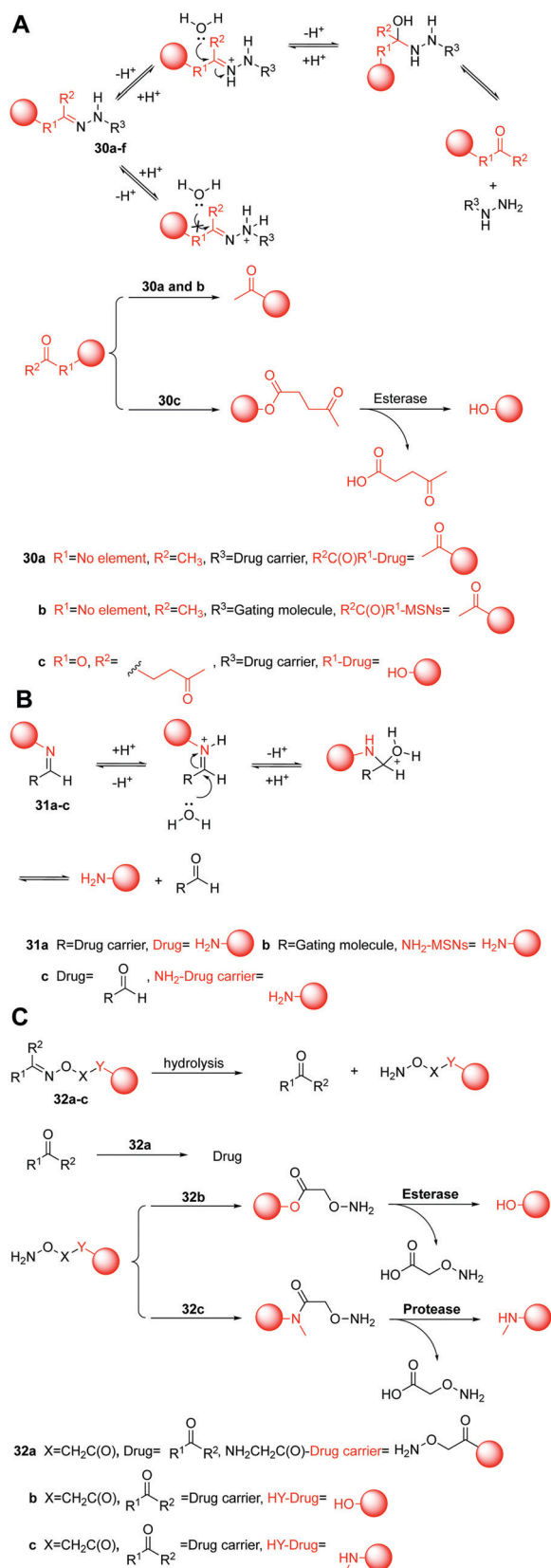


Fig. 16 (A) Chemical structures and cleavage mechanism hydrazone-based linkers triggered by low pH; (B) chemical structures and cleavage mechanism imine-based linkers triggered by low pH; (C) structure of oxime-based linkers and acid-induced hydrolysis of oxime motif.

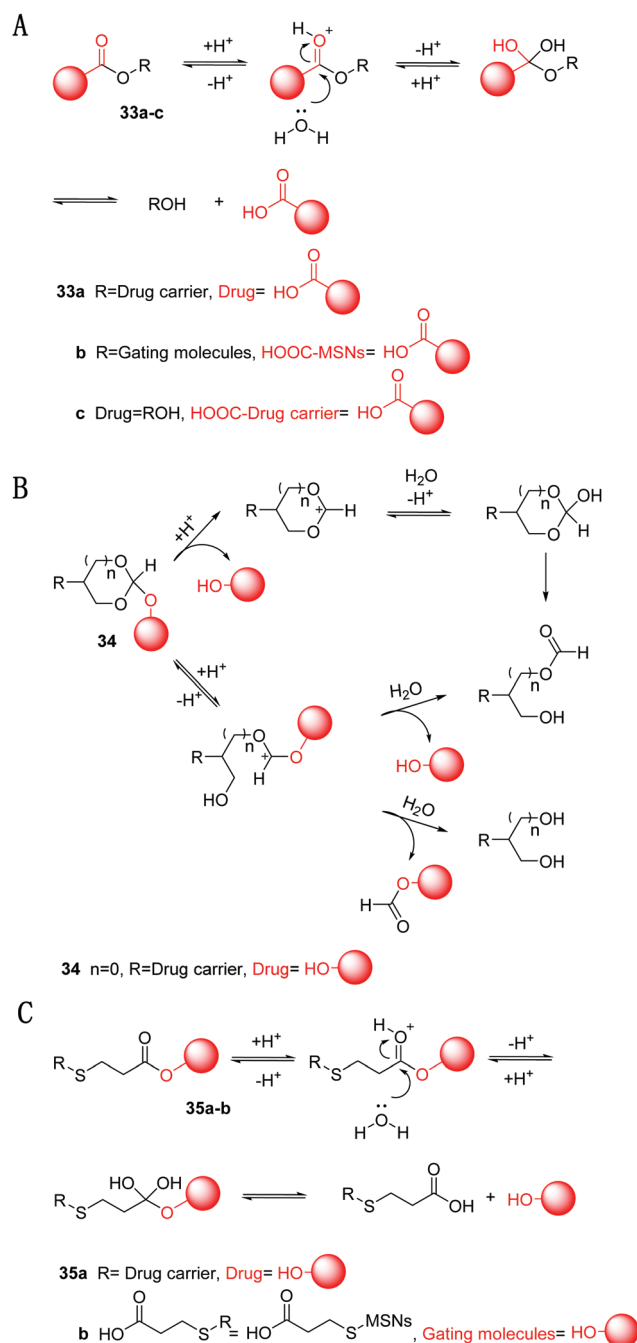


Fig. 17 (A) Schematic illustration of drug release induced by carboxylic ester cleavage in the presence of low pH or corresponding enzymes; (B) proposed cyclic orthoester hydrolytic mechanism under acidic environments; (C) chemical structure and hydrolysis mechanism of β -thiopropionate-based linker, and drug/gating molecule release is realised through the degradation of β -thiopropionate linker.

as CPT (**37a**)³⁵³ and GEM (**37b**)³⁵⁵ with its other side serving as the drug carrier (e.g., antibodies³⁵⁶ and MSNs³⁵³) attaching site (Table 4).

2.4.4 Other pH-cleavable linkers. More recently, phosphoramidate scaffolds have emerged as novel pH-cleavable linkers.³⁵⁷ Having a mildly acidic group (pyridinium or carboxylic acid) proximal to the phosphorus centre, the phosphoramidate linkers

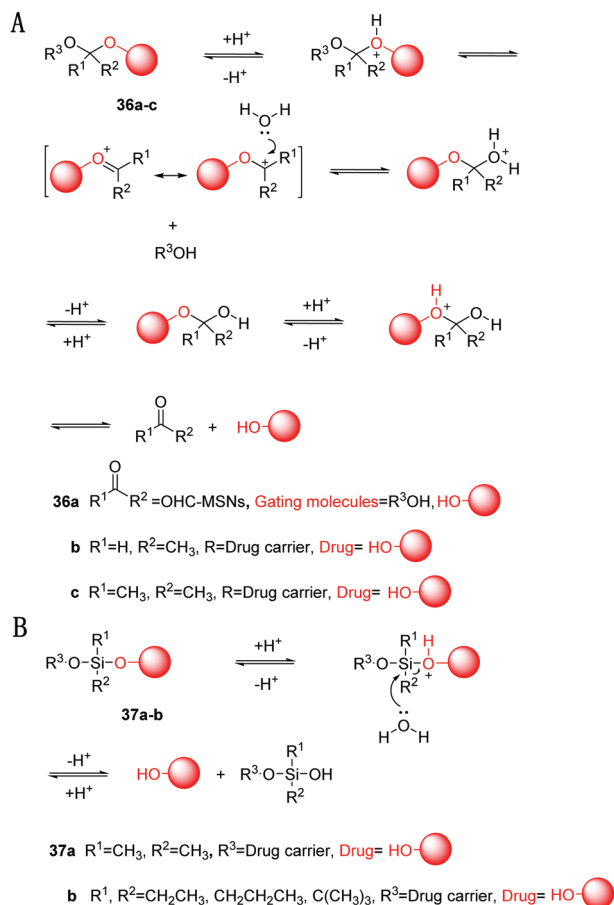


Fig. 18 Hydrolysis of ether linkers. (A) Hydrolysis of acetal or ketal linker for cargo release. (B) Mechanism of silyl ether hydrolysis.

exhibit unusual hydrolytic stability at mild acidic pH values, which can be further tuned through altering the proximity of the pyridinium or carboxylic acid to the core of the phosphoramidate scaffolds (Fig. 19A).^{357,358} In general, the neighbouring proton donor-assisted cleavage of phosphoramidate scaffolds results in the release of the corresponding drug and the linker residue terminated with phosphoric acid (Fig. 19A). Phosphoramidate linkers have been recently utilised for prodrug formations, where amine group-containing drugs including levodopa (**38**)³⁵⁹ and MMAE (**39**)³⁶⁰ were tethered to the linker directly or *via* a self-immolative spacer (Fig. 19B). Although phosphoramidate scaffolds remain rarely pursued for constructing DDSs with pH-cleavable linkers, the R site of compound **39** has been modified with azide group, which opens up the possibility of attaching the linker to the drug carriers based on 'click chemistry'.

As a derivative of natural aconitic acid, the *cis*-aconityl moiety has facilitated controlled drug release from drug carriers like ONPs (dendrimers³⁶¹ and micelles³⁶²), GNPs,⁵⁰ MSNs³⁶³ and antibodies.³⁶⁴ The proposed mechanism of the hydrolysis of *cis*-aconityl-based linkers involves first proton transfer from the carboxylic acid in *cis*-position of the hydrolytic bond to the amide, followed by the formation of tetrahedral intermediate which is further hydrolysed under mild acidic conditions (Fig. 19C).³⁶⁵ Based on the acylation reaction, drug/gating molecules that contain amine

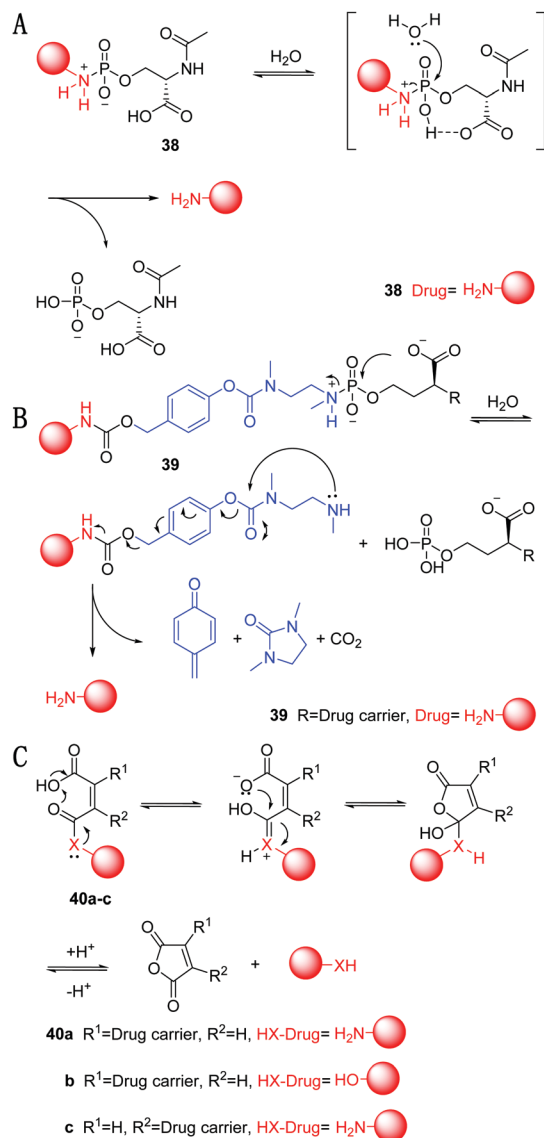


Fig. 19 Structures and drug release mechanisms of DDSs containing novel pH-cleavable linkers. (A and B) Illustration of acid-triggered cleavage of phosphoramidate linker. (C) Proposed mechanisms of hydrolysis of *cis*-aconityl linker.

or hydroxy groups such as DOX (**40a** and **40c**),^{361,363} DAU³⁶⁶ and CPT (**40b**)³⁶² can react with *N*-*cis*-aconityl anhydride to form *cis*-aconityl-based linkers (Table 4). The resulting linkers can be immobilised on amine-functionalised drug carriers.^{362,367} Due to the novel chemical structure, *cis*-aconityl linkage enriches the library of pH-cleavable linkers and provides more options to meet the specific requirement for certain controlled drug release conditions.

2.5 Enzyme linker cleavage

Enzymes are biological molecules (mostly proteins) that are responsible for accelerating virtually all the chemical reactions within cells.³⁶⁸ The high substrate specificities and robust enzymatic reactions of enzymes differentiate them from many

other chemical stimuli, thus allowing for more specific chemical reactions.³⁶⁹ Moreover, dysregulation of certain enzymes is a hallmark of pathology for a variety of diseases such as cancer,³⁷⁰ inflammation³⁷¹ and neurodegenerative diseases.³⁷² Therefore, using enzymes as stimuli offers a great opportunity in the controlled drug release from DDSs.

2.5.1 Hydrolase-cleavable linkers. There are six different types of enzymes that are well studied in the applications of stimulus-induced drug release, including hydrolases, oxidoreductases, transferases, lyases, isomerases and ligases.³⁷³ Among them, the hydrolases are capable of cleaving chemical bonds such as ester³²⁵ and peptide³⁷⁰ bonds using water as the nucleophile. As the subclass of hydrolases, esterases, proteases and glycosidases have been extensively explored as the stimuli of controlled drug release, promoting the development of corresponding enzyme-labile chemistries.

2.5.1.1 Esterase-cleavable linkers

2.5.1.1.1 Carboxylesterase-cleavable linkers. Esterases represent the most common group of hydrolases involved in the catalysis of ester bond cleavage.³⁷⁴ A wide range of esterases exists and can be further classified based on their substrates, such as carboxylesterases, acylesterases, thioester hydrolases, phosphatases, sulfatases and endonucleases.

As an important class of protective enzyme in biological systems, carboxylesterases can detoxify xenobiotics into less toxic hydrolysis products.³⁷⁵ Carboxylesterases can be further classified into two main types referring to non-specific carboxylesterases and lipases, based on their different substrate specificity.³⁷⁶ It was firstly established that non-specific carboxylesterases are usually active in hydrolysing water-soluble carboxylic esters whereas lipases preferentially catalyse the hydrolysis of triglycerides comprising of natural fatty acids.³⁷⁷ The short-chain carboxylic ester (**41a**), as aforementioned in the pH-cleavable linker section, exhibits dual response to both pH variations and non-specific carboxylesterases, the latter can cleave ester bonds using a base-catalysed approach with the assistance of water molecules (Fig. 20A).³⁷⁸ Meanwhile, given the slow hydrolysis rate of alkyl carboxyl esters in the presence of esterases,³⁷⁹ a more active ester bond known as carbonic ester (**41b**), has been more frequently utilised for constructing nanocarrier-based DDSs. Although the expression level of carboxylesterase in several pathological tissues is higher than that in healthy tissues,³⁸⁰ certain level of them has been identified in plasma which may result in premature drug release.³⁸¹ In general, the non-specific response of carboxyl ester bond to both pH variations and enzymes constitutes an obstacle for expanding their applications in the field of drug delivery.

Lipases are serine hydrolases that enable hydrolysis of carboxyl ester of long-chain acylglycerol into fatty acid and glycerol. As the most extensively studied subclass of lipases, phospholipases are capable of cleaving phospholipid substrates at specific ester bonds, which can be further divided into five subfamilies including Phospholipase A1 (PLA1), Phospholipase A2 (PLA2), Phospholipase B (PLB), Phospholipase C (PLC) and Phospholipase D (PLD).³⁸² Among these phospholipases, human secretory PLA2 (sPLA2) has attracted tremendous

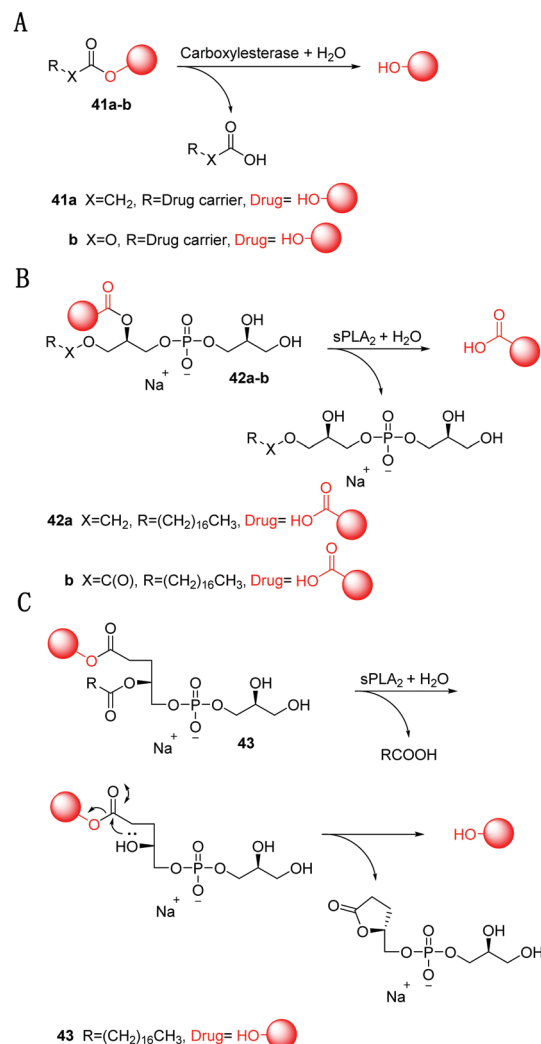


Fig. 20 Drug release triggered by the cleavage of ester linkers. (A) Structure and cleavage mechanism of carboxylic ester/carbonic ester-based linkers upon the exposure to esterases. (B and C) Schematic overview of sPLA2-triggered degradation of glycerophospholipids prodrugs, releasing the lipophilic drug and the cytotoxic byproduct.

attention owing to their abnormal level in pathological tissues such as inflammation and cancer,^{383,384} making this enzyme subfamily a promising target for the construction of DDSs. sPLA2 can preferentially hydrolyse glycerophospholipids at the sn-2 position, affording fatty acids and cytotoxic lysophospholipids (Fig. 20B).³⁸⁵ Glycerophospholipids were reported to conjugate drug molecules bearing carboxylic acid at the sn-1 or the sn-2 position, using the drug's carboxylic acid group. Molecules like chlorambucil,³⁸⁶ prostaglandins³⁸⁷ and all-*trans* retinoic acid (ATRA)³⁸⁸ can be installed at the sn-2 position to form **42a** and **b**, and sPLA2 can directly cleave the ester bond to enable the payload release. In contrast, for molecules such as capsaicin³⁸⁹ installed at the sn-1 position, the resulting compound **43** can be hydrolysed by sPLA2 to afford an active hydroxy group at the sn-2 position, which subsequently undergoes a cyclisation reaction with the ester group at the sn-1 position to release the payload (Fig. 20C). Utilising the sn-1 ester or sn-1 ether

as the backbone, the phospholipid–drug conjugates can be formulated as liposomes, which are widely used DDSs to deliver the lipophilic drugs (Table 5).

2.5.1.1.2 Phosphatase-cleavable linkers. Phosphatases are a subcategory of hydrolases that play a key role in protein dephosphorylation.³⁹⁰ Because of their prevalence in cellular signalling and activity regulation, phosphatases are receiving increasing interest in pharmaceutical research.³⁹¹ Two important phosphatase subfamily members: acid phosphatase (ACP) and alkaline phosphatase (ALP), have been widely studied over the past few decades for their biological roles in metabolic activities. These two enzymes are distinguished by the fact that ACP have an acid pH optimum³⁹² while ALP is optimally active under alkaline conditions.³⁹³

As the substrates of ALP, phosphate monoesters impart a negative charge at physiological pH, owing to their low pK_a .³⁹⁴ Since phosphate monoesters can enhance the water solubility of hydrophobic drug molecules^{395,396} and the cleavage of them is mainly triggered by intestinal ALP, they find broad applications in prodrug formations³⁹⁷ and fluorogenic probe design³⁹⁸ (Fig. 21A), particularly in hepatobiliary and bone diseases where ALP is upregulated. The two hydroxy groups in phosphoric acids can also be simultaneously protected by placing them in a cyclic ring. As the most extensively explored nucleotide masking group, the cyclic 1-aryl-1,3-propanyl ester (HepDirect) has been utilised for prodrug formations of some drugs in clinical trials such as (3,5-dimethyl-4-(4'-hydroxy-3'-isopropylbenzyl)phenoxy)methylphosphonic acid³⁹⁹ and Adefovir⁴⁰⁰ (Table 5). As shown in Fig. 21B, the cleavage of the HepDirect phosphate linker conjugated with hydroxy group-containing drug molecules **45** starts with the cytochrome P450 (CYP)-catalysed oxidative cleavage of 1-aryl-1,3-propanyl ester, where only the phosphates with a *cis*-relationship between the nucleoside part and the aryl group can be oxidised by CYP.⁴⁰¹ A subsequent ring opening reaction occurs after the oxidative cleavage of 1-aryl-1,3-propanyl ester, followed by spontaneous β -elimination reaction to generate the corresponding phosphate monoester and an aryl vinyl ketone byproduct (Fig. 21B). The phosphate monoester can be further hydrolysed by ALP to release the drug.⁴⁰² Although the HepDirect approach serves as a novel extension to the cyclic phosphate prodrugs and is able to conjugate drugs with hydroxy groups, the incorporation of this linker into the construction of DDSs has been rarely reported.

2.5.1.1.3 Sulfatase-cleavable linkers. Sulfatases are enzymes of the esterase class that catalyse the hydrolytic desulfonation of sulfate esters and sulfamates.⁴⁰³ The activity of sulfatases is found to be higher in lysosomes than in plasma,⁴⁰⁴ thus offering an opportunity for site-specific payload release. Moreover, sulfatases are overexpressed in several diseases such as cancer,⁴⁰⁵ giving additional selectivity for tumour targeting DDSs.

The sulfate esters (**46**) as the cleavable linkers have been applied in constructing DDSs,⁴⁰⁶ where sulfate esters were used in combination with self-immolative spacers to realise sulfatase-cleavable payload release. As shown schematically in Fig. 22, enzymatic hydrolysis of sulfate esters leads to the formation of 4-alkoxybenzyl carbamate, which undergoes a

spontaneous 1,6-elimination to release amine-containing payloads such as MMAE. Owing to the utilisation of the aromatic ring-containing self-immolative spacer that can bear versatile functional groups, a variety of drug/gating molecules can be introduced to the linker. Moreover, the versatile chemistry afforded by the self-immolative spacer also enables installation of linkers on an expanded range of drug carriers.

2.5.1.2 Protease-cleavable linkers. Proteases are defined as enzymes that catalyse the hydrolysis of peptide bonds, and have diverse biological functions,⁴⁰⁷ in particular digestion of ingested proteins, protein catabolism, and cell signaling.⁴⁰⁸ Given the vital roles of proteases in the protein metabolism, dysregulation of them can lead to various kinds of diseases including cancer, inflammatory, cardiovascular diseases and neurodegenerative diseases.⁴⁰⁹ Therefore, multiple proteases have become the subject of interest of pharmaceutical research, such as the discovery of protease inhibitors, protease cleavable linkers and diagnostic fluorogenic probes.⁴¹⁰

Based on the structure of catalytic substrates, proteases are classified into six groups including glutamic protease, aspartic protease, serine protease, cysteine protease, threonine protease and metalloprotease.⁴¹¹ It is widely acknowledged that glutamic protease, aspartic protease and metalloproteases cleave peptide bonds with the assistance of water,⁴¹² whereas the other families use the amino acid residues as the nucleophile to complete peptide bond cleavage.⁴¹³ Among these six families of proteases, the serine protease,⁴¹⁴ cysteine protease⁴¹⁵ and metalloprotease⁴¹⁶ have been widely explored as targets for the design of protease-cleavable linkers.

2.5.1.2.1 Serine protease-cleavable linkers. Accounting for over one third of proteolytic enzymes, serine proteases are widely distributed in both eukaryotes and prokaryotes. A unique structure, commonly comprised of two beta-barrel domains, is characteristic for serine proteases.⁴¹⁷ Three typical types of serine proteases which are well studied as the stimuli of cleavable linkers will be further discussed in this section, including prostate-specific antigen (PSA), plasmin (PLA) and neutrophil serine proteases (NSPs).

Prostate-specific antigen (PSA) belongs to the family of serine proteases and has been widely used as a biomarker for the diagnosis of prostate cancer, as the systemic level of it in prostate cancer patients is found to be elevated.⁴¹⁸ Therefore, the specificity of PSA for prostate cancer has gained attention in controlled drug release area. The commonly used PSA-cleavable peptide linkers like Ala-Ser-Chg-Gln sequence and His-Ser-Ser-Lys-Leu-Gln-Leu sequence that contain a Gln residue at P1 position, which imparts PSA specificity since PSA is the only member of kallikrein family bearing a Gln residue in that position to enable proteolysis.⁴¹⁹ In an initial attempt to construct peptide–drug conjugates (peptide–DCs), the amine group-containing drug molecule like L12ADT^{420,421} was directly tethered to the His-Ser-Ser-Lys-Leu-Gln-Leu sequence to form **47a** and **47b** (Table 5). After the PSA-triggered cleavage between the Gln and Leu site, the resulting hydrolysis product with a leu residue requires the assistance of aminopeptidases to remove the leucine residue and

Table 5 Summary of enzyme-cleavable linkers

Core structure	Cargo	Enzymes	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
Glycerophospholipid	Chlorambucil	sPLA2	ONPs (liposomes)	HT-29/MT-3/ES-2 cells	—	386
	Prostaglandins	sPLA2	ONPs (liposomes)	HT-29/Colo205 cells	—	387
	All- <i>trans</i> retinoic acid	sPLA2	ONPs (liposomes)	MT-3/HT-29/Colo205 cells	—	388
	Capsaicin	sPLA2	ONPs (liposomes)	Colo 205 cells	—	389
Phosphate monoester	Benzimidazole derivative	ALP	Prodrugs	MDCK cells	Liver/intestinal S9 fractions of rat/monkey/dog/human; rat/dog/monkey	577
	[5-Chloro-2-(methylthio)-6-(1-naphthylthio)-1 <i>H</i> -benzimidazole]	ALP	Prodrugs	<i>Fasciola hepatica metacercariae</i>	Sheep infected with <i>F. hepatica metacercariae</i>	397
Cyclic 1-aryl-1,3-propanyl ester	(3,5-Dimethyl-4-(4'-hydroxy-3'-isopropylbenzyl)phenoxy)methyl-phosphonic acid	CYP450	Prodrugs	Primary rat hepatocyte cells	Rat/diet-induced obese mouse	399
	Adefovir	CYP450	Prodrugs	Primary rat hepatocyte cells	Rat/portal vein-cannulated rat/monkeys	400
	4-(6-Methoxynaphthalen-2-yl)butan-2-ol	CYP450 + ALP	Prodrugs	Human liver microsomes	Human liver samples	402
Sulfate ester	Withaferin A	Sulfatase	Prodrugs	MCF-7/NCI-H460/PC-3M cells; mouse embryo fibroblasts	—	578
Gln-Ser-Leu	Monomethyl auristatin E	Sulfatase	ADCs	BT474/MCF-7 cells	—	406
	Doxorubicin	PSA	Prodrugs	LNCaP/DuPRO-1/PC3/Hct116/MDA-MB435S/T24/MRC-5/NHEB/NHME cells	LNCaP/CWR22 xenografts in nude mice	579
Hyp-Ser-Ser-Chg-Gln-Ser-Ser-Pro	Desacetyl vinblastine	PSA	Prodrugs	LNCaP/Colo320/T47D/NHEB/NHME cells	LNCaP/Colo320/CWR22 xenografts in nude mice	580
Ser-Ser-Lys-Tyr-Gln-Ser-Leu	TGX-D1	PSA	Peptide-DCs	C4-2 cells; human serum	C4-2 xenografts in nude mice	425
Ser-Ser-Lys-Tyr-Gln-Leu	L12ADT	PSA	Polymer-DCs	CWR22Rv1 cells	CWR22Rv1 xenografts in nude mice	414
His-Ser-Ser-Lys-Leu-Glu	5-Fluoro-2'-deoxyuridine	PSA	Prodrugs	LNCaP/TSU cells	—	581
His-Ser-Ser-Lys-Leu-Gln	Paclitaxel	PSA	Prodrugs	CWR22Rv1/DU145/TSU cells	—	582
	Diethyldithiocarbamate	PSA	Prodrugs	LNCaP/PC3 cells	—	422
	Niclosamide	PSA	Protein-DCs	Metastatic castration-resistant prostate cancer cells	CWR-22 xenografts in nude mice	421
Ala-Ser-Chg-Gln	L12ADT	PSA	Prodrugs	LNCaP/HCT-116 cells	LNCaP xenografts in nude mice	420
	Doxorubicin	PSA	Prodrugs	DU145 cells/human and mice hepatocytes/mouse and human blood	—	423
	Phosphoramidate mustard	PSA	Prodrugs	LNCaP/DU145 cells/human and mice hepatocytes/mouse and human blood	Mice	424
Val-Leu-Lys	Paclitaxel	PLA	Prodrugs	MCF-7/EVSA-T/WIDR/IGROV/M19/A498/H266 cells	—	432
Ala-Phe-Lys	Cytarabine	PLA	ONPs (PNPs)	Human blood plasma	—	431
	Methotrexate	PLA	Proteins	Human blood plasma	OVCAR-3 xenograft in nude mice	434
	Camptothecin and 9-aminocamptothecin	PLA	Prodrugs	MCF-7/EVSA-T/WIDR/IGROV/M19/A498/H266 cells	—	583
	Doxorubicin	PLA	Peptide-DCs	HUVEC/H5V/HT1080 cells	—	430
	Doxazolidine	PLA	Prodrugs	MCF-7/MiaPaCa2/BxPC3/L3.3/L3.5/L3.6PL/nontumour-derived HEK-293T cells; human plasma	—	584

Table 5 (continued)

Core structure	Cargo	Enzymes	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
Leu-Leu-Lys	Monomethyl auristatin E	PLA	ADCs	Pancreatic tumour cells	Pancreatic tumour xenograft in nude mice	433
Asn-Pro-Val DEVD linker	Paclitaxel	NE	Peptide-DCs	786-O cells	—	437
	Doxorubicin	Caspase-3	Protein-DCs/ Peptide-DCs	SCC7/MDA-MB-231 cells	SCC7/MDA-MB-231 xenograft in nude mice; X-ray exposed mice	454 and 456
	Monomethyl auristatin E	Caspase-3	Protein-DCs/ ONPs (PNPs)	MDA-MB-231/HCC70/SCC7 cells	MDA-MB-231/HCC70/SCC7 xenograft in nude mice	455 and 457
	Benzothiazolyl-cyanovinylene	Caspase-3	Fluorescent probes	HeLa cells	—	458
Phe-Arg-Arg-Gly	Doxorubicin	CTSB	ONPs (PNPs)	HT-29/MDA-MB231/A549/MCF7/HDF/H9C2 cells	HT-29 xenograft in nude mice	585
Phe-Lys	Doxorubicin	CTSB	Prodrugs	SGC-7901/AsPC1/MDA-MB-231 cells	GC-7901 peritoneal carcinoma model; Nude mice; MDA-MB-435 xenograft in nude mice	586
Val-Ala/Val-Cit	Monomethyl auristatin E/SN-38/paclitaxel	CTSB	ADCs/ONPs (PNPs)/Polymer-DCs	SK-BR-3/T-47D/HCT-116/HT-29/COLO205/SKOV-3/MGC-803/MCF-7 cells	HCT-116/MCF-7 xenograft in nude mice	472, 473 and 587
					4T1/MDA-MB-231 xenograft in nude mice	470, 471 and 588
Gly-Phe-Leu-Gly	Gemcitabine/paclitaxel/ibuprofen	CTSB	ONPs (dendrimers)	4T1/COS-7/MDA-MB-231/MDA-MB-468/BT-20/T47-D/MDA-MB-435/JEG-3/BUMPT mouse proximal tubule kidney cells	4T1/MDA-MB-231 xenograft in nude mice	470, 471 and 588
Arg-Arg-Ala-Leu-Ala-Leu-Ala	Camptothecin	CTSB	Protein-DCs	—	HT-29/M-3366 xenograft in nude mice	474
Gly-Ile-Val-Arg-Ala-Lys-Glu-Ala-Glu	Safranin O	CTSB	MSNs	HeLa/MEFs cells	—	475
Lys-Lys	Luciferin derivative	CTSB	Fluorescent probes	HeLa/MDA-MB-231/U87MG/HEK-293 cells	—	476
Pro-Val-Gly-Leu-Ile-Gly	Methotrexate	MMP2/9	ONPs (dendrimers)	HT1080 cells	HT1080 xenograft in nude mice	416
Gly-Pro-Leu-Gly-Ile-Ala-Gly-Gln	Paclitaxel/doxorubicin	MMP2/9	ONPs(micelles)/Protein-DCs	A549/H9C2/RENCA cells	A549 xenograft in nude mice	490 and 491
Gly-Pro-Leu-Gly-Val	Doxorubicin	MMP2/9	ONPs (micelles)	LLC cells	LLC xenograft in nude mice	493
Cys-Pro-Leu-Gly-Leu-Ala-Gly-Gly	Doxorubicin	MMP2	ONPs (micelles)	SCC-7/COS7 cells	—	492
Pro-Leu-Gly-Leu-Glu-Glu-Ala	Coumarin 343 and TAMRA	MMP12	Fluorescent probes	Primary macrophages	Mice pulmonary inflammation	589
Gly-Val-Pro-Leu-Ser-Leu-Thr-Met-Gly-Cys	NIRQ820 and Cy 5.5	MMP7	Fluorescent probes	—	HT1080 xenograft in nude mice	590
Gelatin	Doxorubicin	MMP9	MSNs	HT-29/LO2 cells	HT-29 xenograft in nude mice	494
Glucuronide linker	Duocarmycin derivative	GUS	Prodrugs	A549 cells	—	508
	Paclitaxel	GUS	Prodrugs	—	—	509
	9-Aminocamptothecin	GUS	Prodrugs	LS174T/EJ/BT-20/(PLC/PRF/5)/CL 1-5/HepG2/HT29 cells	LS174T xenograft in nude mice	591
	Tacedinaline	GUS	Prodrugs	NCI-H661 cells	—	592
	Monomethyl auristatin E	GUS	Prodrugs/ADCs/Proteins	L540cy/L428/Karpas299/DEL/DEL-BVR/786-O/Ramos/A549/KB/MDA-MB-231/Primary human bronchial epithelial cells	L540cy/KB/LLC xenograft in nude mice	511, 513 and 593
	Doxorubicin	GUS	Prodrugs/ONPs (micelles)/Proteins	OVCAR-3/UM-SCC-14C/H290/LLC/U87-MG/MDA-MB-231 cells	OVCAR-3/LLC xenograft in nude mice	510, 514 and 594

Table 5 (continued)

Core structure	Cargo	Enzymes	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
Galactoside linker	8-Hydroxyquinolines	GLB1	Prodrugs	A2780/A549/MDA-MB-231 cells	—	521
	5-Fluorouridine	GLB1	Prodrugs	LS174T/LOVO/A375 cells	—	595
	Doxorubicin	GLB1	Prodrugs	OVCAR-3/HeLa/MCF-7 cells	—	523 and 596
	Daunorubicin	GLB1	Prodrugs	OVCAR-3 cells	—	525
	Pyrrolo[2,1- <i>c</i>][1,4]benzodiazepines	GLB1	Prodrugs	HepG2/A375 cells	—	597
	Monomethyl auristatin E	GLB1	ADCs/SMDCs	SK-BR-3/MDA-MB-231 cells	BT-474 xenograft in nude mice	524 and 527
	Dicyanomethylene-4H pyran derivative	GLB1	Fluorescent probe	293T/OVCAR-3 cells	LoVo xenograft in nude mice	598
	Hydroxymethyl- <i>N,N</i> -diethylrhodol	GLB1	Fluorescent probe	HEK/lacZ cells	<i>D. melanogaster</i>	599
	Salicylaldehyde azine	GLB1	Fluorescent probe	C6-LacZ/HeLa cells	—	600
	[Ru(bipy) ₃]Cl ₂	GLB1	MSNs	—	—	522
Indoquinone scaffold	SN-38	DTD	Peptide-DCs	KB cells	—	537
	Oridonin derivative	DTD	Prodrugs	A549/H596/HT-29 cells	—	601
	Nitric oxide	DTD	Prodrugs	T-24/HeLa cells	—	539
	Naloxone	DTD	Prodrugs	Human plasma	—	541
	Camptothecin	DTD	Prodrugs	HT-1080/EMT6/KU cells	—	540
	Phosphoramidate derivative	DTD	Prodrugs	—	—	542
	Triptolide	DTD	Prodrugs	HepG2 cell	Mice; HepG2 xenograft in nude mice	546
Quinone propionic acids	<i>N,N</i> -Bis(2-chloroethyl)-1,4-benzenediamine	DTD	Prodrugs	H460/BE colon cells	—	547
	Melphalan and Acivicin	Bioconversion	Prodrugs	—	Rat brain perfusion	602
	5-Fluorouracil	DTD	Prodrugs	A549/LO2 cells	A549 xenograft in nude mice	548
	Camptothecin	DTD	Prodrugs	A549/L929 cells	—	549
	Cucurbitacin B	Bioconversion	Prodrugs	MCF-7/Vero cells	4T1 xenograft in nude mice	603
	SN-38	DTD	Prodrugs	A549/HeLa/WI-38/BJ cells	A549 xenograft in nude mice	550
	Coumarin derivative	DTD	Fluorescent probes	—	—	551
	Eneidyne	NTRs	Prodrugs	AA8/UV4/SKOV-3/WiDr/T78-1 cells	—	604
	Phosphoramidate derivative	Bioconversion	Prodrugs	HT29 cells	—	556
	Camptothecin	NTRs	Prodrugs	HT-1080/EMT6-KU/HT29 cells	—	605
4-Nitrobenzyl scaffold	Doxorubicin	NTRs	Prodrugs	SKOV3/WiDr/V79 ^{puro} /EMT6 cells	Nude mice	606
	Nitric oxide	NTRs	Prodrugs	DLD-1 cells	—	607
	Checkpoint kinase 1 (Chk1) inhibitor	NTRs	Prodrugs	RKO/A549/H1299/H1975/WI-38 cells	—	557
	6-Thioguanine	NADPH: cytochrome P450 reductase	Prodrugs	A549 cells	—	558
	<i>O</i> (6)-Benzylguanine	xanthine/xanthine oxidase and NADPH: cytochrome P450 reductase	Prodrugs	EMT6/DU145 cells	—	559
	Coumarin derivative	NTRs	Fluorescent probes	—	—	560
	4-Amino- <i>N</i> -(2-(2-acetoxyethoxy)ethyl) naphthalimide	NTRs	Fluorescent probes	A549 cells aerobic and hypoxic model	—	561
	2-Nitrophenylalkanoyl moiety	NTRs	Prodrugs	Human serum	—	608 and 609
	5-Fluoro-2'-deoxyuridine	NTRs	Prodrugs	Human serum	—	608 and 609

Table 5 (continued)

Core structure	Cargo	Enzymes	Application	<i>In vitro</i> models	<i>Ex vivo/in vivo</i> models	Ref.
2,6-Dinitrophenyl unit	4-AAM and ACBIAU	Chemical reduction	Prodrugs	Irradiated AA8/MET6/SKOV3 cells	—	610

Note: Ala means alanine; ALP means alkaline phosphatase; Arg means arginine; Asn means asparagine; CG means cathepsin G; Cit means citrulline; CTSB means cathepsin B; Cys means cysteine; DEVD means comprising of tetrapeptide sequences (Asp-Glu-Val-Asp) exhibits as a specific substrate of for caspase-3; DTD means DT-diaphorase; GLB means β -galactosidase; Gln means glutamine; Gly means glycine; GUS means β -glucuronidase; Ile means isoleucine; Leu means leucine; Lys means lysine; MMP means matrix metalloproteinase; NE means neutrophil elastase; NTRs means nitroreductases; Phe means phenylalanine; PLA means plasmin; Pro means proline; PSA means prostate-specific antigen; Ser means serine; sPLA2 means secretory phospholipase A2; Thr means threonine; Met means methionine; Val means valine; — means not mentioned; ONPs means organic nanoparticles; ADCs means antibody–drug conjugates; MSNs means mesoporous silica nanoparticles; SMDCs means small molecule drug conjugates; polymer–DCs means polymer–drug conjugates; peptide–DCs means peptide–drug conjugates; protein–DCs means protein–drug conjugates; TGX-D1 means a phosphoinositide 3-kinases (PI3Ks) inhibitor; L12ADT means a cytotoxic thapsigargin analogue; Coumarin 343 and TAMRA mean fluorophores; NIRQ820 means quencher while Cy 5.5 means fluorophore, 4-AAM and ACBIAU means 4-aminoaniline mustard and 5-amino-1-(chloromethyl)benz[e]indoline alkylating units; PNPs means polymeric nanoparticles; — means not mentioned.

to produce the drug molecules in their original structure (Fig. 23A). To introduce drug molecules bearing other functional groups, such as thiol,⁴²² hydroxy groups⁴²³ and phosphoric acid⁴²⁴ to the PSA-cleavable linker, self-immolative spacers (**48**, **49a** and **49b** in Fig. 23B and C) were used, which afford amine terminated analogues of those drug molecules for the construction of amide linkage. So far, the application of PSA-cleavable linkers for constructing DDSs has been realised in an array of drug carriers, including proteins⁴²¹ and peptides⁴²⁵ (Table 5).

Plasmin (PLA) is a two-chain serine protease present in plasma and lysosomes. PLA can degrade blood plasma proteins such as fibrin clots. PLA plays a vital role in multiple physiological processes including macrophage invasion in inflammation,⁴²⁶ angiogenesis⁴²⁷ and breakdown of follicular wall for ovulation.⁴²⁸ Moreover, PLA has also been recognised as a mediator in

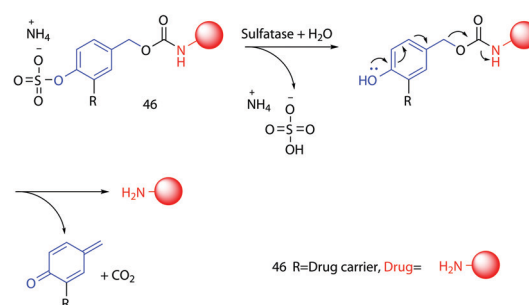


Fig. 22 Sulfatases-triggered linker cleavage.

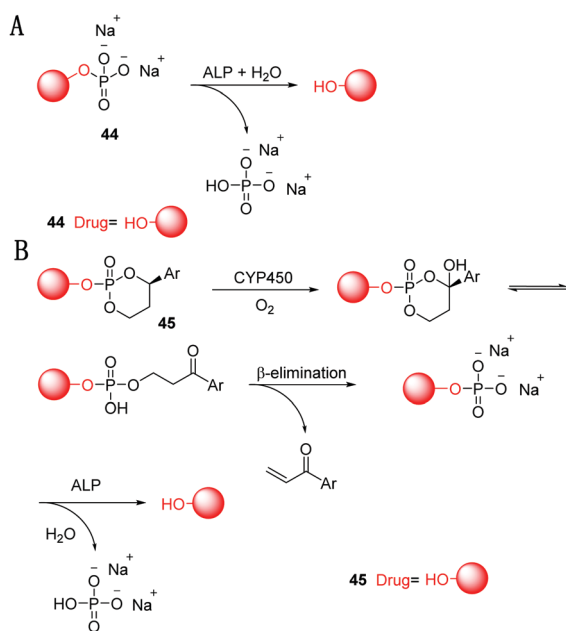


Fig. 21 Phosphatases-triggered linker cleavage. (A) Phosphate mono-ester cleavage upon phosphatases exposure. (B) Mechanism of cleavage of HepDirect derivative.

pathological process of cell migration associated with tumour invasion and metastasis.⁴²⁹ The most commonly used PLA-cleavable linkers refer to a series of tripeptide terminated with Lys or Arg, which remain extensively pursued in the drug delivery field.⁴³⁰ For amine group-containing drug molecules like cytarabine,⁴³¹ they can be directly conjugated to the C terminus of Val-Leu-Lys sequence to form **50**, and the PLA-triggered amide bond cleavage can enable the drug release (Fig. 24A). Likewise, drug molecules like PTX⁴³² and DOX⁴³⁰ can be introduced to self-immolative spacers prior to the PLA-cleavable linker attachment (Table 5), in order to ensure that the drug molecules are released in their original structure (**51a** and **b** in Fig. 24B). Incorporation of PLA-cleavable linkers into DDSs has already been achieved in drug carriers like antibodies,⁴³³ ONPs,⁴³¹ peptides⁴³⁰ and proteins⁴³⁴ (Table 5).

Neutrophil serine proteases (NSPs) involve three main enzymes, referring to neutrophil elastase (NE), proteinase 3 (PR3) and cathepsin G (CG).⁴³⁵ NSPs are centrally involved in multiple inflammatory human diseases especially chronic lung diseases.⁴³⁶ Among the NSPs, NE has been exploited as a potential target for constructing DDSs, which is based on a tripeptide Asn-Pro-Val (NPV) linker. As a substrate of NE, NPV was utilised to bridge PTX to targeting peptide to obtain **52**. This strategy opens up the possibility of conjugating a broad range of drug and gating molecules to various drug carries for NE-induced drug release due to the application of self-immolative spacers (Fig. 25A).⁴³⁷ NE is also explored as a target for fluorogenic

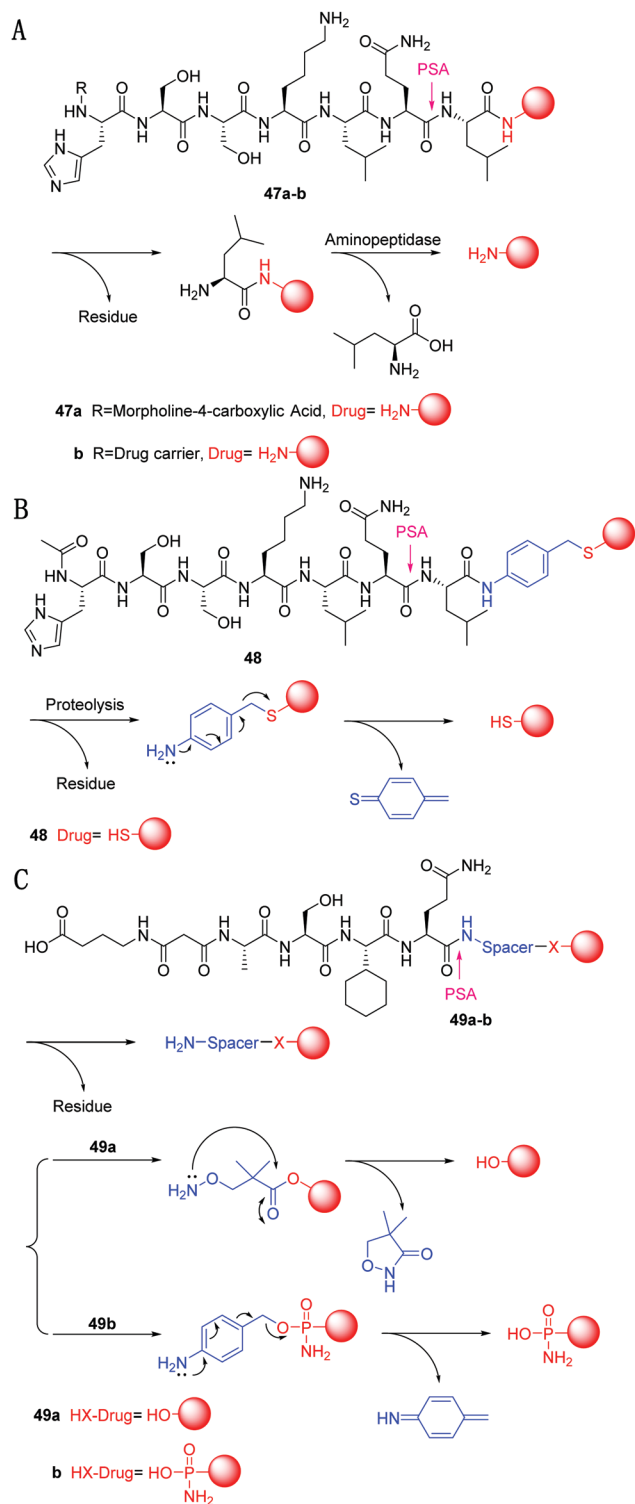


Fig. 23 PSA-triggered linker cleavage. (A) Chemical structures of PSA-cleavable peptide sequences, where controlled release of drug is achieved in two steps: cleavage of the linker by PSA and subsequent amino acid trimming by non-specific aminopeptidases. (B and C) The application of two different PSA-cleavable peptide sequences in prodrugs where self-immolative spacers enabled drug release either by elimination or by cyclisation. The arrow indicates the site of PSA hydrolysis.

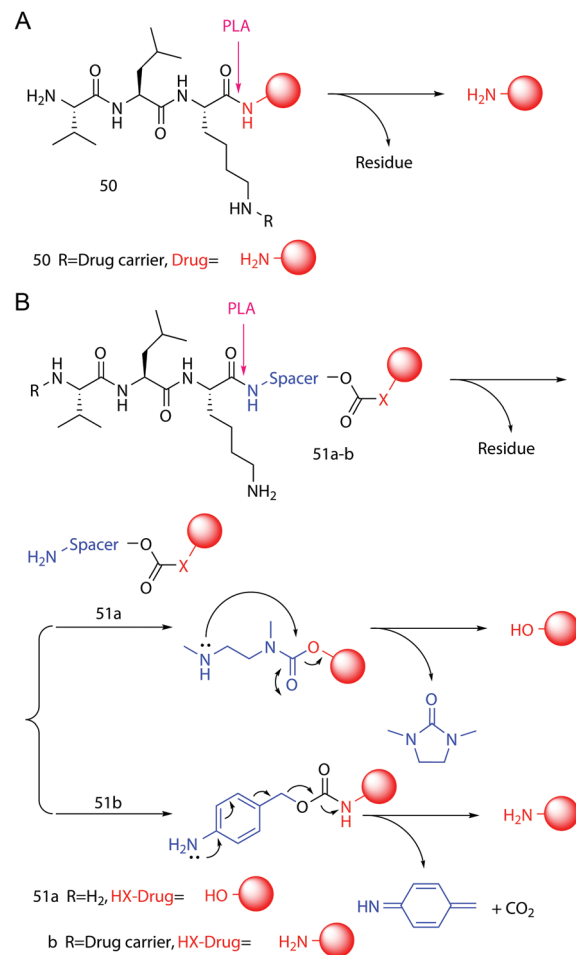


Fig. 24 Plasmin (PLA)-triggered linker cleavage. (A) PLA-induced bond cleavage to enable amine-containing cargo release. (B) Mechanism of payload release from spacer-involved PLA-cleavable linker.

probe design (53 in Fig. 25B).⁴³⁸ CG and PR3 have not been explored as targets for SCLs.

2.5.1.2.2 Cysteine protease-cleavable linkers. Cysteine proteases, also known as thiol proteases, are regarded as one of the four representative proteases. Cysteine proteases are widely distributed in all life kingdoms, possessing a cysteine-histidine-asparagine triad at the active site that predominantly determines their substrate specificity.⁴³⁹ Cysteine proteases have been traditionally viewed as human lysosomal enzymes that play key roles in senescence, apoptosis, prohormone processing and extracellular matrix remodeling,⁴⁴⁰ but recent studies suggest that they have an expanded distribution range and biological function.^{441,442} Based on their structures, cysteine proteases are divided into two distinct superfamilies, including interleukin 1 β converting enzymes (ICE) and the papain superfamily.⁴⁴³

The ICE superfamily is largely known for its unique cleavage site at the C-terminal site of aspartate,⁴⁴⁰ which plays a role in regulating cell death and inflammation.⁴⁴⁴ As one of the first identified mammalian ced-3 homologs, ICE was renamed caspase-1, the name of which is derived from cysteinyl aspartate-specific protease.⁴⁴⁵ Human caspases are classified into three groups

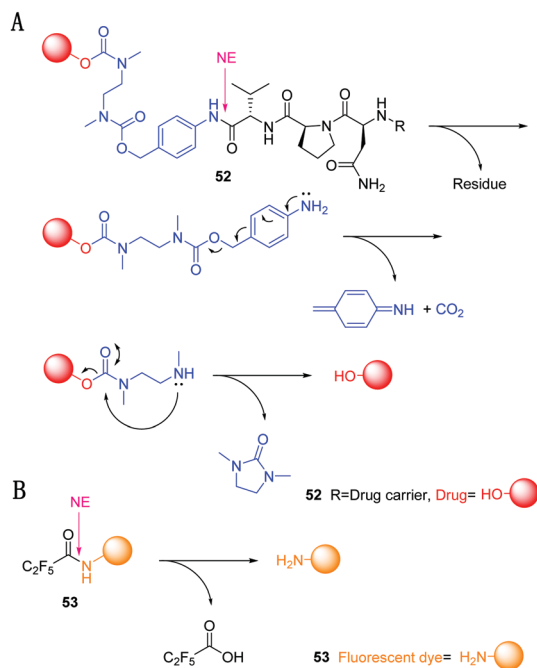


Fig. 25 Structures and cleavage mechanisms of NE-cleavable peptide linkers. (A) Cleavage mechanism of self-immolative spacers in NE-cleavable linkers which involve elimination and cyclisation reactions. (B) Sensing mechanism of fluorogenic probes for NE detection. The arrow indicates the site of NE hydrolysis.

based on sequence similarity and biological functions, including inflammatory caspases (caspase-1, -4, -5, -11, -12), initiator caspases (caspase-2, -8, -9, -10) and executioner caspases (caspase-3, -6, -7).⁴⁴⁶

Although the expression of proapoptotic caspases are reported to be downregulated in various cancers,⁴⁴⁷ certain types of caspase (*e.g.* caspase-3) are upregulated by apoptotic cells showing response to radiotherapy,^{448,449} thereby promoting the development of radiation-induced apoptosis-targeted chemotherapy (RIATC).⁴⁵⁰ RIATC takes advantages of continuous upregulation of caspase-3 during the therapy: radiotherapy induces caspase-3 upregulation and facilitates the delivery of cytotoxic drug, which further promotes the activation of caspase-3 and the release of more cytotoxic drugs around the irradiated area.⁴⁵¹ For linker selection, a DEVD linker comprising of the tetrapeptide sequences (Asp-Glu-Val-Asp) acts as a specific substrate for caspase-3 (Fig. 26).^{452,453} Accordingly, multiple drug molecules like DOX (54a)⁴⁵⁴ and MMAE (54b)⁴⁵⁵ have been attached to the C terminus of the DEVD linker through a self-immolative spacer to realise caspase-3-controlled drug release (Table 5). Moreover, the DEVD linker has been attached to multiple drug carriers such as peptides,⁴⁵⁴ proteins⁴⁵⁶ and ONPs.⁴⁵⁷ In addition, the DEVD linker also finds applications as the central moiety for fluorogenic probe design to detect the activity of caspase-3.⁴⁵⁸ Therefore, the progress made in the field of both caspase-3-induced drug release and fluorophore activation has seen the birth of theranostic applications targeting caspase-3.⁴⁵⁹

As a widely studied member of the papain superfamily, cathepsin B (CTSB) is expressed constitutively and associated with general protein turnover in lysosomes.⁴⁶⁰ Interestingly, the

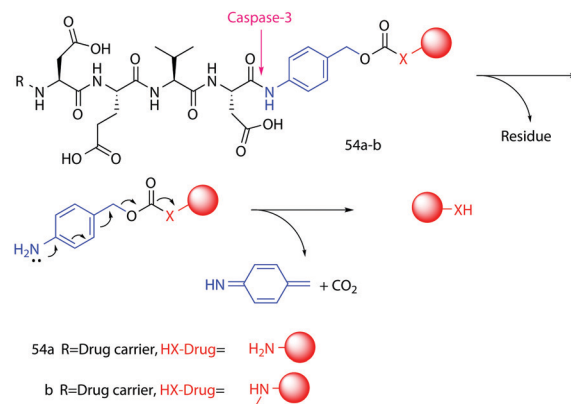


Fig. 26 Structure and cleavage mechanism of caspase-3 cleavable linkers, also known as DEVD (Asp-Glu-Val-Asp) linker. Upon the recognition by caspase-3, the proteolytic site is hydrolysed and the subsequent elimination reaction occurs on the self-immolative spacer to enable drug release. The arrow indicates the site of caspase-3 hydrolysis.

level of CTSB in normal physiological conditions is well regulated, whereas the upregulation of CTSB is observed in many malignancies including brain,⁴⁶¹ lung,⁴⁶² and prostate cancers.⁴⁶³ This discrepancy of the CTSB level between the circulation and certain tumour sites thus makes CTSB an ideal stimulus for the development of the corresponding SCLs. Compared with other cysteine cathepsins of the papain-like family members, CTSB possesses both the endopeptidase and exopeptidase activities,^{464,465} owing to a unique structural element known as occluding loop.⁴⁶⁶ For this reason, a variety of substrates have been shown to be responsive to CTSB, including Arg-Arg,⁴⁶⁷ Ala-Leu,⁴⁶⁸ Ala-Phe-Lys⁴⁶⁹ *etc.* (Fig. 27). Exploiting CTSB-cleavable linkers in controlled drug release has been quite successful where the amine or carboxylic acid group-containing drug molecules like gemcitabine (GEM)⁴⁷⁰ and ibuprofen⁴⁷¹ can be directly attached to either the C terminus or N terminus of the Gly-Phe-Leu-Gly peptide sequence to form 55 and 56 (Fig. 27A and B). Moreover, drug molecules bearing amine groups (*e.g.*, MMAE⁴⁷²) can also be introduced to the CTSB-cleavable linker (Val-Cit sequence) through self-immolative spacers to produce 57a, while the utilisation of spacers further expanded the scope of applicable molecules such as PTX⁴⁷³ (57b in Fig. 27C and Table 5). Numerous efforts have been made to tether CTSB-cleavable linkers to the vast majority of drug carriers including ONPs (*e.g.*, dendrimers⁴⁷⁰), antibodies⁴⁷² and proteins.⁴⁷⁴ Of note, CTSB-cleavable linkers (Gly-Ile-Val-Arg-Ala sequence) were also reported to tether gating molecules like peptides to the surface of MSNs (58),⁴⁷⁵ regulating the drug release in a CTSB-controlled manner (Fig. 27D). Apart from their application in the field of drug delivery, CTSB-cleavable linkers also gained importance in the design of fluorogenic probes for CTSB sensing.⁴⁷⁶

2.5.1.2.3 Metalloprotease-cleavable linkers. Metalloproteases are a class of enzymes that involves a metal ion at the active site which plays an essential role in catalysing peptide hydrolysis.⁴⁷⁷ The most common metal found in metalloprotease is zinc, a stable metal ion that allows for the accommodation of a broad range of coordination numbers and geometries when forming

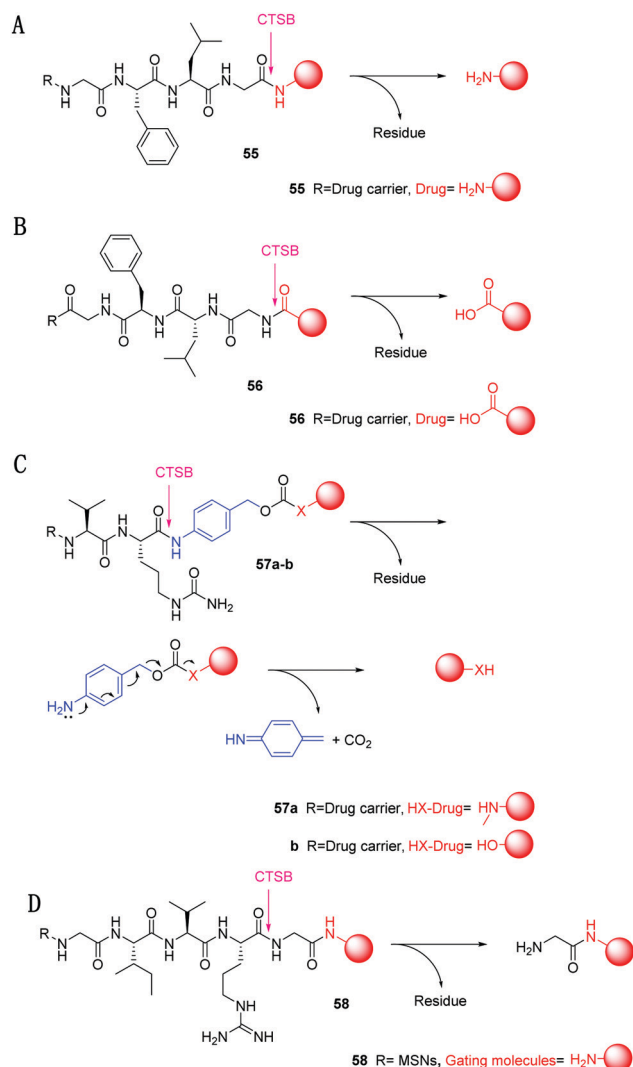


Fig. 27 Structures and drug release mechanisms of CTSB-cleavable linkers. (A and B) CTSB-triggered linker cleavage to enable amine or carboxylic acid-containing drug release. (C) Self-immolative spacer involves in the controlled release from CTSB-cleavable linker. (D) The application of CTSB-cleavable linkers in MSNs, drug release is enabled upon CTSB-triggered proteolysis. The arrow indicates the site of CTSB hydrolysis.

the complexes.⁴⁷⁸ However, other transition metals such as cobalt and manganese have also been found in certain enzymes.⁴⁷⁹

Matrix metalloprotease (MMPs), also known as matrixins, refer to an important family of metalloproteases, of which the enzymatic activity is determined by calcium and zinc ions.⁴⁸⁰ Historically, MMPs were thought to play a key role in the degradation of extracellular matrix (ECM), yet new physiological functions have been revealed in regulating the release and activation of nonconventional substrates like cytokines, chemokines, growth factors and antimicrobial peptides.⁴⁸¹ Despite the fact that MMPs are centrally involved in physiological processes including tissue modelling, for instance, angiogenesis and wound healing,⁴⁸² there are also interests in exploiting them as therapeutic target for specific diseases like cancer,⁴⁸³ arthritis⁴⁸⁴ and inflammatory diseases⁴⁸⁵ owing to their overactivity in these pathological conditions.

MMPs, especially MMP-2 and MMP-9, have been widely utilised as prodrug stimuli.^{486–488} Regarding the utilisation of MMP-cleavable linkers in DDSs, synthetic MMPs substrate based on short linear peptide are more favourable compared with natural MMP substrate that always are large proteins like collagen, gelatin and fibrinogen.⁴⁸⁹ Consequently, controlled release of DOX (59a)⁴⁹⁰ and PTX (59b)⁴⁹¹ have been demonstrated using the MMP-cleavable linker (Gly-Pro-Leu-Gly-Ile-Ala-Gly-Gln sequence) from drug carriers including proteins (albumin)⁴⁹⁰ and ONPs⁴⁹¹ (Fig. 28A and Table 5). While carboxylic acid-containing drug molecules like MTX⁴¹⁶ was conjugated to polymers to afford **60** (Fig. 28B). The incorporation of MMPs-cleavable linkers into DDSs has also been reported in drug carriers such as dendrimers⁴¹⁶ and micelles^{492,493} (Table 5). In the meanwhile, gating molecules like gelatin was also demonstrated to be conjugated to the surface of MSNs *via* the MMP-cleavable linker, which can be further cleaved in the presence of MMPs to release

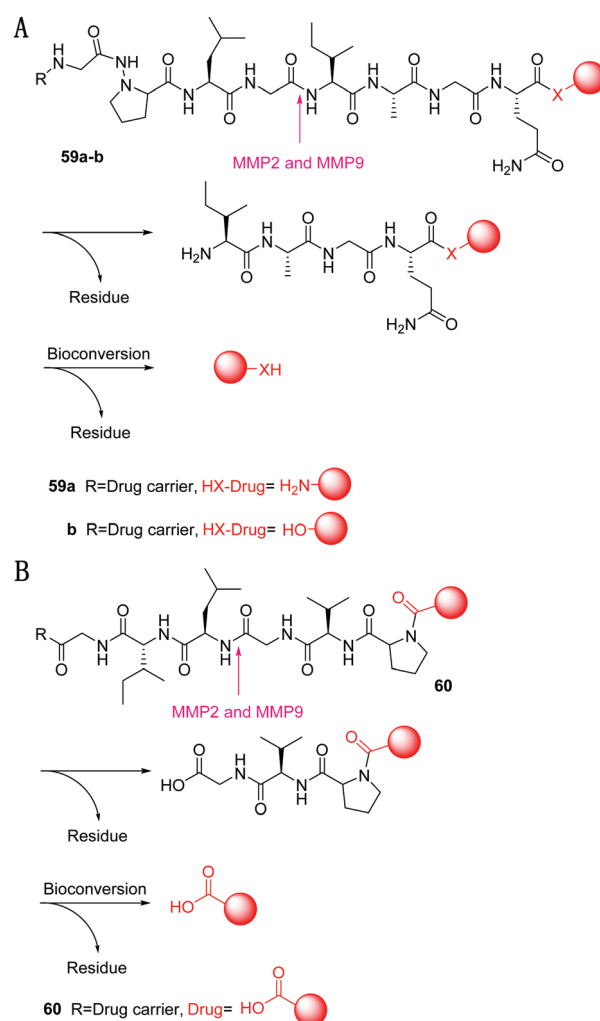


Fig. 28 (A) Schematic illustration of peptide cleavage triggered by MMPs to enable the release of drug molecules bearing amine or hydroxy group; (B) MMPs-induced linker cleavage between the carboxylic acid-containing drug molecules and drug carriers. The arrow indicates the site of MMPs hydrolysis.

the drug loaded in the pores of MSNs.⁴⁹⁴ Apart from the applications in drug delivery field, MMPs have long been of interest as biomarkers. An array of fluorogenic probes comprising of an MMP-cleavable peptide and a fluorophore were discovered to monitor the specific MMP activity under disease conditions.⁴⁹⁵

2.5.1.3 Glycosidase-cleavable linkers. Glycosidases, also known as glycoside hydrolases (GHs), refer to a large family of enzymes responsible for hydrolytically cleaving the glycosidic bonds in glycoconjugates, oligo- and polysaccharides to form a sugar hemiacetal/hemiketal and the corresponding aglycon.⁴⁹⁶ Traditionally, these enzymes are divided into two main groups, including the exo-glycosidases that release a specific monosaccharide from the nonreducing terminus of an oligosaccharide, and the endoglycosidases capable of cleaving the glycosidic bonds in the carbohydrate polymer chain.⁴⁹⁷ The GHs are mainly involved in the glycometabolism, during which they perform both important extracellular and intracellular degradative functions based on two distinct mechanisms.⁴⁹⁸ One of the roles for GHs in the targeted drug delivery field is related to their elevated level in cancer cells in contrast to normal cells,⁴⁹⁹ where GHs have thus gained significant attention in the study of tumour-specific drug release.

Among the broad spectrum of exo-glycosidases, β -glucuronidase (GUS) is notable for stimulus-induced drug release/activation due to its elevated activity in various tumours including breast,⁵⁰⁰ colon,⁵⁰¹ lung⁵⁰² and leukaemia.⁵⁰³ Human GUS is located in lysosomes, where the acidic microenvironment contributes to its optimal activity, and most interestingly, GUS exhibits unusual resistance to thermal denaturation, corresponding to their stability up to 70 °C.⁵⁰⁴ Structurally, human GUS is a tetrameric glycoprotein, which is comprised of four identical monomers.⁵⁰⁵

The main role of GUS in biological processes is the deconjugation of β -D-glucuronides from a variety of natural substrates, such as the deconjugation of chondroitin sulfate to produce β -glucuronic acid and the corresponding aglycone.⁵⁰⁶ Similar to phosphate monoester, the high polarity of glucuronic acid offers advances in improving the hydrophilicity of poorly water-soluble drug molecules, thus greatly contributing to the prodrug formations.⁵⁰⁷ Although glucuronic acid can be directly conjugated to the phenol alcohol-bearing molecules like duocarmycin derivatives **61** (Fig. 29A),⁵⁰⁸ self-immolative spacers have been more frequently used. Accordingly, the elongated linkers, such as *N*-phenyl β -O-glucuronyl carbamates, nitrobenzylphenoxy carbamate *etc.*, showed high susceptibility to GUS and have been utilised to conjugate various drug molecules like PTX (**62**),⁵⁰⁹ DOX (**63a**)⁵¹⁰ and MMAE (**63b**)⁵¹¹ (Fig. 29B and Table 5). Likewise, these glucuronide-drug conjugates have been attached onto carriers like antibodies,⁵¹² proteins⁵¹³ and ONPs (*e.g.*, micelles⁵¹⁴). However, the main shortcoming of glucuronide-based linkers is their tedious (usually 10+ steps) synthetic route, which hinders the broad application of these linkers in more material- or biology-oriented labs.^{515,516}

β -Galactosidase (GLB1) is an exo-glycosidase which exists abundantly within spleen, liver, intestinal cells and lymphocytes. Three types of enzymatic reactions of GLB1 are reported, including the cleavage of disaccharide lactose, catalysis of

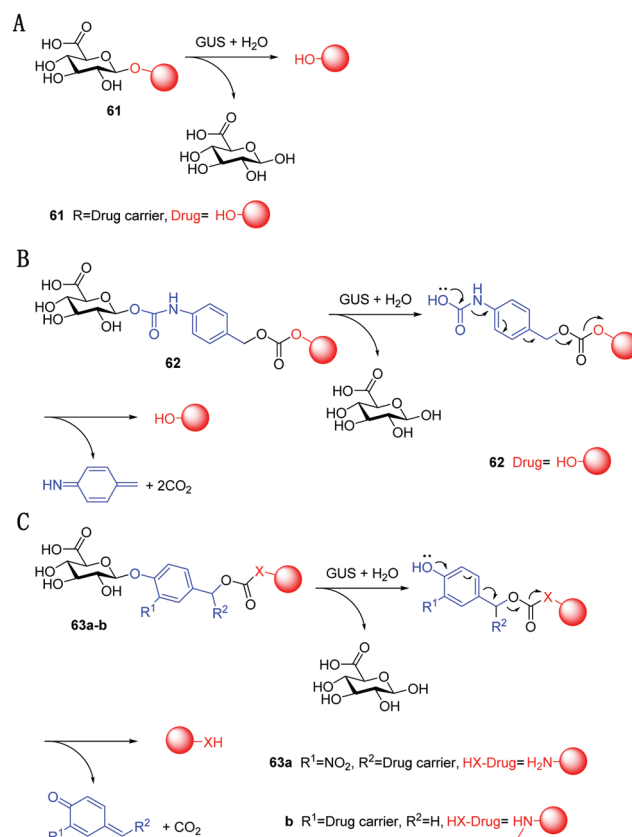


Fig. 29 The principle of prodrugs or DDSs targeting GUS for controlled drug release. (A) Mechanism of GUS-mediated drug release from the glucuronide linkers. (B and C) Structure and release mechanism of GUS-responsive DDSs, where the application of self-immolative spacers enables the conjugations of drug molecules bearing various functional groups.

transgalactosylation of lactose to allolactose and the cleavage of allolactose to form monosaccharide.⁵¹⁷ Apart from preferentially recognising and cleaving the oxygen glycosidic bond of β -D-galactopyranoside with high catalytic efficiency,⁵¹⁸ GLB1 can also hydrolyse other glycosidic linkages containing nitrogen, fluorine and sulfur but with lower efficiency.⁵¹⁹ Due to the overexpression of insulin-independent glucose transporters (GLUT-1) in a variety of cancers, the glycoside-modified prodrugs or DDSs can allow for tumour-specific drug release in principle.⁵²⁰ The pioneering work in the galactoside-based prodrug of 8-hydroxyquinolines (**64a**)⁵²¹ further inspired comprehensive investigation on other drug and gating molecules. Galactose⁵²² that contains glycoside linker (**64b**) can be anchored to the surface of MSNs, acting as the gatekeeper to enable drug release in the presence of GLB1 (Fig. 30A). For drug-linker conjugates like **65a**⁵²³ and **65b**,⁵²⁴ self-immolative linkers were utilised to broaden the selection of drug molecules (Fig. 30A, B and Table 5). Of note, galactoside prodrugs⁵²⁵ exhibit improved cell permeability compared with glucuronide prodrugs,⁵²⁶ suggesting that the additional biological functions/properties of SCLs need to be taken into consideration in the design of the linker. In addition, the substitution on the aromatic ring in self-immolative spacer opens up the

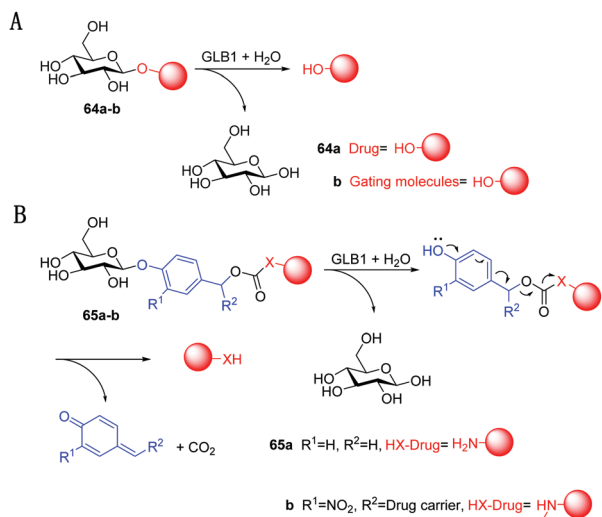


Fig. 30 (A) Structure and cleavage mechanism of galactoside-based linkers cleaved by GLB1. (B) Cleavage mechanism of self-immolative spacer-assisted drug release from galactoside-based linkers.

possibility to further installation of the linker onto antibodies⁵²⁴ and folic acid.⁵²⁷

2.5.2 Oxidoreductase-cleavable linkers. Oxidoreductases are defined as a type of enzyme with the capability to catalyse the electron transfer from one chemical species, known as electron donor, to another, *i.e.* electron acceptor, and the enzymatic activity of most oxidoreductases are NAD(P)H-dependent.⁵²⁸ Among NAD(P)H-dependent oxidoreductases, the nucleotide binding domain is structural conserved which plays an essential role in recruiting NAD(P)H/NAD(P)⁺ cofactors for their activity.^{529,530} The catalytic domain determines the substrate specificity and the catalytic mechanism.⁵³¹

2.5.2.1 NAD(P)H:quinone reductase-cleavable linkers. NAD(P)H:quinone reductases1 (NQO1), also known as the DT-diaphorase (DTD), is a two-electron quinone reductase mainly located in the cytosolic fraction of cells, and there is a small portion of DTD distributed in the endoplasmic reticulum⁵³² and mitochondria.⁵³³ Due to its possible role in detoxification, the upregulation of DTD is thought to be related with early carcinogenic process, corresponding to its overexpression in many cancer types like hepatomas,⁵³⁴ colonic carcinomas⁵³⁵ and breast carcinomas.⁵³⁶ Therefore, the significantly high levels of DTD in cancerous tissues as compared with healthy tissues has been widely pursued as an endogenous stimulus to trigger tumour-specific drug release.

The rational design of DTD-responsive chemistry started with the study on the mechanism of action of mitomycin C (MMC), a bio-reductive prodrug, where the indoquinone functions as an excellent substrate of DTD to trigger the subsequent removal of methoxide group at 3-position, which attracts significant interest in substituting the methoxide functionality with drug molecules to realise DTD-responsive drug release (Fig. 31).⁵³⁷ A proposed mechanism of DTD induced indoquinone-based prodrug uncaging is shown in Fig. 31, where the reduction of indoquinone

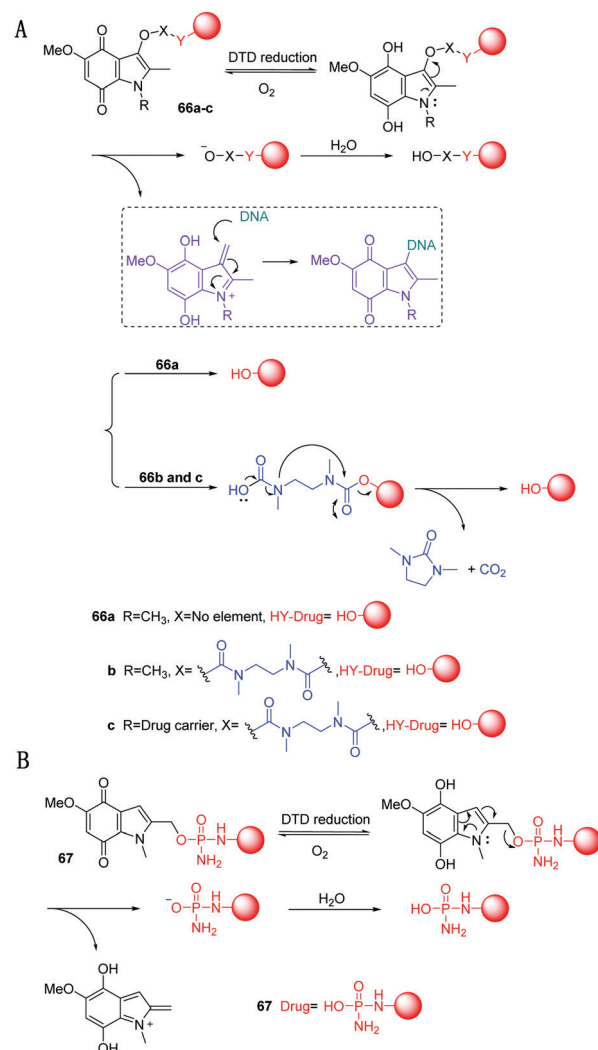


Fig. 31 (A) Structure and cleavage mechanism of incorporation of indoquinone-based linkers upon the treatment of DTD. The bioreduction of indoquinone motif not only releases the drug attached to its C-3 position, but also generates a cytotoxic byproduct. (B) Structure and cleavage mechanism of DTD-mediated drug release from the C-2 position of indoquinone linker.

increases the electron density at indole nitrogen, making the leaving group at 3-position more easily to escape. In addition, the indoquinone-based DDSs displayed independent toxicity other than the therapeutic effect of carried drugs, which is viewed as a unique feature. Evidence suggests that upon the cleavage of the linker, the alkenyliminium electrophile byproduct serves as a potent DNA ligand, and this DNA-alkylation is thought to be cytotoxic to cells.⁵³⁸ Accordingly, numerous drug molecules have been introduced to the 3-position of indoquinone framework. For instance, molecules like nitric oxide (NO)⁵³⁹ can be directly attached to the indoquinone linker through the formation of ether bond (66a). As for drug molecules such as CPT⁵⁴⁰ and SN38,⁵³⁷ a self-immolative spacer was used to bridge them to indoquinone linker to afford 66b and 66c (Fig. 31A and Table 5). Amine and azide functional groups can be introduced to the C-1 position of indoquinone linker,⁵⁴¹ thus allowing for the attachment of the

linker to drug carriers such as peptide.⁵³⁷ Apart from C-3 position, the C-2 position on indoquinone linker has also been explored for drug attachment. Accordingly, phosphoramidate mustards has been installed at the C-2 position of indoquinone linker (**67**), and the corresponding derivatives exhibited promising selectivity toward DTD to enable drug release (Fig. 31B).⁵⁴²

Benzoquinones can be used as DTD-responsive linkers as well.⁵⁴³ As shown in Fig. 32, the quinone propionic acids (QPAs) moiety is rapidly reduced by DTD to form hydroquinone which subsequently undergoes an intermolecular lactonisation reaction to liberate the attached cargo from the carboxylic site.⁵⁴⁴ Different from reduction of indoquinone structure, the cyclised lactone byproduct showed no toxicity to cells.⁵⁴⁵ A direct drug conjugation strategy is available to introduce hydroxy or amine group-containing drug molecules to QPAs linker, affording **68a**⁵⁴⁶ and **68b**⁵⁴⁷ (Fig. 32A). However, attaching drug molecules to QPAs *via* different self-immolative spacers is more common, such as in the case of the conjugations of 5-FU (**69a**)⁵⁴⁸ and CPT (**69b**)⁵⁴⁹ (Fig. 32B and Table 5). The aromatic ring in self-immolative spacers also enables versatile chemistry, thus allowing for the attachment of QPAs to drug carriers *via* biotin (**69c**).⁵⁵⁰ Meanwhile, the susceptibility of QPAs to DTD also attracts considerable attention in the design of fluorogenic probe for detection of the DTD.⁵⁵¹ However, the synthesis of QPA derivatives remains the challenge for expanding the scope of their application, since the synthetic route involves multiple complicated synthetic steps under harsh conditions.

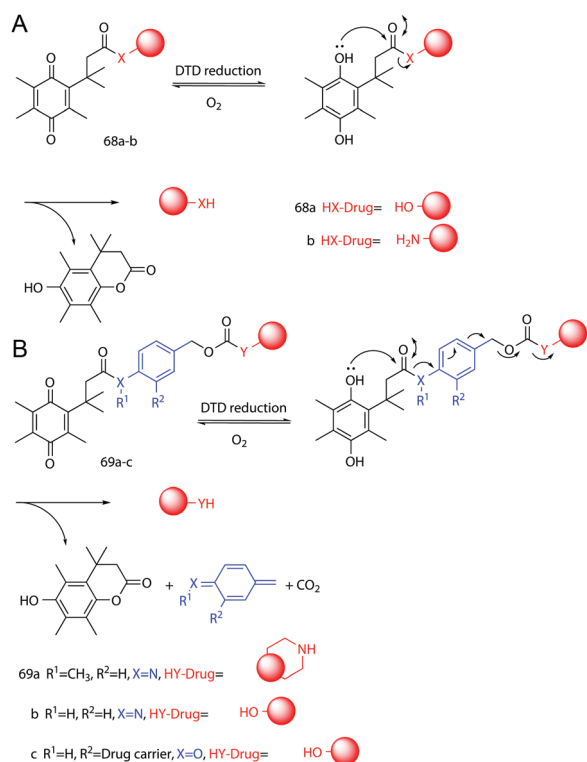


Fig. 32 Chemical structure and cleavage mechanism of (A) quinone propionic acids (QPAs) and (B) QPAs conjugated with self-immolative spacers for DTD-responsive drug release.

2.5.2.2 Nitroreductase-cleavable linkers. Nitroreductases (NTRs) refer to a class of enzymes capable of reducing nitro functional groups on aromatic rings, and they can be subclassified into two groups based on the oxygen sensitivity.^{552,553} Mainly distributed in bacteria, type 1 NTRs are oxygen-insensitive enzymes with flavin mononucleotide (FMN) in the active centre.⁵⁵⁴ Whereas type 2 NTRs display oxygen-sensitivity with flavin adenine dinucleotide and/or FMN as the cofactor.⁵⁵⁵ The nitroaromatics can be the substrate of both types of NTRs, and in most cases nitro groups are reduced to hydroxylamines, which causes the electron redistribution within the aromatic backbone (Fig. 33). A number of chemical structures were applied *via* this feature to realise drug release following the

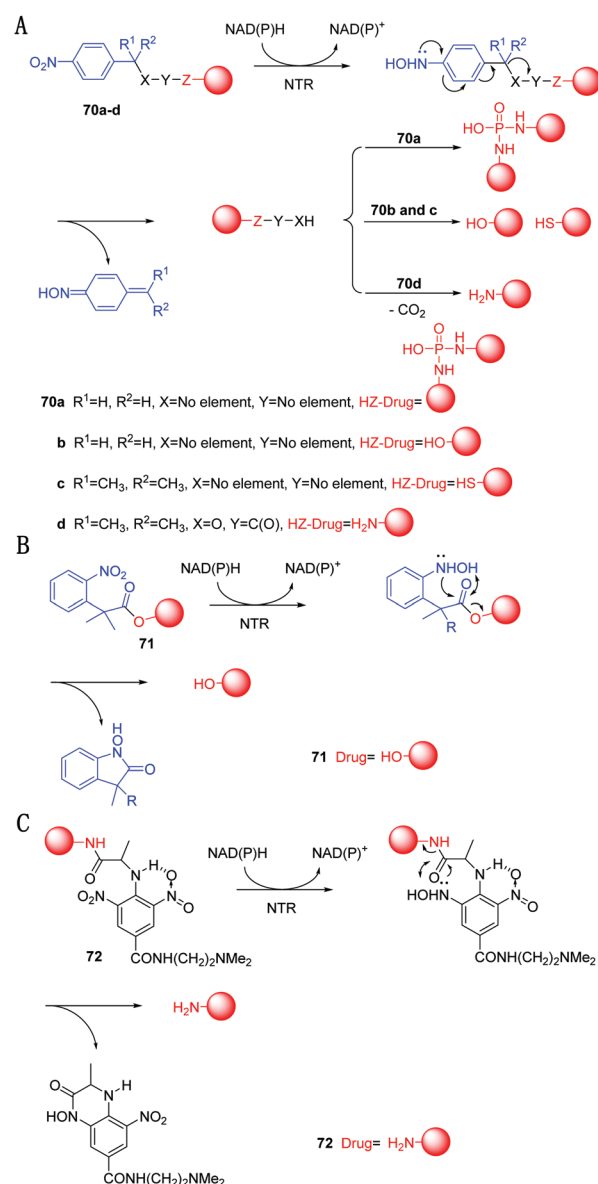


Fig. 33 Illustration of NTRs-mediated drug release from *p*-nitrobenzyl linker, 2-nitrophenylalkanoyl linker and 2,6-dinitrophenyl linker. (A) The application of *p*-nitrobenzyl motif in prodrug formation. (B) The structure and cleavage mechanism 2-nitrophenylalkanoyl-based linkers. (C) The utilisation of 2, 6-dinitrophenyl linker for improved controlled drug release.

NTR-triggered nitro reduction, such as *p*-nitrobenzyl, 2-nitrophenylalkanoyl and 2,6-dinitrophenyl linkers.

As the most extensively investigated NTR-cleavable scaffold, *p*-nitrobenzyl group finds broad applications in delivering phosphoramidate drugs (**70a**), a class of alkylating anticancer drugs.⁵⁵⁶ Following the NTR-catalysed reduction of nitro group, the corresponding hydroxylamine intermediate will undergo a spontaneous 1,6-elimination reaction to release the drug at *para*-position (Fig. 33A). Drug molecules bearing amine, thiol and hydroxy group can be conjugated to the *p*-nitrobenzyl scaffold, therefore the scope of this methodology has been successfully expanded to the prodrugs of checkpoint kinase 1 (Chk1)/Aurora A inhibitor (**70b**),⁵⁵⁷ thioguanine (**70c**)⁵⁵⁸ and *O*⁶-benzylguanine (**70d**)⁵⁵⁹ (Table 5). The *p*-nitrobenzyl scaffold is also attracting interest in the field of fluorogenic probes, where different fluorophores were exploited to detect the activity of NTRs.^{560,561} Based on a similar mechanism of drug release from 4-nitrobenzyl scaffolds, 2-nitroimidazole-5-ylmethyl units⁵⁶² and 5-nitrofuran-2-ylmethylidene scaffolds⁵⁶³ have also been used as a nitroreductase cleavable linker in prodrug formations.

Apart from elimination reactions, cyclisation is another strategy to promote drug release by using the increased electron density on the nitroaromatic ring after the bio-reduction. This method enables the cleavage of an amide or ester bond through the intramolecular nucleophilic attack of amine group formed by the reduction of nitro group (Fig. 33B). The 2-nitrophenylalkanoyl moiety was first developed in an attempt to prepare prodrugs of FUDR (**71** in Fig. 33B).^{608,609} Further investigations found that the hydrophilicity of the amine group and the geometry of derivative play a predominant role in determining the cyclisation rate.⁶¹¹ However, insufficient enzymatic reduction of 2-nitrophenylalkanoyl moiety has severely hindered its application, which is probably due to its low reduction potential or steric hindrance. To alleviate this problem, a 2,6-dinitrophenyl conjugate with increased reduction potential was developed (Fig. 33C). Noteworthy, the increased reduction potential is due to the intramolecular hydrogen bonding between the aniline and 2-nitro group, which provides additional conformation restriction.⁶¹² So far, amine-containing drug molecules have been conjugated to the 2,6-dinitrophenyl group and tested against cancer cells.⁶¹⁰

In conclusion, specific enzymes involved in the oxidoreductases family can enable reduction reaction under hypoxic conditions that exists as a unique characteristic of solid tumours, therefore the corresponding SCLs are capable of releasing the drug in the absence of oxygen. Despite the fact that the reduction also occurs in aerobic cells, the reduced intermediates can be rapidly oxidised by molecular oxygen to their original structure, which ensures the stability of prodrug or DDSs in normal cells and offers high selectivity to hypoxic environment.

2.6. Self-immolative spacers

As mentioned in the previous sections, cleavage or degradation of the SCLs should give rise to the drug molecule in its original structure, in order to guarantee the therapeutic effect. However, the application of direct conjugation strategy to introduce the drug molecule to the linker moiety is limited by various factors,

the main of which are related to the lack of suitable functional groups of the drugs to conjugate with the specific SCLs.⁶¹³ To address this issue, spacers, commonly referred to self-immolative spacers, are introduced between the drug payload and SCL to accommodate drug molecules.⁶¹⁴ There are two widely accepted self-immolation processes, including the electronic delocalisation and cyclisation.⁶¹⁵ The first one relies on the electron cascade to release the payload with the formation of quinone or azaquinone methide intermediate, whereas the second process involves intramolecular cyclisation reaction to form the oxazolidinone, imidazolidinone or 1,3-oxathiolan-2-one ring structure intermediates. The initiating condition of both processes requires the generation of nucleophilic functional groups like hydroxy, amino or thiol groups after exposure to the stimulus, which subsequently triggers the self-immolation to release the drug payload.

2.6.1 Self-immolation by 1,4-, 1,6- or 1,8-elimination. The electronic delocalisation to trigger the drug release usually occurs on the aromatic ring containing a hydroxy,^{616,617} thiol⁷⁹ or an amino group,^{189,618} the nucleophilicity of which is masked or protected by different stimulus-sensitive groups. Upon the recovery of the nucleophilic functional groups by the stimulus, spontaneous and irreversible self-immolation initiates and finally release the drug. The most widely used electronic delocalisation reactions rely on the 1,4-⁶¹⁹ and 1,6-elimination¹⁸⁹ (**73** in Fig. 34 A and **74** in Fig. 34B), whereas 1,8-elimination reaction (**75** in Fig. 34 C and **76** in Fig. 34D) was only observed in coumarinyl alcohol or *p*-amino/hydroxy cinnamyl alcohol.^{618,620}

The dissociation rate of electronic delocalisation-based self-immolation is believed to be affected by the structural properties of spacer, including the length of the spacer,⁶²¹ substituents on the aromatic ring⁶²² and leaving groups.⁶²³ As the core structure of the self-immolative spacers whose degradation rely on the electronic cascade, the aromatic ring plays a vital role in affecting the disassembly kinetics since its nature allows for flexible proximity of the leaving group to the activation site, which is independent of the electronic effect. Although the disassembly times for 1,4-elimination and 1,6-elimination were shown to be

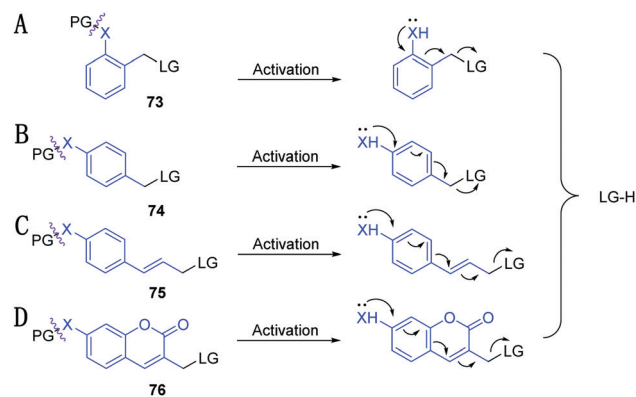


Fig. 34 Self-immolative spacers relying on an electronic cascade (*ortho* or *para* delocalisation) for disassembly. Reaction mechanism for (A) 1,4-elimination. (B) 1,6-Elimination; (B and D) 1,8-elimination. PG = protecting group; X = O, NH or S; LG = leaving group, i.e. desired product.

the same,¹⁸⁹ in certain cases where a dual release is involved, faster release was observed for the leaving group on the para site than the ortho one.⁶²¹ When it comes to the substituents on the aromatic ring, it is believed that electron-withdrawing substituents can decrease the electron density of the aromatic ring to destabilise the partial positive charge at benzylic position in the transition state of spacer disassembly. Consequently, the increased kinetic stability of the activated spacer contributes to the prolonged half-time, which was observed in self-immolative spacers bearing nitro or methoxycarbonyl group.⁶²⁴ In contrast, substitution at the benzylic position with electron-donating group such as methyl group was reported to accelerate the release of the leaving group, while electron-withdrawing group will therefore prevent the release.^{619,625} Apart from the aromatic core, the leaving group also affects the disassembly rate of self-immolative spaces with its nucleofugacity as well as the steric hindrance.⁶²⁶

2.6.2 Self-immolation by cyclisation. Compared with elimination by electronic cascade, cyclisation proceeds more slowly in most cases, where the disassembly of self-immolative spacers affords cyclic esters, carbamates, ureas or thiocarbonates (from **77** to **80** in Fig. 35).^{627,628} Two main factors are considered to govern the disassembly based on cyclisation, they are Thorpe–Ingold effect⁶²⁹ and reactive rotamer effect.⁶³⁰ Specifically, the

introduction of an electron-donating group such as a hydroxy group to the α position of an ester can boost the release of the leaving group, indicating the importance of substituents on the spacer. Moreover, the pK_a of the leaving group in the cyclisation system is strongly related to the disassembly rate, as reported by Hanusek *et al.*⁶³¹ As the cyclisation process is commonly described as a series of equilibria and proton transfer, the environmental pH has also been proven to be linked to cyclisation rate.^{632,633}

3. General conjugation strategies of SCLs to nanoparticles

Nanoparticles (NPs) as drug carriers exhibit ubiquitous size-related optical and magnetic properties as compared with those of their bulky materials,⁶³⁴ while they can be generated in various shapes like tubes, rods and spheres. They provide key advances in the field of drug delivery, including improved bioavailability, increased half-time in human body and enhanced targeting ability.⁶³⁵ More importantly, the high surface area to volume ratio of NPs allows for versatile chemical surface modification, therefore opening up the possibility of covalently incorporating the SCLs with the NPs.⁶³⁶ But this combination is also highly dependent on the inherent physicochemical properties of NPs. Here we will briefly introduce the corresponding surface modification strategies of several types of NPs which have been extensively investigated as platforms for drug delivery, including porous silicon nanoparticles (pSiNPs), mesoporous silica nanoparticles (MSNs), gold nanoparticles (GNPs), metal organic frameworks (MOFs) and organic nanoparticles (ONPs).

3.1 Surface modification of nanoparticles

3.1.1 Surface modification of porous silicon nanoparticles.

Porous silicon nanoparticles (pSiNPs) have gained increasing popularity for their potential applications in drug delivery field. Firstly discovered in 1956, pSiNPs were found to be biocompatible and biodegradable, which is considered to be one of its most appealing features as a drug carrier.⁶³⁷ Other characteristics of pSiNP include high porosity, large surface area, high loading capacity and tunable pore size. Those features make this type of nanomaterial a promising carrier for encapsulating the therapeutic payload for drug delivery purpose.^{638,639} pSiNPs used for biomedical applications are commonly prepared through electrochemical etching of silicon wafers in aqueous or ethanolic HF electrolytes, also known as the anodisation approach. The nanostructure of pSiNPs can be manipulated by adjusting the fabrication parameters like current density, level of doping and electrolyte composition to meet specific requirement.^{640,641}

Although the electrochemical etching of silicon wafers provides pSiNPs with unique and tuneable properties, the freshly etched pSiNPs are highly reactive due to their hydride termination (Si-H , Si-H_2 , Si-H_3) surface.⁶⁴² The as-anodised pSiNPs undergo slow oxidation upon its exposure to oxygen or water, and form the water soluble orthosilicic acid $[\text{Si}(\text{OH})_4]$ which is an essential silicon source for human body.⁶⁴³ Although this

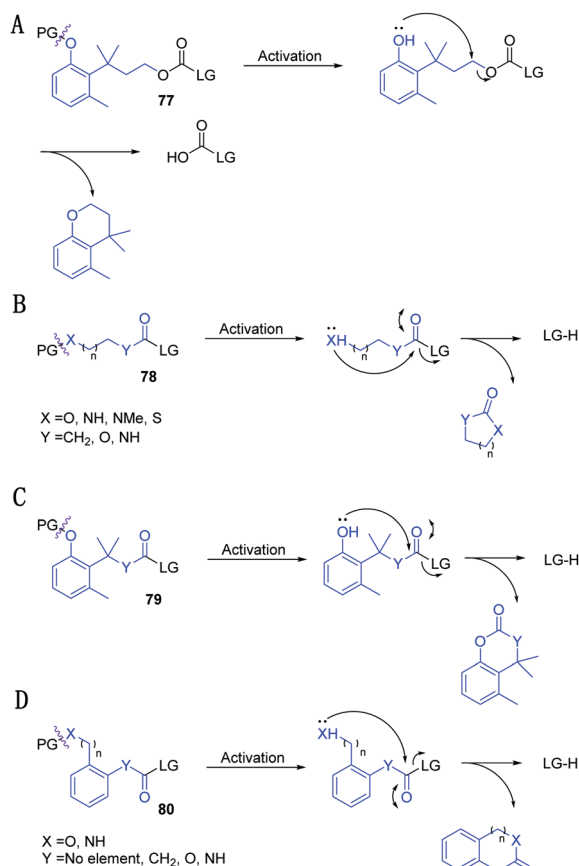


Fig. 35 (A–D) Self-immolative spacers relying on 1,5- or 1,6-cyclisation. PG = protecting group; LG = leaving group, i.e. desired product.

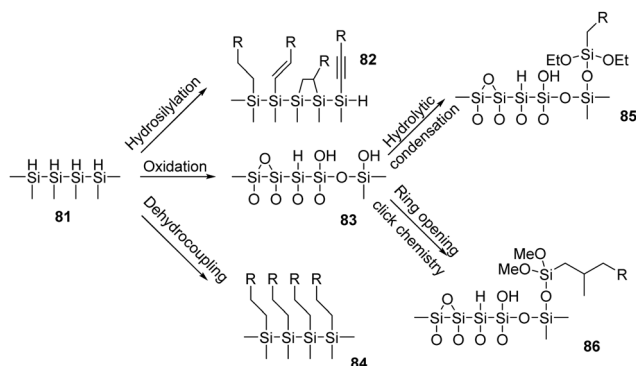


Fig. 36 Different surface chemistries on the freshly etched pSiNPs.

property provides excellent biodegradability, this can result in the instability of pSiNPs and premature drug release. In order to make pSiNPs a better candidate for drug delivery, further surface modifications to stabilise the pSiNPs are required.

Conventional surface modification methods for pSiNPs involve oxidation, hydrolytic condensation and hydrosilylation reactions. Among the three modification methods, the oxidation remains as the most commonly used one to stabilise the surface of pSiNP. By breaking the silicon-hydride (Si-H) and silicon-silicon (Si-Si) bond (81 in Fig. 36) on the surface of pSiNPs to produce hydrated silicon oxide (Si-OH) and silicon oxide (Si-O-Si) bonds (83 in Fig. 36). Given the low bonding energy of Si-Si bond as compared with Si-H bond, the addition of a mild oxidant like DMSO⁶⁴⁴ and pyridine⁶⁴⁵ is able to break Si-Si bond, resulting in the generation of Si-O-Si bond at room temperature. The increase of oxidant reactivity can lead to the increase of oxidation degree, for instance, ozone (O₃)⁶⁴⁶ can break both the Si-Si and Si-H bonds to generate Si-O-Si and Si-OH at room temperature. Apart from the chemical-based oxidation, thermal oxidation serves as another popular method to stabilise the surface of pSiNPs. In the presence of dry oxygen, the back-bond oxidation mainly occurs at 300–400 °C (Fig. 36), whereas the oxidation at 600 °C and above completely converts the pSiNP into silicon oxide (Fig. 36).⁶⁴⁷ As a result of oxidation, the hydrophobic nature of pSiNPs changes to hydrophilic.

The oxidised pSiNPs can serve as a promising platform for further modification and the most commonly used method refers to hydrolytic condensation. The main efforts have been focused on two silanes, 3-aminopropyltriethoxy-silane (APTES)⁶⁴⁸ and 3-mercaptopropyltriethoxy-silane (MPTES),⁶⁴⁹ for hydrolytic condensation with oxidised pSiNP materials, producing the primary amine and thiol terminals on their surface, respectively (85 in Fig. 36). Both the amine and thiol groups allow for further conjugation with other molecules *via* the amide coupling⁶⁵⁰ or thiol-Michael addition⁶⁵¹ with high yield.

Apart from surface oxidation, the hydrosilylation reaction is another strategy to stabilise the surface of pSiNPs, where Si-H reacts with unsaturated compound such as alkynes, alkenes and aldehydes to form the silicon-carbon (Si-C) bond (82 in Fig. 36).⁶⁵² Owing to the low electronegativity of carbon, the Si-C bond possess greater kinetics stability as compared with Si-O

bond, thus leading to more stable pSiNP surface.⁶⁵³ Hydrosilylation can be catalysed through a variety of means including thermal, photon, chemical catalysts and by microwave radiation. More importantly, hydrosilylation is able to provide versatile functional groups on the pSiNP surface for further modification.^{654–656} However, one of the disadvantages of hydrosilylation refers to the production of silicon oxide during the modification process, therefore it usually proceeds in the absence of oxidants. A replacement of hydrosilylation has been recently proposed, namely dehydrocoupling, which can occur under mild thermal conditions through dehydrogenative coupling with a trihydrosilane reagent (84 in Fig. 36).⁶⁵⁷ The dehydrocoupling shows tolerance to oxidants like oxygen and water, while various functional groups such as bromide, azide and amine are allowed to be installed on the surface of pSiNP materials.

As aforementioned, the Si-H bond on the pSiNP surface is usually oxidised to Si-O bond, which can be further modified through hydrolytic condensation. However, there are several drawbacks of this modification method, including long reaction time, production of byproduct due to cross-linking and low coupling efficiency.⁶⁵⁸ To overcome these limitations, a novel alternative of hydrolytic condensation was introduced, known as ring-opening click chemistry.⁶⁵⁹ The general mechanism is illustrated in Fig. 36, where the Si-OH on the pSiNP surface attacks the silicon core of cyclic-silane reagent to break the Si-X (X = S or N) bond, giving rise to the new terminals like primary amines and thiols on the pSiNP surface (86). The ring-opening click chemistry provides an advance in bio-conjugation applications due to its high yield without by-product and mild reaction conditions and has been applied to conjugate various functional groups and linkers.^{659,660}

3.1.2 Surface modification of mesoporous silica nanoparticles.

As the counterpart of pSiNPs, mesoporous silica nanoparticles (MSNs) also gained interest as promising carriers for drug delivery, owing to their characteristic features including large surface area and pore volume,⁶⁶¹ tuneable morphology,^{662,663} simple synthesis method and versatile surface functionality.^{664,665} Moreover, as a subgroup of inorganic materials, the inherent properties of MSNs such as chemical and biological stability can avoid the degradation of payload during administration, thus ensuring the therapeutic effect.⁶⁶⁶ Compared with pSiNPs, the uniform pore size of MSNs allows the combination of molecular gating mechanism and SCL design to realise controlled drug release, which is considered to be a unique advantage of MSNs.^{667,668}

The majority of MSNs are synthesised through the modification of sol-gel method with added surfactants such as cetyl trimethylammonium bromide (CTAB), polymers and other dopants that serve as the structure-directing agents for pore formation,⁶⁶⁹ and this method depends on the electrostatic interactions between the cationic surfactant template and the silica species. Comprehensive reviews have published, providing the readers with deeper understanding of the MSNs synthesis procedures.⁶⁷⁰

The common strategies regarding the surface functionalisation of MSNs refer to the 'grafting' approach and co-condensation method.⁶⁷¹ Among the two surface modification methods, the 'grafting' approach uses a two-step procedure to functionalise the

inner surface of MSNs with organic groups. Based on the reaction between the organosilanes, chlorosilanes and silazanes with the free silanol groups (**87**) on the surface of MSNs (Fig. 37A), this method is desirable to install various functional groups on accessible silica surface, such as aminopropyl,⁶⁷² carboxy⁶⁷³ and thiol groups⁶⁷⁴ *etc.* (**88**), allowing for further modification with linkers. Although introduction of SCLs to directly bridge drug molecules to the surface of MSNs is possible,²⁹³ these linkers are more commonly conjugated with gating molecules as the gate-keepers which can be cleaved or isomerised in the presence of corresponding stimuli to enable drug release from the pores of MSNs. Several reviews have covered the molecules uses as gates, opening mechanism and controlled release behavior.^{45,74} However, the main drawback of 'grafting' approach is that the functional groups may preferentially reacts at the channel opening, which can impair the further diffusion of grafting reagents, resulting in nonhomogeneous distribution of organic groups within the pores or even the pore blocking.⁶⁷⁵

In contrast, co-condensation, also known as one-pot synthesis, has proven advances in overcoming the pore blocking problem during the grafting process, since the organic reagents used in this method are the direct components of the silica matrix.⁶⁷⁶ This process can be catalysed by structure-directing agents, where tetraalkoxysilanes (**89**) react with organosilanes (**90**) to bridge the organic functional groups such as thiol group⁶⁷⁷ to the pore wall (Fig. 37B). Apart from providing more uniform spreading of organic functionalities, the co-condensation also simplifies the synthesis, making it more preferred over the 'grafting' method.⁶⁷⁸ This strategy has demonstrated the possibility to incorporate SCLs such as boronate esters²²⁹ into the MSN framework for drug delivery purpose. However, a significant disadvantage of this method is a loss of mesoscopic order in the structure of the MSNs.⁶⁷⁹

3.1.3 Surface modification of gold nanoparticles. Gold nanoparticles (GNPs) have recently gained increased interest

as carrier for delivery a variety of payloads,⁶⁸⁰ due to their unique physicochemical properties such as chemically inertness, good biocompatibility, ease of synthesis and optical properties.⁶⁸¹ An array of methods have been reported to synthesise the GNPs, the majority of which are based on the reduction of chloroauric acid with a stabilising agent to prevent the aggregation of GNPs. Among these fabrication approaches, the citrate synthesis method remains as the most extensively investigated synthetic method by using the trisodium citrate to reduce the chloroauric acid, where the size of GNPs can be tuned by altering the salt concentration, temperature and addition of other structure directing agents.⁶⁸²

Surface chemistry not only stabilises the GNPs, but also provides organic functionalities for further conjugation with payload. The commonly used technique to tailor the metal NPs surface chemistry is the post-synthesis ligand exchange method, referring to the replacement of a labile ligand molecule on the metal NPs surface with another ligand possessing higher binding affinity.⁶⁸³ When it comes to GNPs, due to the dynamic nature of the thiol-gold (S-Au) bond, thiol-containing agents (**93**) are usually utilised to substitute the ligands (**92**) like the citrate which is used in GNP preparation (Fig. 38A).⁶⁸⁴ The thiol chemistry on GNPs has been broadly investigated for therapeutic purposes, including drug delivery and medical imaging.^{685,686} In particular, several SCLs like the ONB linker⁹⁷ and hydrazone linker²⁹² have been successfully tethered to the surface of GNPs to bear different drugs, providing GNPs with controllable drug release behaviour.

Another approach to functionalise the GNP surface is to graft functional groups to the gold surfaces through the formation of C-Au bond, which is mainly achieved by using organic reagents bearing aryl-diazonium (**94** in Fig. 38B).⁶⁸⁷ The aryl-diazonium group can be easily reduced, either electrochemically or chemically, to afford the highly reactive aryl radical that reacts with the gold surfaces.⁶⁸⁸ Since a variety of functional groups like amine, carboxylic acid and thiol groups can be placed at the *para*-position of aryl-diazonium group, this grafting method allows

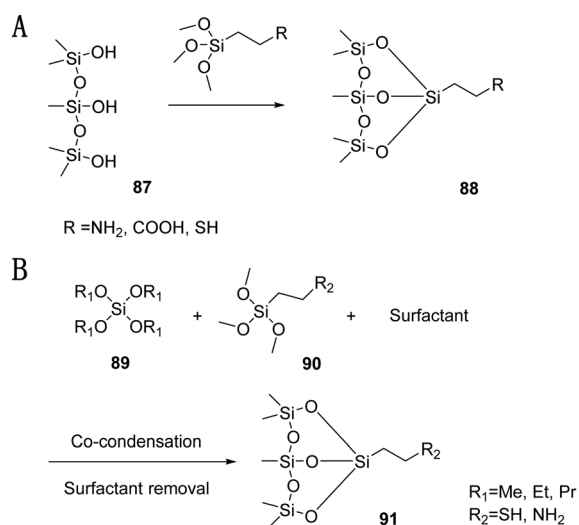


Fig. 37 Surface modification of MSNs. (A) Illustration of 'Grafting' approach to attach linker system to the surface of MSNs. (B) Schematic representation of surface modification of MSNs through co-condensation method.

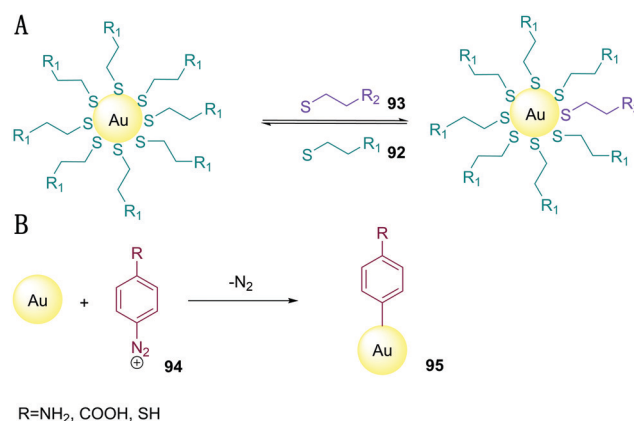


Fig. 38 Surface modification of GNPs. (A) Mechanism of reaction between thiol-containing organic ligands and GNP surface. (B) Schematic representation of surface modification of GNPs through grafting aryl-diazonium reagents to GNPs surface.

for versatile surface chemistry (95).⁶⁸⁹ However, this strategy suffers from a main disadvantage that refers to lack of control of layer thickness, as the generated aryl radicals showed high susceptibility to those grafted groups.⁶⁸⁹ To address this issue, calix[4]arene-tetradiazonium-based reagents have been used as building blocks to form closely packed and stable monolayers.^{690,691}

3.1.4 Functionalisation of organic nanoparticles. Organic nanoparticles (ONPs) refers to particles dispersions varying in size from 10 to 1000 nm, which can be prepared from natural or synthetic organic or polymeric molecules.⁶⁹² Here, we classify several widely utilised ONPs, such as dendrimers, micelles and polymeric nanoparticles (PNPs) *etc.* as ONPs. ONPs exhibit improved efficiency and engineered specificity to deliver a higher concentration of drug to the target site, compared with traditional drug administration methods.⁶⁹³

The most commonly used method to equip ONPs with stimulus-responsiveness refers to the incorporation of SCLs into the backbone of ONPs during the synthesis process. Drug molecules can either be encapsulated into the cavity of ONPs⁶⁹⁴ or be covalently attached to the SCLs.⁶⁹⁵ Various aspects of stimulus-responsive ONPs have been reviewed, including their design, synthesis and drug loading methods.^{696,697} Consequently, the ONPs have been engineered to show response to various stimuli like light,⁶⁹⁸ pH,⁶⁹⁹ enzyme,⁷⁰⁰ redox⁷⁰¹ and so on, enabling release of numerous drugs including not only small molecules⁷⁰² but also biomolecules.⁷⁰³

Since foreign nanoparticles without modification can be rapidly cleaved by the mononuclear phagocytic system (MPS), surface modification plays an important role in achieving longer circulation time of ONPs.⁷⁰⁴ Surface modification of ONPs can be achieved by physically adsorbing,⁷⁰⁵ entrapping or covalently grafting⁷⁰⁶ hydrophilic polymers onto the surface, which can repel the adsorption of opsonin protein to prevent the ONPs from being recognised by MPS. Comprehensive reviews have been published, elaborating the grafting method to form a protective layer around the ONPs.^{707,708} The most preferred polymers used for surface modification of ONPs is poly-ethylene glycol (PEG), owing to its hydrophilicity and biocompatibility.⁷⁰⁹ After the PEG coating, the ONPs showed higher spleen uptake than conventional non-PEGylated ONPs.⁷¹⁰ Aside from PEG, poly-ethylene oxides (PEO),⁷¹¹ polysorbate⁷¹² and dextran⁷¹³ also find broad applications in coating NPs.

3.1.5 Functionalisation of metal-organic frameworks. Metal-organic frameworks (MOFs) refer to a novel type of porous crystalline hybrid materials, also known as porous coordination networks (PCNs) or polymers (PCPs), which have attracted increasing popularity over the past few years for their applications in drug delivery.⁷¹⁴ MOFs are constructed through building the covalent linkage between metal-based node-centres (ions or clusters) and organic ligands, where the shape, size and surface nature of the pores can be easily tuned to provide chemical versatility and morphology.^{715–717}

Since the synthesis of MOFs has been reviewed extensively,^{718,719} here we will only briefly list several novel synthetic and surface functionalisation methods in MOF preparation. MOFs have been conventionally synthesised either at room

temperature or at high temperatures, and the fabrication process usually proceeds for several days.⁷²⁰ By offering the phase selectivity, fast crystallisation and control during the crystal formation, microwave processing has been recently utilised for the synthesis of MOFs.⁷²¹ Furthermore, the electrochemical synthesis has been reported, it is based on the reaction between the dissolved bridging ligands and nodic dissolution derived metal ions.⁷²² Other than the abovementioned methods, sonochemistry using ultrasound has also been proved to be a simple and efficient method to form MOFs, taking advantage of ultrasound power.^{723,724} A typical method to equip MOFs with stimulus responsiveness occurs during the synthesis process, where the bridging ligands containing SCLs such as disulfide bonds⁷²⁵ and azobenzenedicarboxylate (AZB) linker⁷²⁶ were incorporated into the MOFs (Fig. 39A). Following the exposure to corresponding stimuli, the SCLs can either be cleaved or isomerised to induce the drug release from MOFs.

Similar to other as-synthesised nanomaterials, MOFs require further functionalisation because of their instability in physiological conditions.⁷²⁷ Two main strategies have been proposed so far to functionalise the surface of MOFs, including coordination modulation and post-synthetic surface modification.⁷²⁸ The coordination modulation relies on the introducing a monodentate ligand with similar chemical functionality to the bridging ligand during the synthesis of MOFs, where the crystal growth can be either promoted or inhibited in the presence of the modulator (Fig. 39B). By using this method, SCLs like disulfide linkers⁷²⁹ have been incorporated into the MOFs *via* the coordination modulation, while this strategy also enables control over size of MOFs.

Meanwhile, the post-synthetic method exists as a robust and versatile approach to install the organic functionalities onto MOFs, which can be further subclassified as the surface coating, ligand exchange and covalent surface modification.^{730,731} The surface coating is applicable to various MOFs by using differing coating agents such as silica,⁷³² polymers⁷³³ and lipids,⁷³⁴ where the surface modification mainly protects the MOFs from degradation. Although further functionalisation that occurs on the coating materials is possible, the main purpose of it is for attaching targeting units such as peptides⁷³⁵ to the surface of coated MOFs (Fig. 39C).

Apart from the ligand or metal exchange method, covalent bonding also serves as an effective method to introduce SCLs into the MOFs. The bridging ligands in the MOFs is the main target for this modification, where versatile covalent transformations (Fig. 39D) can be applied to build SCLs like imine⁷³⁶ and ester⁷³⁷ between the MOFs and drugs. It is worth mentioning that novel covalent bonding methods, especially click chemistry,⁷³⁸ has the advantages such as mild conditions and high yield, which opens up the possibility to graft more functional groups to the surface of MOFs.

3.2 Conjugation chemistries

Although the surface modification of NPs provides a convenient means of conjugating the SCLs with the NPs, this procedure is highly NP-dependent where the physicochemical properties of

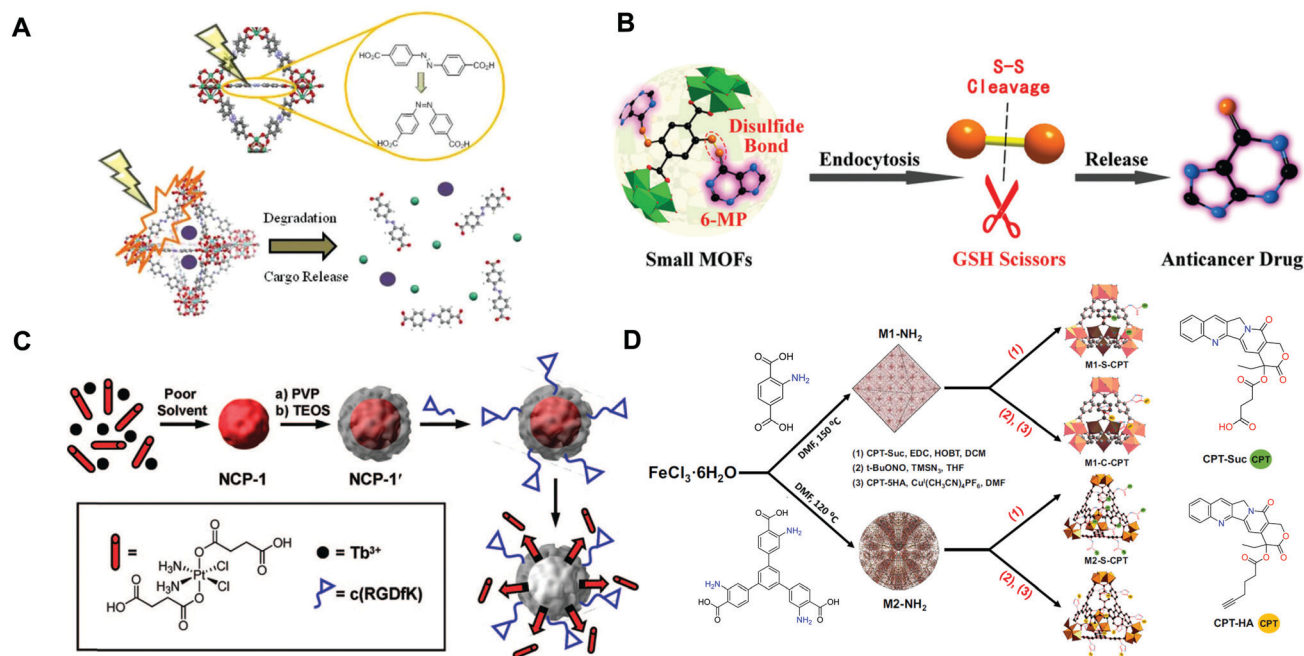


Fig. 39 Examples of different strategies to functionalise the MOFs with stimulus-responsiveness. (A) Schematic Illustration of drug release from MOFs in response to UV light, where the bridging ligand of the MOFs, AZB linker, underwent photoisomerisation from the *trans*-form to the *cis*-form, enabling drug release; (B) mechanism of drug release from GSH-responsive MOFs, which is prepared via the coordination modulation method. Benzoic acid (BA) was used as a modulator to compete with 2,5-disulfanyltterephthalic acid during the preparation of MOFs. (C) Illustration of surface coating during the synthesis of MOFs, where the targeting peptide was tethered to the silica shell to provide MOFs with tumour targeting ability. (D) Schematic illustration of covalent binding drug molecules to the bridging ligands of MOFs through the construction.

NPs predominantly determine the experimental conditions such as temperature, humidity, solvent and the corresponding reagents. Therefore, conjugation strategies based on more general mechanism are required to eliminate the inherent NP factors in order to offer expanded applicability. Normally, the initial surface modification of NPs can stabilise their surface while providing a chemical handle, where the general conjugation chemistry is performed to eventually incorporate the linker systems into the NPs. So far, several conjugation strategies have been extensively applied in the construction of DDSs, including amide coupling, maleimide-based reaction, condensation and click chemistry.

3.2.1 Amide coupling reaction. As one of the ubiquitous conjugation chemistries, the amine-carboxylic acid coupling reaction has been extensively applied in the conjugation of SCLs to drug carriers. The resulting amide bond has been shown to be stable against chemical hydrolysis under physiological conditions, and can only be cleaved under harsh conditions like high temperature and strong base or acid.⁷³⁹ Therefore, this characteristic feature makes the amide bond attractive for conjugation of amine- or carboxylic acid-containing SCLs to drug carriers.

Amide bonds can be formed through a variety of methods, which mostly focus on the activation of the carboxylic acid group prior to the addition of amine group. There are numerous outstanding reviews that systemically introduced the amide bond formation strategies.^{650,740} Among all the reported methods, carbodiimide coupling has been frequently used in the formation

of amide bond between numerous drug carriers and organic linkers.^{741–743} Carbodiimide can be either water-soluble or water-insoluble, the choice of which depends on the optimum solution providing necessary solubility of NPs or the stimulus-responsive moieties. For carbodiimides, *N*-(3-dimethylaminopropyl)-*N*'-ethylcarbodiimide (EDC) has been frequently utilised for amide coupling reaction along with *N*-hydroxysuccinimide (NHS) or sulfo-NHS as an additive.^{744,745} As shown in Fig. 40A, the amide coupling starts with the nucleophilic attack of carboxylic acid (96) to carbodiimide (97) to form an *O*-acylisourea (98) followed by reaction with NHS (99), the resulting intermediate (101) can react with amine (102) to form the final product (103). The reaction rate of EDC coupling is shown to be affected by pH value, where the ideal yield can be obtained at pH ranging from 4.5–7.5.⁷⁴⁶ Noteworthy, the amide coupling reaction using carbodiimide as coupling agent can decrease the negative surface charge of NPs, resulting in aggregation. The most effective method to address this issue is lowering the amount of coupling agent utilised for the reaction.⁷⁴⁷

Additionally, other amide coupling agents like 1-[bis-(dimethylamino)methylene]-1*H*-1,2,3-triazolo[4,5-*b*]pyridinium 3-oxid hexafluorophosphate (HATU) can also be used to construct amide linkage between the drug carriers and SCLs. HATU has been proven to be efficient in sterically hindered couplings and the most important merit of this coupling agent is the ability to prevent the racemisation during the synthesis, leading to its wide applications in peptide synthesis.⁷⁴⁸ The mechanism of HATU catalysed amide coupling is shown in Fig. 40B, where the

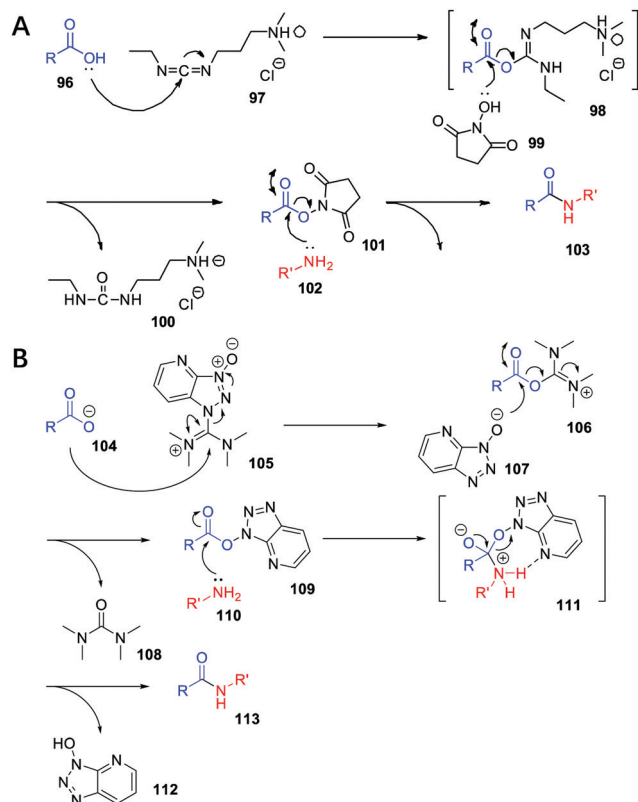


Fig. 40 Schematic representation of amide coupling reaction. (A) Mechanism of EDC/NHS assisted amide coupling reaction. (B) Attachment of linker system to the drug carrier through the formation of amide bond, which is catalysed by HATU.

carboxylate anion (104) attacks HATU (105) to form the active intermediate, *O*-acyl(tetramethyl)isouronium salt (106), which is then attacked by the other product of carboxylic acid activation by HATU, referring to 1-hydroxy-7-azabenzotriazole (HOAt) anion (107), to form the OAt-ester (109). Following the carboxylic acid activation step, addition of amine (110) to the OAt-ester results in the amide linkage (113). In terms of amide coupling reactions occurring on NPs, HATU may serve better than EDC/NHS in some cases where the steric hindrance predominantly affects the yield of the reaction.^{749,750}

3.2.2 Thiol-Michael chemistry. The second most common conjugation reaction is the thiol-maleimide reaction. Among the possibilities, the nucleophilic addition of thiols to C=C bond remains as one of the most commonly utilised strategy for conjugating small molecules or biomolecules to drug carriers owing to its facile and powerful nature (Fig. 41).⁷⁵¹ This nucleophilic addition reaction is also known as thiol-Michael addition, the reaction rate of which increases along with the increment of pH value.⁷⁵² The thiol-Michael addition has gained increasing interest for tethering organic linkers to numerous drug carriers such as antibodies and NPs, which is mainly because the reaction proceeds without the formation of significant side products like radical-radical termination products.⁷⁵³

To enable the thiol-Michael addition, an electron withdrawing activating group, also referred to Michael acceptor, is required.

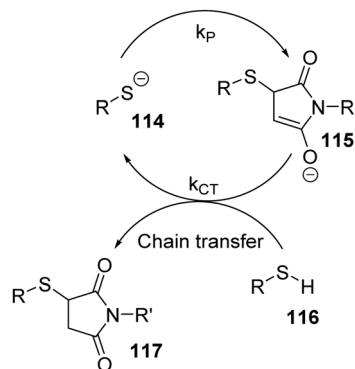


Fig. 41 In the thiol-Michael nucleophilic reaction mechanism, the nucleophile undergoes conjugate addition to the activated C=C bond, generating the strong intermediate carbanion, which, in turn, deprotonates the thiol, subsequently undergoing thiol-Michael addition again.

Different kinds of Michael acceptors have been reported so far, including cyanoacrylates, acrylate esters, acrylonitrile, maleimides and so on. In particular, maleimides display unique reactivity in thiol-Michael addition owing to the synergistic lowest unoccupied molecular orbital (LUMO) lowering effects, making it more reactive than other typical Michael receptors.⁷⁵⁴ The mechanism of thiol-Michael addition reaction is shown in Fig. 41, where the thiolate anion (114) attacks the π -bond of maleimide to form a basic enolate intermediate (115), which subsequently deprotonates an additional equivalent of thiol (116) to give rise to the addition product (117) and another equivalent of thiolate that perpetuates the catalytic cycle.

Consequently, the thiol-Michael addition has been widely utilised in conjugating various kinds of organic functionalities to drug carriers, covering antibodies,⁵⁶⁷ ONPs,⁷⁵⁵ GNPs,⁷⁵⁶ silica nanoparticles (SiNPs)⁷⁵⁷ and so on. Recently, several novel maleimide scaffolds have been introduced, including *N*-aryl maleimides,⁷⁵⁸ β -amino maleimides,⁷⁵⁹ acetal-containing maleimides⁷⁶⁰ and exocyclic maleimides.⁷⁶¹

3.2.3 'Click chemistry'. The concept of 'click chemistry' was first introduced in 2001. Its characteristics include high yield, less toxic side products and high stereospecificity.⁷⁶² In addition, click chemistry proceeds under mild conditions to couple small elements with heteroatoms in the form of R-X-R, and most importantly, the reaction can occur in both aqueous media and organic solvents.⁷⁶³ Therefore, click chemistry has been widely applied to modify the surface chemistry of various drug carriers for bioimaging⁷⁶⁴ and drug delivery⁷⁶⁵ purposes.

The Cu(I)-catalysed azide alkyne cycloaddition (CuAAC) remains as the most extensively applied click reaction, probably because of its high orthogonality, reliability and most importantly, the experimental simplicity for researchers without synthetic chemistry background.⁷⁶⁶ The mechanism of CuAAC reaction is shown in Fig. 42A, where the 1,3-dipolar addition occurs between the terminal alkynes (118) and azides (120) to afford the 1,2,3-triazole (124).⁷⁶⁷ Bearing high tolerance to aqueous solution with a broad range of pH level, CuAAC chemistry has been widely used for conjugating numerous organic functionalities with drug carriers like antibodies,⁷⁶⁸

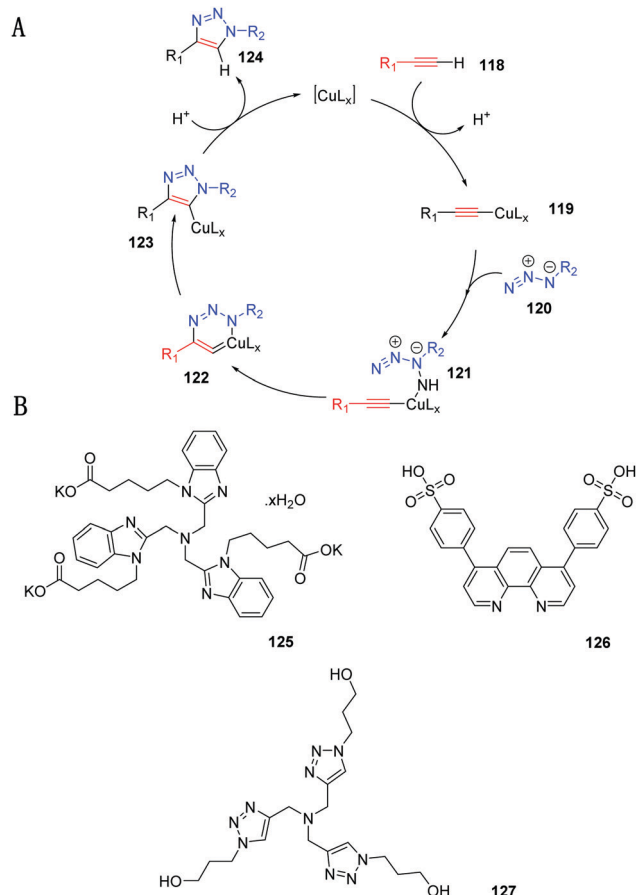


Fig. 42 (A) Mechanism of Cu(I)-catalysed azide-alkyne cycloaddition (CuAAC) reaction. (B) Chemical structures of Cu(I) conjugating ligands (BimC4A)₃, BPDS and THPTA.

ONPs,^{769,770} MSNs⁷⁷¹ and GNPs.⁷⁷² Nevertheless, the Cu(I) used for CuAAC reaction may cause structural damage to biomolecules under standard experimental condition, and the lack of adequate kinetics in some cases also inhibits its further applications in bioconjugation. To accelerate the CuAAC process, numerous Cu(I) conjugating ligands, such as tris(benzimidazole) (BimC4A)₃ (125),⁷⁷³ sulfonated bathophenanthroline (BPDS) (126),⁷⁷⁴ tris(hydroxypropyltriazolylmethyl)amine (THPTA) (127)⁷⁷⁴ (Fig. 42B) were introduced into the reaction to stabilise the oxidation state of Cu^I and thus increase its catalytic efficiency.

To improve the biocompatibility of CuAAC, the best way is to avoid using Cu(I) catalyst. Therefore, a novel strain-promoted azide-alkyne cycloaddition (SPAAC) method is proposed to optimise the conventional CuAAC, based on the robust ligation between azides (128) and the inherent ring strain of cyclooctynes (OCT) (129) to generate final product 130.⁷⁷⁵ Although the SPAAC eliminates the requirement of Cu(I) catalysis, it seems to proceed at a relatively slow rate compared with CuAAC, where the second-order rate constant of the reaction is only 0.0024 M⁻¹ s⁻¹.⁷⁷⁶ For this reason, significant efforts have been dedicated to the improvement of OCT reagents used in SPAAC (Fig. 43A), resulting in several modified OCTs which can accelerate the process, including difluorinatedcyclooctyne (DIFO) (130), dimethoxyazacyclooctyne

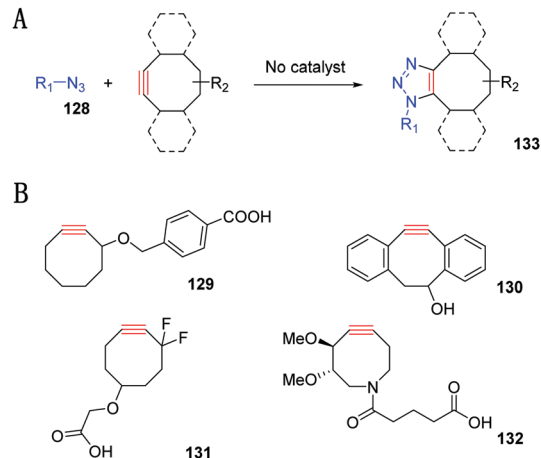


Fig. 43 (A) Schematic representation of strain-promoted azide-alkyne cycloaddition (SPAAC) reaction between azide and OCT. (B) Chemical structures of modified OCT covering DIFO, DIMAC and DIBO.

(DIMAC) (131) and dibenzocyclooctyne (DBCO) (132) (Fig. 43B).⁷⁶² This low toxic and quicker conjugation strategy has soon found wide applicability in attaching various organic functionalities to drug carriers, including ONPs⁷⁷⁷ and antibodies.⁷⁷⁸ The main disadvantage of SPAAC is the steric hindrance and hydrophobic nature of OCTs, which may impair the binding efficiency in some cases where large biomolecules like proteins are used.⁷⁷⁹

In another attempt to obtain faster reaction rates, an inverse electron demand Diels-Alder (IEDDA) method involving *trans*-cyclooctene (TCO) 134 derivatives and *s*-tetrazine 135 was recently developed (Fig. 44A).⁷⁸⁰ The TCO-tetrazine ligation is one of the most reactive biorthogonal reaction, and proceeds at the spectacular rate of 10⁶ M⁻¹ s⁻¹ without the addition of any catalyst.⁷⁸¹ The biorthogonal reaction has been optimised in different ways to improve product stability and reaction kinetics,^{782,783} and most importantly, these reagents showed

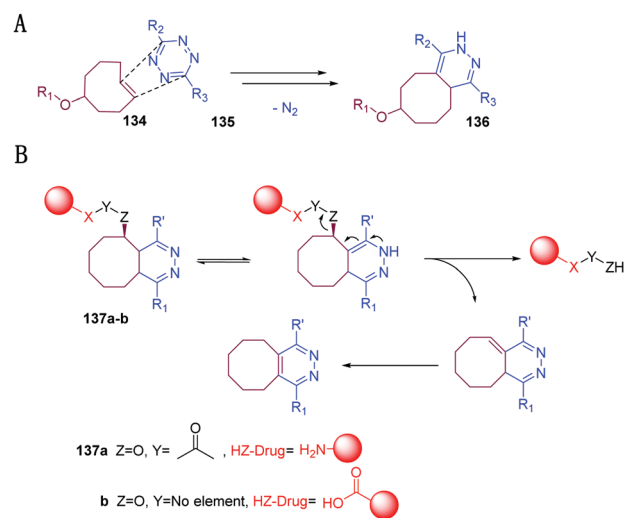


Fig. 44 (A) Schematic representation of strain-promoted azide-alkyne cycloaddition (SPAAC) reaction between azide and OCT. (B) Chemical structures of modified OCT covering DIFO, DIMAC and DIBO.

low or no toxicity to cells at normal conditions. Interestingly, the 4,5-dihydropyridazine product formed by reacting tetrazine with TCO can convert to its 1,4-tautomers, which undergoes subsequent electronic cascade to release the payload (shown in Fig. 44B). This phenomenon has attracted significant attention for drug delivery purpose, by offering a novel drug release mechanism known as ‘click and release’. Consequently, the iDEEA reaction has been well studied for prodrug formations⁷⁸⁴ and DDS construction,⁷⁸⁵ where the drugs bearing amine (**137a**) and carboxylic acid group (**137b**) are tethered to TCO scaffold. Moreover, the cycloaddition partners was not limited to TCO/tetrazine pair but expanded to TCO/azide,⁷⁸⁶ 3-isocyanopropyl (ICP)/tetrazine⁷⁸⁷ and vinyl ether/tetrazine⁷⁸⁸ pairs, the application of which are mainly in prodrug formations.

4. *In vitro* biocompatibilities, *in vivo* biological models and applications of stimulus-cleavable linkers

In Tables 1–5, we have summarised all biological applications of SCLs in DDSs. Here, we will further discuss their *in vitro* biocompatibility, *in vivo* biological models and bioapplications in details.

4.1 *In vitro* biocompatibilities of stimulus-cleavable linkers

The *in vitro* biocompatibilities and aforementioned SCLs have been summarised in Fig. 45. Immortalised cell and tumour cell lines are frequently used cell lines for *in vitro* studies. This is not only because they were easy to culture/handle, but also due to different cell lines providing various stimuli for cell-based validation of the cleavable reaction.⁷⁸⁹ But several publications have also used more difficult to handle primary cells.^{226,399,400,438,490,575,593} From simple to complicated, four levels of biological issues need to be addressed *in vitro* based on our literatures review.

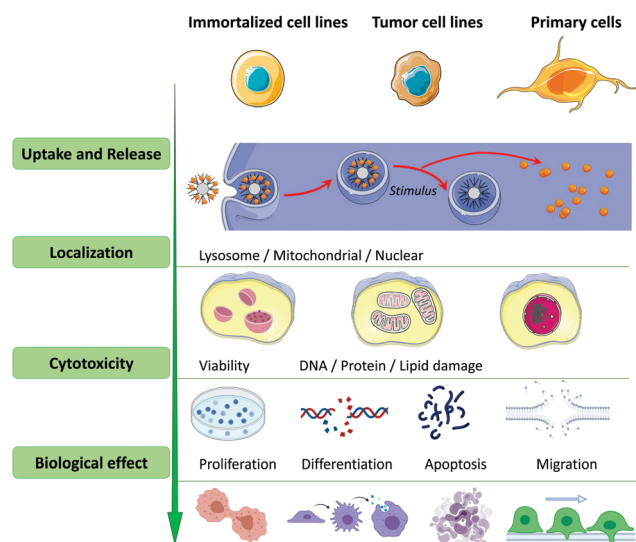


Fig. 45 Summary of SCLs for *in vitro* application from references cited.

Cellular uptake and release are a prime concern in the whole process since SCLs may be attached to big/hydrophobic cargos (drugs, fluorophores or even proteins), which often do not easily cross the plasma membrane. So, scientists have been designing quite a lot of successful cellular delivery strategies to release cargos at suitable sites, such as endocytosis free methods, subcellular targeting strategies and so on. This cell permeability is a necessary precondition of cell-based or even *in vivo* applications.

The next issue is cellular localisation. There are three mainly targeted subcellular organelles, namely lysosomes, mitochondria and nuclei. Lysosomes receive and degrade macromolecules from the secretory, endocytic, autophagic and phagocytic membrane-trafficking pathways. The main function of lysosomes is the digestion of internal (non-functional cell organelles) and external (food, bacteria, leukocytes, debris) material. Consequently, most of the cargos may be trapped/localised in lysosomes after crossing cell membrane. This provides convenience for the study of lysosome function, which is closely related to the development of disease.⁷⁹⁰ Mitochondria and nuclei are also popular subcellular targeted organelles for SCL design and bio-applications. Not only because mitochondria and nuclei are the “power house” and “headquarters” of the cell, respectively, but also due to them being closely related to cell fates in many physiological and pathological processes.⁷⁹¹ Confocal fluorescence microscopy imaging has been the common method to study localisation of SCLs. Nevertheless, super-resolution fluorescence imaging and electron microscopy imaging have been involved in those investigation in increasing numbers recently.

Cytotoxicity has been used to evaluate the bio-function/compatibility of SCLs together with their cargos in almost all the references. Viability assays which use MTT, XTT or PrestoBlue as reagents are the most commonly applied *in cellulo* analysis. However, in addition to viability assays, deeper biological analysis such as DNA/protein/lipid damage are also key biological parameters for cytotoxicity evaluation after SCLs being cleaved. PCR and Western Blot (WB) are meaningful experiments for DNA copy numbers and protein expression, respectively. At the same time, there are numerous assay kits/probes for functional detection of different proteins before/after SCLs being cleaved in cells. Finally, the thiobarbituric acid reactive substances (TBARS) assay is a common way to measure lipid damage in cells.

The last issue of investigating SCLs *in vitro* are biological effects on proliferation,^{355,356} differentiation, apoptosis²²⁸ and migration of different cells. Cell proliferation is defined as the growth or production of cells by multiplication of parts. Cell differentiation is a process in which a generic cell develops into a specific type of cell in response to specific triggers. Apoptosis is the term used to describe the generally normal death of the cell. Cell migration is the process by which cells move from one location to another by adopting different motility modes.⁷⁹² Over the past few decades, SCLs have contributed important progress in understanding the mechanisms of those biological effects in cells. However, results obtained *in vitro* often poorly correlate with *in vivo* performance, partly because the intracellular and extracellular environments *in vitro* have little physiological relevance.

4.2 *In vivo* biological models for bio-applications of stimulus-cleavable linkers

Although some SCLs show excellent performance *in vitro*, they fail to deliver cargos in the more complex *in vivo* models. For instance, ONB groups and coumarin esters are able to release drugs under UV irradiation. However, due to the poor tissue permeability of the UV light, these linkers cannot be used in most *in vivo* models. In this section, the *in vivo* applications of SCLs are summarised. In addition, an outlook and guidance of the use of these linkers in biomedicine and clinics were provided as well. As shown in Fig. 46, we have categorised *in vivo* animal models in biological applications of SCLs as non-mammals, small mammals and large mammals reported in the field of oncology, embryology,¹⁰⁰ nerve systems,¹⁶² cardiovascular,¹⁹⁶ gastrointestinal systems,²⁰⁶ orthopedics,²³³ wound healing,²⁴³ obesity,³⁹⁹ infection,³⁹⁷ irradiation,⁴⁵⁴ ischemia/reperfusion,^{400,602} inflammation,⁵⁸⁹ and so on. The selection of which organisms to use as a research model depends on the nature of the disease or mechanism being studied.

C. Elegans,⁵⁵⁹ fly/*Drosophila*⁵⁹⁹ and Zebrafish¹⁰⁰ are three commonly used non-mammalian model organisms to study SCLs. Compared to traditional mammalian models, they provide numerous experimental advantages for study of SCLs including easy genetic processing, low experimental cost, and high experimental throughput.⁷⁹³ *C. elegans* is a transparent, free-living, non-parasitic nematode. Its neurons have been identified and the

connections between them are well known. Therefore, by using fluorescent markers, processes such as axon growth, embryogenesis and fat metabolism can easily be studied in the living worm. *C. elegans* is also a powerful model system for the study of aging, because of its genetics and relatively short lifespan (20 days).⁷⁹⁴ *Drosophila*, with a fully sequenced genome,⁷⁹⁵ is a valuable model to define gene function, reveal the molecular bases of physiological processes, and study mechanisms underlying human diseases. *Drosophila* has differentiated organs that are functionally equivalent to the mammalian ones, making it a powerful model to study complex biological processes such as organogenesis, learning and behaviour. Zebrafish has many features that make it an excellent animal model for developmental research in vertebrates.⁷⁹⁶ Early-stage zebrafish embryos develop externally to the mother and are transparent, making it relatively easy to be manipulated and monitored in real time using high-quality imaging.

Mammalian model organisms, in particular rodents, are essential for scientists to evaluate the biological efficacy of SCLs and further improve design strategies. As shown in Tables 1–5, both small mammals (mouse/rat/rabbit) and large mammals (dog/sheep/monkey) have been used in the context of SCLs. Among them, the most widely used animal models are rodents, including mice and rats.⁷⁹⁷ In terms of oncology research, over 95% of models were conducted in mice. Subcutaneous xenograft are the most commonly used models to determine the efficacy of prodrugs delivery. In these subcutaneous models, the tumour is grown under the skin of a nude or

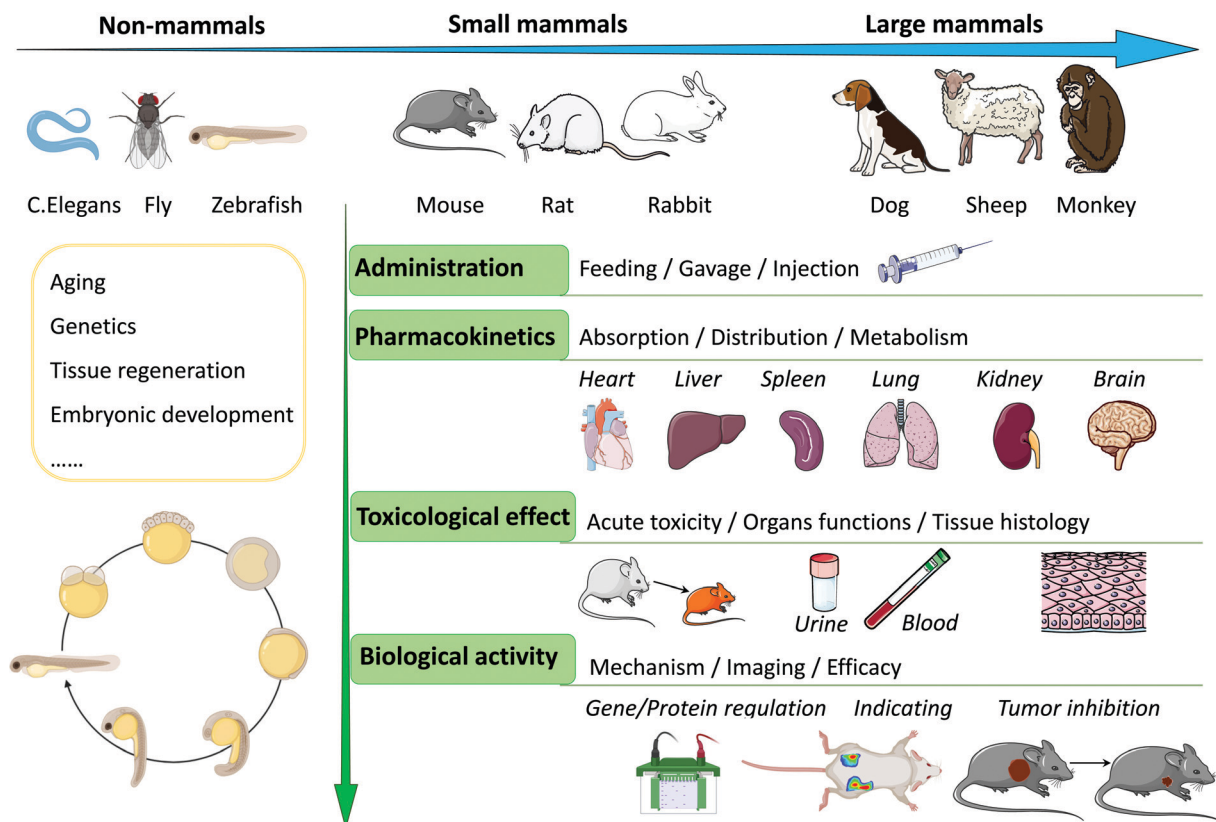


Fig. 46 Summary of SCLs for *in vivo* application from references cited.

severe combined immunodeficiency mouse. Rats were also used in some biomedical models such as wound healing²⁴³ and ischemia,⁴⁰⁰ perfusion.⁶⁰² Rabbits are a standard ophthalmic animal model. Due to the unique anatomy of the rabbit eye, ocular hypertension and retinal or macular damage have been successfully induced by drugs or lasering. In the context of SCLs, rabbit has also been used to study the delivery of antihypertensive drugs.^{310,312,569} Although rodent models have been invaluable in furthering the mechanistic understanding of human disease, there are circumstances in which choosing a large animal species in place of a rodent model is scientifically justifiable.⁷⁹⁸ For instance, when investigating the delivery of triclabendazole, a medication used to treat fascioliasis, sheep would be the best animal model to be infected by *F. hepatica metacercariae*.³⁹⁷ When it comes to the prodrug delivery for HIV treatment, monkeys were considered to be better than other animals.^{400,577} Despite the numerous advantages of large animal models, the high cost of animal maintenance and husbandry are key limitations that have impeded their widespread use in the application of SCLs.

What are next focal areas following the publication of numerous designs of SCLs in the biological field? From simple to complicated, four levels of biological issues need to be addressed *in vivo* in our view, such as administration, pharmacokinetics, toxicological effect and biological activity (Fig. 46). In administration, injection of SCL containing DDSs is more commonly used in mammalian model organisms than oral gavage/feeding. Although the oral route is a preferred way of administration, drug delivery efficiency and off-target side effects are often challenging.⁷⁹⁹

For pharmacokinetics, absorption/distribution/metabolism are three key parameters have been involved in the studying of *in vivo* biological applications of SCLs. Nearly all organs (heart/liver/spleen/lung/kidney/brain) have been involved in these studies.

In terms of toxicological effects, acute toxicity, organ function and tissue histology are considered. The body weight, IC₅₀ value, urine/blood test and hematoxylin-eosin staining (H&E) are usually considered in the literatures. For biological activity parts, scientists mainly focused on mechanism exploration based on gene/protein regulation, imaging whole biological process and treatment effectiveness/efficacy of SCLs. In this case, gene/protein regulation related experiments are critical. Meanwhile, quite a lot of strategies have been devised for imaging the chemical/biological processes involving SCLs.

Although numerous SCLs have been developed to investigate various biological problems such as biomarker specific labelling and efficient drug delivery, the available SCLs are limited to bench use only and have barely met the baseline criteria for clinical use. To address this problem, one solution might lie in the standardisation of biochemical and physiological research step-by-step to evaluate the influence of SCLs on biological parameters, *e.g.* standardisation of pharmacokinetics and toxicological evaluation for clear demonstration of the safety and efficiency. To this end, we need contributions from not only chemists/biologists/physicists, but also clinical doctors/users/manufactures.

Through the standardisation process, we envision that more robust reagents based on next-generation SCLs could be developed for *in vitro* diagnostics or clinical therapies, which can bring notable advancement to research and health care in the near future.

4.3. Biological applications of stimulus-cleavable linkers

SCLs exhibit properties that make them ideal candidates for biosensing and bioimaging.⁸⁰⁰ When these linkers are specifically designed to respond to biomarkers, stimuli-responsive probes which are capable of recognising and transducing analyte signals, can be used to diagnose and treat disease. Those markers are traditionally analysed by sophisticated instrumentation and detailed protocols that limit their applications. But simple, fast, sensitive and low-cost detection strategies can be developed with the SCLs system. Although the current investigations have shown the promise and success of the SCLs in preclinical tests for diagnostic applications, most of the technology cannot meet the requirements for clinical translation. This is due to a great diversity of compounds in the real samples harvested from the patients of various conditions which significantly interfere with the accuracy and sensitivity of the measurement. Besides detecting the level of the clinically relevant analytes, their local distribution in the human body and continuous monitoring are also problems to be solved. Thus, the SCL-based diagnostic systems remain in the early stage.

Besides diagnostics, another important application is disease treatment where SCLs have also exhibited great promise.⁸⁰¹ DDSs and pharmaceutical formulations are very important for effective, safe, and convenient administration of drugs. SCLs are used in the systems and formulations for interaction with the surface of the body, organs, tissues, and cells to improve efficacy, safety, and adherence. Based on our knowledge, most SCLs systems focus on controlled drug release for cancer treatment. This is predicated on the fact that the microenvironment of tumour tissues differs significantly from normal tissues. The characteristics of low pH value, ROS overproduction, high GSH level and hypoxia can provide good choices for endogenous stimulus. Thus the pH-, ROS-, redox- and enzyme-responsive drug carriers have been widely investigated to achieve different effects such as better tissue or membrane penetration, reduced toxicity, extended circulation time, improved accumulation, and controllable release of active therapeutics. Although there is much excellent research, most of it focuses on one point of the challenges and remains at the early stage of concept-proof.

In the past decade, SCLs have been extensively investigated in preclinical tests. Nevertheless, few have been commercialised let alone entered into clinical trials. The transformation of SCLs is often limited by the complexity of their structures and mechanisms. The design of bio-responsive materials should focus on easy conversion, rather than modifying complex structures or using new methods. More importantly, a majority of publications are based on simple nude mouse models, which cannot provide sufficient evidence for clinical translation. Thereby, it is urgent to explore models that more accurately mimic the human body and clinical diseases.

5. Conclusions and future prospects

In this review, we systematically introduced various types of SCLs, the cleavage of which is triggered by either exogenous or endogenous stimuli. The development of SCL technology has primarily focused on cancer treatment, where numerous hallmarks of tumour cells such as upregulated expression of catalytic enzymes and higher acidity in microenvironment, have been extensively investigated as stimuli to trigger the cleavage of the corresponding SCLs. In addition, the utilisation of exogenous stimuli like light can also promote linker cleavage to enable the release of therapeutic agents or fluorophores. Therefore, the variety of cancer-related stimuli provides researchers with flexible choice of cleavable linkers, while the application of them has also been expanded to treatment of other diseases.

Apart from controlling the release site, the cleavage rate and drug candidates of SCLs can also be adjusted by introducing structural variations to the linker system. For some SCLs with limited available drug candidates, we also discussed the possibility to expand their application to a wide range of drug molecules by introducing self-immolative spacers into the linker system. With the details given in this review, readers even without extensive chemistry background could gain a comprehensive understanding of inherent properties of these SCLs covering their stimulus sensitivity, drug candidates and synthesis difficulty, which will be useful when choosing the appropriate linker system for DDS construction.

Moreover, we introduced three different ways to provide the drug carriers with stimuli-responsiveness in terms of controlled drug release, depending on the drug carrier type. For drug carriers like antibodies, pSiNPs and GNPs, utilising the SCLs to bridge the drug molecules to the drug carriers remains as a direct and convenient way to achieve the goal, and numerous successful cases using this method can be found. In addition, SCLs can also be incorporated into the frameworks of drug carriers to offer stimulus responsiveness. This strategy is commonly used when preparing the ONPs such as dendrimers and micelles, where the SCLs can be either cleaved or isomerised after stimulus exposure to enable drug release.

As a unique strategy to realise controlled drug release on MSNs, gating chemistry has attracted significant attention over the past few years.⁷⁴ A molecular gate can be formed by tethering the SCLs to the pore openings of MSNs, preventing the drug molecules entrapped into the pores from leaking, and stimuli-induced linker cleavage or isomerisation leads to the drug release.

Since there are numerous outstanding reviews summarising the preparation of stimulus-responsive NPs (*e.g.* ONPs), we only focus on the strategies to install the SCLs to the drug carriers. Although the surface modification strategies vary according to drug carrier type, in this article we have tried to summarise some general conjugation methods to tether the SCLs to the surface of drug carriers. These methods refer to the combination of conventional surface modification and popular bioconjugation strategies, where the surface of the drug carrier is

modified with chemical handles such as amine or azide group that serve as the attaching sites for further conjugation like amide coupling or 'click' reaction. Since the bioconjugation strategies often occur under mild conditions with high yield, and most importantly, they do not require professional training in organic chemistry, these characteristics seem to be appealing for researchers looking for simple and efficient conjugation strategies to attach SCLs to drug carriers.

However, challenges still remain regarding the optimal design of DDSs. To begin with, the synthesis of some SCLs with complicated structure appears to be challenging for material scientists and biologists, especially during the scale-up process. Additionally, attention needs to be paid to drug molecules attached to SCLs in order to retain their biological activity, in particular when introducing a new drug molecule to an already reported linker structure. Likewise, the interaction between the drug carriers and linker-drug conjugates also needs to be taken into consideration. Although we proposed an array of general conjugation strategies to install the assembled linker-drug conjugates onto the drug carriers, these reactions may suffer from the undesired problems like steric congestion, which is particularly evident in cases where co-attachment of drug molecules with targeting ligands on the drug carriers' surface is pursued.^{802,803} Another challenge is to minimise the toxicity of the drug carriers, especially when assembled NPs are used. Such a problem may be derived from many aspects, including the NPs, linker and drug. In this article, we summarised the potential toxicity of some model drugs and SCLs but did not mention the toxicity of NPs. Generally speaking, the physicochemical properties of NPs like size,⁸⁰⁴ shape⁸⁰⁵ and surface charge⁸⁰⁶ can affect their toxicity by influencing how they interact with cells. Dose⁶⁸¹ and chemical composition⁸⁰⁷ are additional factors in nanotoxicity. All of these need to be considered in an effort to construct safer nanomaterial-based DDSs.

In summary, we have tried to provide the readers with a deeper understanding of SCLs by not limiting their availability to a specific drug carrier or drug molecule, and other aspects important to DDS design. And this article is expected to spark new ideas on addressing the problems faced by the current SCLs and develop novel multi-stimuli system for drug delivery purpose.

Conflicts of interest

There are no conflicts to declare.

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